



International Technology Roadmap for Wide Bandgap Power Semiconductors

2019 Edition

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Chapter 1: Foreword

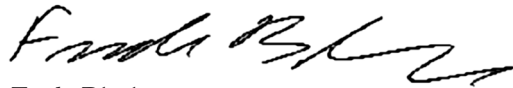
1.1 Foreword from the IEEE Power Electronics Society President

The International Technology Roadmap for Wide Bandgap Power Semiconductors (ITRW) has been developed as an IEEE Power Electronics Society Initiative to inspire leadership in the emerging and fast moving field of wide bandgap power electronics. The impact of wide bandgap power electronics has been nothing short of revolutionary across all aspects of power electronics giving dramatic changes in efficiency, performance and physical size.

In 2015, the IEEE Power Electronics Society initiated the ITRW project to develop a strategic roadmap to provide vision and leadership to academia, industry and consumers as to where the wide bandgap technology was moving. Several working groups were formed as well as an industry advisory board and a global steering committee to ensure that the resulting work would be genuinely global in nature, and build on the IEEE's role as a neutral forum for advancing humanity. Meetings have been held across the world since the inaugural meeting in December 2015 at TU Delft, including North America, Europe, and Asia to allow as many stakeholders as possible to influence the work of the roadmap and provide their valuable input.

The publication of this inaugural ITRW roadmap is a direct result of the sponsorship and leadership provided by the IEEE Power Electronics Society, with a long-term commitment of several presidents supporting this initiative including Prof. Bram Ferreira, Prof. Alan Mantooth and Prof. Frede Blaabjerg. As can be seen from the extensive list of volunteers who have contributed to the roadmap, much time and effort has been provided by the PELS community at large and this is reflected in the diversity of technology (both fundamental and applications oriented) presented in the ITRW roadmap. The team has also highlighted key technological successes to illustrate how wide bandgap power electronics have made a major impact across the world from grid-connected systems, to mobile communications and computing and transport electrification.

I hope you enjoy reading the ITRW roadmap and find the material contained within it interesting, inspiring, valuable and thought provoking.



Frede Blaabjerg
President, IEEE Power Electronics Society

1.2 Foreword from the ITRW Advisory Board Chairman

Over the last decade, Wide Bandgap Power Semiconductors (ITRW) left the initial niche of being used in isolated applications only. With the availability of more products and especially transistors by various suppliers world wide the penetration increased significantly. Connected to this development there is an increasing interest of all participants worldwide to get guidance and confidence about future developments and the ultimate potential of the new technologies. The industry is excited to be a part of IEEE's initiative to offer such a program in form of the ITRW roadmap. This is reflected by the constitution of the IAB itself where leading representatives from the major players in the field agreed to serve voluntarily as reviewers for the roadmap drafts.

Thanks to the enormous efforts of Bram Ferreira as the face and heart of the initiative, supported by a powerful international team of experts, we can now present the first edition of the WBG roadmap. We are convinced that the document is able to support all aspects of the new technology today and in the near future, beginning from materials over devices until the ultimate end application. Similar to previous IEEE roadmap efforts we believe that the ITRW document will become one of the key information sources for stakeholders in the wide band gap area.

A special thanks again is given to the working groups which spent an tremendous amount of time in collecting information, conducting interviews and evaluating questionnaires. The result can be judged as the best available essence of the global view and future of wide band gap technologies.

Of course roadmaps are always living documents, so the ITRW team will be excited to get feedback and further inspiration. Provided you are inspired by the content, feel free to join us in the working groups or the advisory bodies in order to work on next editions of the ITRW roadmap.



Peter Friedrichs
Chair of the ITRW IAB

1.3 Acknowledgements

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Chapter 2: Executive Summary

2.1 Motivation

The world is changing rapidly with ever-increasing demands of reliable energy supply and increased electrification of devices and transportation. This is giving a compelling market pull for power electronics products to support expanding markets related to e-mobility and renewable energy. This market is expected to exceed **1000 TW-units per year in 10 years**. (For comparison, the current market for computer and laptop power supplies is **2 TW-units**). On the one hand, there is this market pull of substantial growth related to the energy transitions and on the other hand, there is in the technology push of the potential transition from silicon device to new wide bandgap (WBG) semiconductors. The new generation of WBG semiconductor power electronic devices are potentially **100 – 1000 times faster** and potentially have **100 – 1000 times lower losses** than their silicon counterparts. The latest report from market research firm Yole Développement SA suggests that total market for **SiC power devices by 2023 will be over \$1.5 billion** with a compound annual growth rate (CAGR) of **31% for next five years**, and with rising GaN adoption in a variety of applications, Yole is projecting the **GaN device business** to reach around **\$423 million by 2023 with a CAGR of 55%**. Projections by PowerAmerica are that the total market for WBG devices in power electronics will be **over \$11 billion by 2027**.

There is, however, a cost related to this new technology. Existing silicon technology is well established, with proven reliability and lower costs generally per unit than the newer wide bandgap technologies. There is therefore a trade-off between the **superior characteristics** in many regards of the wide bandgap devices compared to their silicon equivalents, **a positive market outlook** (which will eventually drive down the cost of wide bandgap technology) against the status quo of silicon based devices and systems. Several regional roadmaps exist that are already exploring how this new technology will impact on their regional markets: USA (Power America), Europe (ECPE and NEREID), China (CASA) and Japan (SiC Alliance); anticipating opportunities for their industry when suitable technology readiness levels will be achieved.

A key role of this roadmap, the International Technology Roadmap for Wide bandgap Power Semiconductors (ITRW) is to facilitate an acceleration in the R&D process for this new technology to fulfil its potential. ITRW provides a pre-competitive, embracing platform for entities to:

1. **Share** R&D progress and identify opportunities and bottlenecks,
2. **Identify** most effective paths for technology development,
3. **Develop** technology specific content within working groups,
4. **Create** a reference framework for regional roadmaps.

As the ITRW falls within the IEEE, it has a unique position to present a global perspective and to provide a neutral forum that is not dictated by national or regional market or regulatory demands.

2.2 A Market View

It is clear that wide bandgap power semiconductors are already gaining in initial markets where there is a **clear technological advantage** without an excessive cost to entry. One of the earliest WBG devices to enter the marketplace is the Silicon Carbide (SiC) power diode, with broad penetration into specific sectors including PV converters and electrical machine drives. These devices offer **immediate improvements in efficiency, higher voltages and thermal behavior**. Gallium Nitride (GaN) devices are also becoming commonly used in applications such as low voltage lighting, where regulatory constraints on the sale of incandescent lighting has provided an opportunity for higher cost products to gain a foothold where they would normally be uncompetitive, and consumer electronics applications.

An important question is why would the market accept these new devices over an established silicon based system, with an established eco-system of supply and production? There are **several compelling arguments** in favor of WBG device technology for power electronics applications that provide an opportunity for these devices to become prevalent in at least **50% of the existing market in power electronics**.

1. **Reduced power losses** and increased efficiency: typical 50% improvement is possible,
2. **Reduced complexity** due to reduced power conversion stages made possible by high voltage performance,
3. **Improved power density** by increasing the switching frequency,
4. **Reduced acoustic noise or EMI signature**, by increasing the switching frequency beyond audible range and conducted EMI band.

The most important market categories for market penetration are the growing established markets that would benefit from the adoption of WBG technology for power electronics including **PV converters, automotive hybrid and pure electric drivetrains, aerospace and data centers**, and new applications such as **wireless power and electric grid components**. There are also some niche applications exist where WBG has a very clear benefit, for example radiation hardened electronic systems for **space and medical MRI equipment**.

2.3 Looking in the Crystal Ball

The mechanism by which the **ITRW can facilitate an acceleration of the R&D process** is to signpost paths of technology development, using a series of cause and effect events that lead to a specific destination. Providing a list of metrics and benchmarks is not enough in itself, as this is essentially static, and does not describe adequately the path to the desired outcomes. **The participants in the ITRW process come from the world leading R&D groups** that are intimately involved in the dynamic path that the technology development has and will be following.

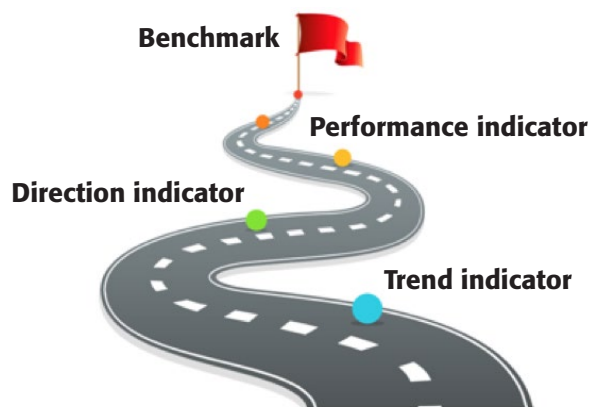
As active experts, we can try to map some future paths, taking history and experience into account, and noting carefully current developments. These dynamic paths have a series of indicators based on the horizon distance, with **short-term performance indicators and benchmarks** being appropriate for existing products and devices, particularly useful for independent and objective comparison of technology.

Direction indicators of **medium term innovations of proven research** show how these technologies could transition into the marketplace and become products of the future.

Finally, indicators that are more “blue sky” in nature show **longer term trends** and the extent of the research landscape and **highlight the potential for research that could lead to exciting new technology in the future**.

ITRW participation comes from the world leading R&D groups that intimately involved in the dynamic path of technology development. The following are potential high impact paths that could unfold during the next 10 years:

- Wide bandgap power device robustness defined in converter context. SiC and GaN do not necessarily need to on par with Si, but the overall converter robustness should be able to meet stringent requirements. For this new test, procedures and equipment could be developed to be used in new standards.
- Heterogeneous integrated power circuits with small footprint on organic substrate using GaN devices. The eminent very large production numbers in existing and new market products makes the required investment in fabs viable.
- Smaller footprint ceramic substrate based power modules with higher level of integration sophistication, made possible by the large market and the need for better EMC management.
- Solid State Transformers in power systems that can outperform conventional transformers with the similar power capacity and voltage class.



- Effective solutions for simplifying the engineering effort development in custom designs. Wide Bandgap power electronic circuits are more complex, but this complexity cannot be passed on the design engineer; standardized building blocks and design automation are two possible approaches.

The International Technology Roadmap for Wide bandgap Power Semiconductors (ITRW) is the only global roadmap that will facilitate an acceleration in the R&D process for this new technology to fulfil its potential.

Chapter 3: Introduction

3.1 Why do we need a Power Electronics Roadmap on Wide Bandgap Semiconductors?

The growth of global electric power usage has continued to accelerate. The electrification of so many aspects of normal life make the modern world reliant on electricity. This has taken place in a dramatic increase in awareness of the human impact on global climate change, particularly from hydro-carbon emissions. This has been a factor in an ever-increasing demand for electricity from fossil fuel generation. Regulatory changes worldwide and concerns about air quality generally are also driving changes in automotive platforms from gasoline and Diesel to the increased use of hybrid and pure electric propulsion systems. There has also been a significant increase in renewable energy generation from photovoltaics and wind generation. Power electronics play a key role in converting these different and diverse forms of energy generation into electricity for use by the consumer as either AC or DC supplies. Power electronics as a discipline covers the whole range of power conversion from grid level transmission and distribution, to power conversion in individual consumer products, and myriad of applications and systems across the range from high voltages at the grid level to less than 1V for the smallest integrated circuits. Power Electronics is at the heart of all these systems, with a key driver in all cases being to maximize efficiency of conversion at every stage.

3.2 Expected Application Growth Spurt of Wide Bandgap Power Electronics

During the last three decennia, power electronics has been undergoing a sequence of growth spurts driven by new applications and societal needs. The first notable technology driver was the rise of the computer industry accompanied by need for computer power supplies. Annually about the combined rating of produced power supplies is about 2.1 TW.

Another important technology driver is a growing awareness about the role that power electronics can play to improve energy efficiency. This increased awareness and sensitivity to climate change and its contributing factors has stimulated the introduction of more power electronic systems in a diversity of applications. A European study played an important role making system integrators and policy makers aware of the significant contribution role that power electronics can make to reduce carbon emissions. Efficiency improvement also plays an important role to introduce WBG because it makes 99%+ electronic conversion efficiency possible.

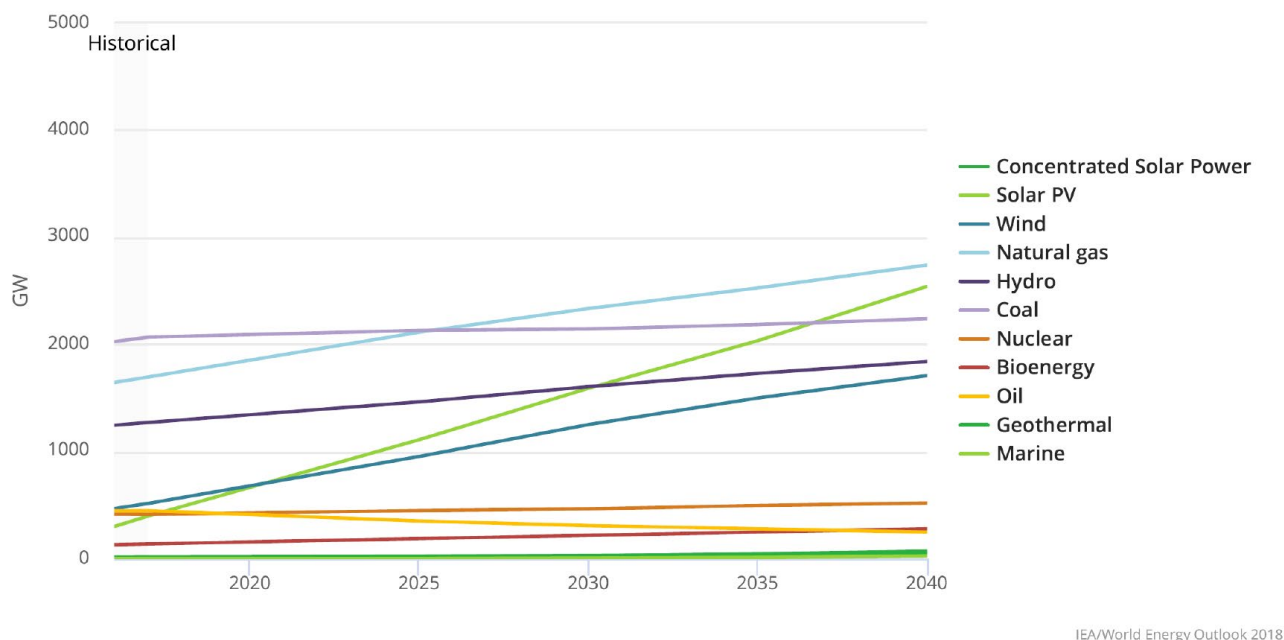


Figure 1: Installed power generation capacity by type (<https://www.iea.org/weo>)

The growing awareness about the impact of CO₂ emissions, triggered by the alarming effects of climate change, is resulting in dedicated efforts to increase the production of renewable energy, as can be seen in Graph 1, as published by IEA. Of significance to ITRW is the additional power conversion capacity. It can be roughly estimated that 2.1 TW and 4 TW of electronic power conversion for PV will be installed in wind-turbines and PV systems respectively during the period 2018 thru 2030.

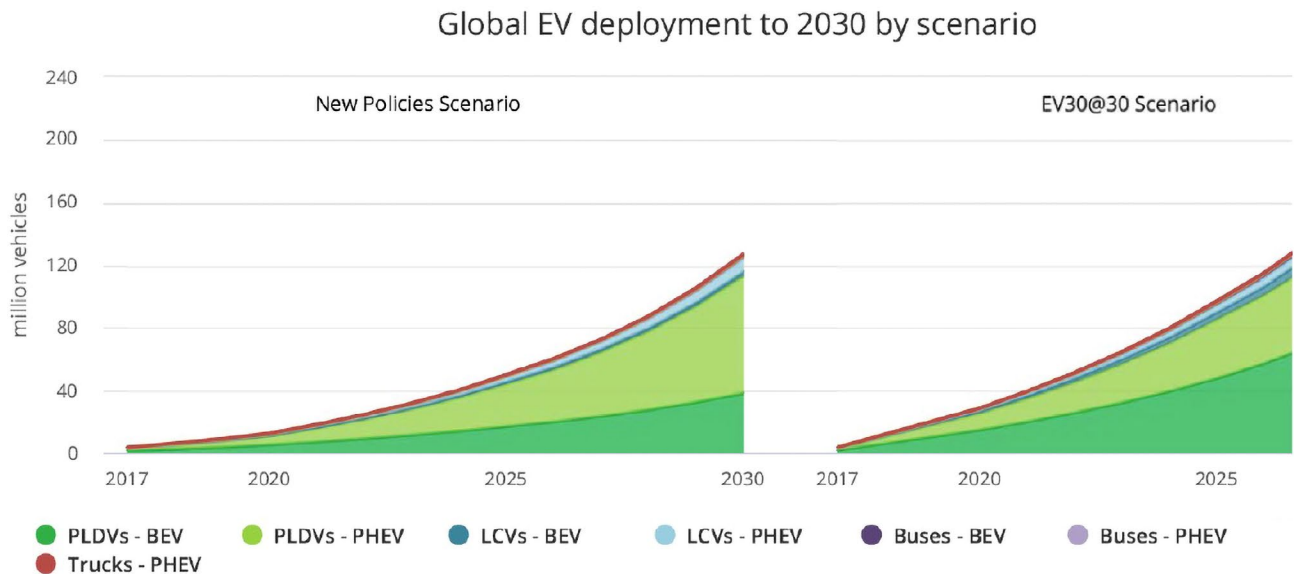


Figure 2: Global EV deployment to 2030 by scenario. (<https://www.iea.org/gevo2018>)

The largest market for power electronic converters, based on installed power conversion capacity, will be electrification of transport. **If 150 million electric vehicles are to be manufactured, and a 100kW power conversion rating is associated with every vehicle, accounting for the drive and battery charging, then during 2018 thru 2030 a staggering 15000 TW of power electronics conversion equipment need to be produced.**

The largest market for power electronic converters, based on installed power conversion capacity, will be electrification of transport. **If 150 million electric vehicles are to be manufactured as indicated in Graph 2, and a 100kW power electronic conversion is associated with every vehicle, accounting for the drive and battery charging, then during 2018 thru 2030 a staggering 15000 TW of power electronics conversion equipment need to be produced.**

Table: Expected production-relative numbers per application

Annual production volumes	Number of power semiconductor chips	Number of circuit assemblies
PV solar converters	+++++	++++
Wind turbine converters	+++	+++
Grid converters	+++	+
Electric Vehicle drives	+++++++	+++++
Computer power supplies	+++	+++++++
Battery chargers and management systems	+++++++	+++++
Medical	+	+
Aerospace	++	++
Consumer electronics	+++	+++++++

Customization to meet different specifications is a characteristic of the power electronics industry. The circuit construction needs to be flexible, assemblies are done with discrete components, a large variety of components exist to accommodate different power ratings and as a result advantage of scale is rarely possible. Low production volumes and large diversity of applications made it difficult in the past to find sufficient investment for technology development. Now new application markets are opening up and large production numbers are forecasted. Relevant for future R&D focus and investment is spread in production numbers, and in the above table expected production numbers projected. Electric vehicle drives and battery charging systems stand out as potential movers of the future.

3.3 Motivation to Form the ITRW

The **I**nternational **T**echnology **R**oadmap for **W**ide bandgap power semiconductors (ITRW) was formed in December 2015 by the IEEE Power Electronics Society (PELS) with the overall goal of publishing a clear technology roadmap every two years, white papers setting out clear technology statements, position papers, defining a strategic research agenda, coordinating information dissemination and community building events and providing operational support to the wide band gap power semiconductor community.

Clear needs have been identified across industry, academia, education and public authorities to have a reliable and comprehensive view on the strategic research agenda and technology roadmap for wide bandgap power devices. There have been many roadmaps over the years in a variety of technical fields, perhaps the most famous being the International Technology Roadmap for Semiconductors (ITRS) – which has been primarily driven by the deep sub-micron silicon based industry, with Moore’s law at its heart.

The expansion of Silicon Carbide (SiC) for higher voltage, high-speed power electronics in the last 5 years has been dramatic, and the technology is moving from the research lab to mainstream products rapidly. The impact on power electronics design has been smaller, more compact designs, with smaller passives due to higher frequencies. SiC has excellent thermal properties and tolerance making it ideal for extreme environment applications. Issues remain with gate drivers and speed of operation – with a focus on packaging and layout particularly important for the successful application of SiC. Gallium Nitride (GaN) devices have expanded from previous large markets such as LED lighting, into consumer applications where the dramatically higher switching frequencies possible with GaN power devices has led to significant reductions in size of devices such as mobile phone chargers and lower voltage devices (less than 600V).

This roadmap encapsulates the current state of the art and looks forward to the exciting future of wide bandgap power electronics over the next two decades and beyond, with specific horizons of :

Short Term: 0-5 years
 Medium Term: 5-10 years
 Long Term: 10-20+ years

3.4 Focus and Scope

The primary technology focus of the ITRW is on the application of power electronics developments using wide bandgap power semiconductors, identifying the materials, design, and manufacturing approaches that will support the thermal, electromagnetic, and through-life management of power electronic converters at power levels from hundreds of watts to hundreds of kilowatts and beyond. A secondary focus is on technologies enabling operation at higher temperatures and other harsh environments.

The roadmap targets technological developments over three distinct time horizons:

1. short-term (commercial realization within 5 years)
2. medium-term (commercial realization within 5-15 years)
3. long-term (commercial realization at 15 years and beyond).

ITRW is unique in that it brings together multiple technical communities from fundamental material scientists and engineers, device specialists and application focused engineers. Each group has its own demands and criteria for success and one of the challenges for any broad based roadmap is addressing those sometimes conflicting requirements. Even the SiC and GaN application groups are very different in terms of end user applications, device requirements and key issues. ITRW encompasses all these different perspectives and uniquely looks for both the commonalities between them, and also the distinct aspects of each.

ITRW also does not just investigate what might be possible in the future, but a key part of the roadmap is to use case studies at different stages of development, from fundamental research all the way through to product applications to showcase the best application of wide bandgap power electronics globally, but also to highlight the range of research and development that is taking place in this rapidly expanding and exciting field.

ITRW has also encouraged, and will continue to be a mechanism for, the working groups to be able to publish review papers and articles to support the roadmap, engender discussion and debate, and provide data to the wider power electronics community in a coordinated and open manner. This data is then available for discussion in between the main roadmap publication dates and enables both feedback and dissemination to take place rapidly.

3.5 What is ITRW and How is it Organized?

Any road mapping exercise that is intended to be genuinely global and representative of the whole community it represents must ensure that it has clear and transparent governance structures, represents all the stakeholders in the field and is global in nature. To this end, the ITRW has been formed with all three of these fundamental tenets at its heart. The ITRW has a steering committee that consists of leadership from societies, industry, government and academia bodies. The participation and leadership of industry is at the heart of the ITRW process and therefore in addition to key industry people involved on the steering committee there is a wider industry advisory board (IAB) that has a specific role of ensuring that the ITRW is relevant and technology driven.

The steering committee also has strong technical representation from the specific technical working groups, ensuring that a broad participation of individuals across all aspects of wide band gap power semiconductor technologies are represented. As with the steering committee, there is extensive industry participation in the individual working groups.

The structure of the ITRW as shown in Figure 1 has been designed to be inclusive and participation is open, to ensure that decisions taken are fair and neutral, the voting membership of the steering committee broadly follows the conventional IEEE approach taken for standards activities in that no one company, geographical grouping or constituency can dominate. In the early stages of the ITRW existence, the steering committee was formed based on interest and knowledge of the field, however over time it is envisaged that the future executive officers will be elected as the organization becomes self-sustaining.

As well as the broad technical groupings shown, there will be underpinning technology interests shared across the technical working groups such as reliability, data sheets, testing, design techniques and these will be developed across the working groups where appropriate.

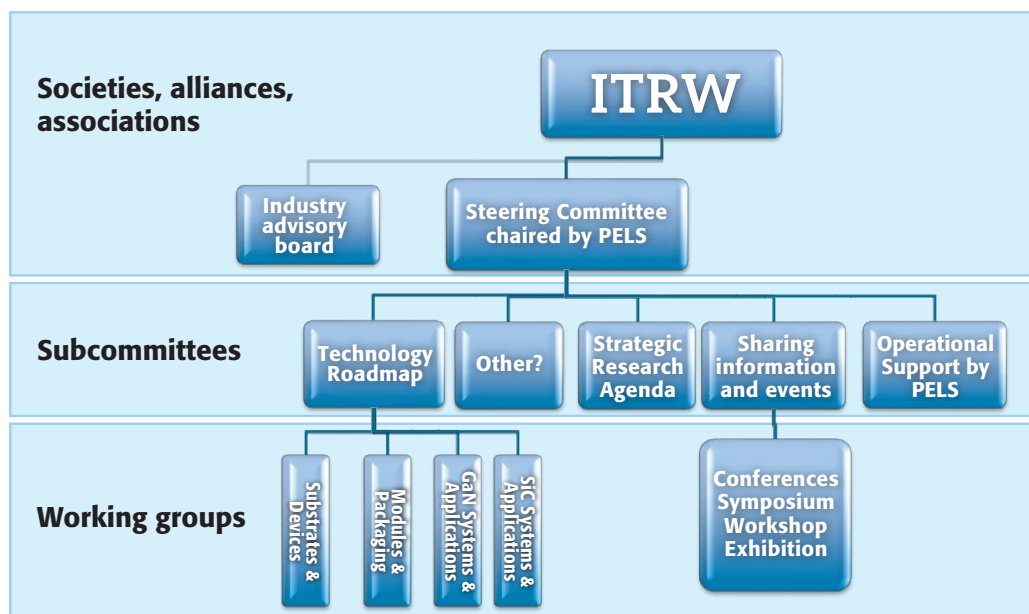


Figure 3: ITRW Organizational Structure

3.6 The Role of Volunteers

Traditionally the IEEE and PELS (the IEEE Power Electronics Society) offer a platform for disseminating research and development (R&D) results in the form of conferences and publications helping their career development of academics. Industry on the other hand, finds more value in standards activities involving technologies higher up on the technology readiness level (TRL) ladder. PELS has introduced ITRW, a roadmap activity to cover the middle ground and create an opportunity for better collaboration between academics and industry and to facilitate and shape the future of an important technology development in the PELS field of interest.

ITRW has been the product of more than three years of collaborative work and is based on the enthusiastic participation of industry and academic members alike. PELS also invested resources by appointing a staff member to support the activities. The main challenge, in our view is how to make it worthwhile for volunteers to continue to put in a lot of effort in the roadmap activities. The best solution would be if their contributions to the roadmap activities would be viewed by their employers as part of their job, as is currently the case with R&D dissemination and standards development. The combination of workshops, meetings, publications and collaborations directly as a result of the ITRW process has been of significant value to those participating and has had a positive bearing on many individual careers, and in turn, their employers.

The heart of the ITRW process is the engagement of PELS members in the four technical working groups which encompass the four areas given below, each of which have contributed articles for publication (including in the PELS magazine) and ultimately as part of the ITRW roadmap itself.

- Devices and Materials
- Integration and Packaging
- SiC Applications
- GaN Applications

The breadth and depth of the content captured in this roadmap illustrate the commitment of time and expertise by key stakeholders in the power electronics community across academia and industry, with dedicated reports from each of these technical working groups. Without these volunteers, this activity would simply not be possible. We would therefore like to gratefully acknowledge this commitment, and encourage other PELS members to engage with the ITRW as it grows to make their own contribution in the future.

The other significant contribution to the work of the ITRW is the industry advisory board, whose expertise and experience are vital in ensuring that the roadmap is relevant and focused on the correct path for industry to be able to leverage the leadership given by the ITRW roadmap.

3.7 ITRW as Gateway to Other Roadmaps

The transition to WBG devices coincides with the global Energy Transition, geared to transform the global energy sector from fossil-based to zero-carbon by the second half of this century. All over the world, programs are being introduced to stimulate and finance technologies that can be used to reduce the carbon footprint of energy usage. As a result, several roadmap initiatives exist in America, Europe and Asia. These roadmaps reflect the perspectives, strengths and ambitions of these regions. The position of ITRW in relation to other roadmaps can be summarized as follows:

- An independent Neutral Forum
- A global effort
- Includes Industry, Academic and Policy
- Pre-Standards, complementary to JEDEC

A characteristic of the regional roadmaps is that they are mostly not open access. Due to their strategic and/or commercial value, the readership is restricted. Since the IEEE is an international volunteer driven organization, ITRW is an independent roadmap available to members. Since ITRW floats on efforts of volunteers and the time that they can set aside after doing their day jobs, it does not make sense to duplicate the regional roadmap efforts in ITRW.

In ITRW, we provide global perspectives combining the insights and views of participating volunteers around the world. It would be good to put ITRW alongside the other roadmaps, as summarized in the tables on the remaining pages of this section of the roadmap.

	PowerAmerica Roadmap version 4, published Jan. 2019
About the organization	PowerAmerica is a member of Manufacturing USA, an organization that was formally established in 2014 as the National Network for Manufacturing Innovation. Manufacturing USA brings together industry, academia and federal partners within a growing network of advanced manufacturing institutes to increase U.S. manufacturing competitiveness and promote a robust and sustainable national manufacturing R&D infrastructure. With the backing of the U.S. Department of Energy and engagement of top researchers, we exchange knowledge and processes and provide access to an educated workforce. This enables American industry to develop more innovative power electronics products and systems.
Purpose of roadmap	The PowerAmerica roadmap outlines key markets and application areas for SiC and GaN power electronics, performance targets for competitive SiC and GaN technologies, and technical barriers to achieving those targets. The primary purpose of the roadmap is to inform PowerAmerica activities needed to overcome those barriers and accelerate adoption of WBG power electronics. Guidance from the roadmap will formulate PowerAmerica's efforts to facilitate coordination across industry, academia, and national labs to execute the priority activities identified in the roadmap, and to make strategic investments in technology development, workforce training, and WBG manufacturing synergistically culminating in large-scale WBG power technology adoption.
Accessibility	The full roadmap is available to PowerAmerica members only. A light version of the roadmap will become available to the public and will be accessible through the PowerAmerica website. https://poweramericainstitute.org/

	ECPE Wide Bandgap Roadmap on Lead Applications of SiC and GaN To be published
About the organization	ECPE European Center for Power Electronics e.V. was founded in 2003 on the initiative of leading power electronics industries as an industry-driven Research Network to promote education, innovation, science, research and technology transfer in the area of Power Electronics in Europe. In the meantime, the network comprises 180 network members. The main objectives are pre-competitive research, education and advanced training as well as public relations for power electronics in Europe.
Purpose of roadmap	Research and Technology Roadmaps are an important strategic tool to identify and guide a mainstream for medium to long-term research. A key objective is to provide input and industrial guidance to research programs on European and on national level. ECPE Member companies will reflect their own company roadmap vs. the ECPE roadmaps and ECPE Competence Centre can use the roadmaps when they define new research directions.
Accessibility	The full information is restricted to ECPE members but parts will be open access (publicly available on the ECPE web site) https://www.ecpe.org/

	SiC Alliance Road Map ~Future Society based on Power Electronics~ First version, published 16th Oct. 2018
About the organization	SiC Alliance was founded in 2010 in order to strengthen the network of researchers in SiC. At that time, there were 3 national research projects on SiC R&D in Japan. The members of SiC Alliance were mainly the members of these 3 national projects. The main objectives are promoting the R&D regarding SiC and contributing to realize the “Low –carbon society”. Recently, we also started to support the commercializing the systems which involve the SiC devices by providing the useful information to use the SiC devices properly. SiC Alliance comprises 32 companies, 19 universities, and 9 research centers.
Purpose of roadmap	Main purpose of our roadmap is to guide the R&D plan of each member. Our SiC roadmap consists of 7 parts. They are Vision, Auto Mobil, Super Express Train, Industrial Inverter, Switching Device, Parts, and Wafer. We made these roadmap slides by the specialists of each area. And after making drafts, we gathered these roadmap slides and discussed the consistence of these roadmap slides each other. In order to make the useful roadmap, we balanced the technology push and the market pull properly.
Accessibility	The full information is restricted to SiC Alliance members but parts will be open access (publicly available on the SiC Alliance web site) http://www.sicalliance.jp/data/doc/1539751125_doc_2_0.pdf

	China Advanced Semiconductor Industry Innovation Alliance
About the organization	China Advanced Semiconductor Industry Innovation Alliance (hereinafter referred to as CASA) was initiated by research institutes, universities and leading enterprises, which related to wide band gap semiconductors with the support of the Ministry of Science and Technology, Ministry of Industry and Information Technology and the Beijing Municipal Government. It has been formally registered in the Civil Affairs Department as a corporation. It is a new organization providing comprehensive innovation services for wide band gap semiconductors and related emerging industries. By integrating and sharing innovative resources around the world, CASA is dedicated to building an market oriented industrial innovation system which deeply integrates R&D, industry and capital to lead the cross-regional, interdisciplinary, cross-industrial and coordinated development of wide band gap semiconductors.
Purpose of roadmap	This roadmap describes the development trends of SiC substrates and epitaxy, SiC power devices and its applications, GaN epitaxial and power devices, diamond materials and devices, Ga₂O₃ epitaxy and devices in the next 30 years (2018~2048). It addresses the key aspects of the wide bandgap semiconductor technology electrical properties, packaging, heat dissipation, reliability, and the cost. The goal is to encourage Chinese organizations and companies to participate in Technology Roadmap, to exchange knowledge of wide bandgap semiconductor and to provide guidance to the Chinese industry.
Accessibility	The full roadmap in Chinese is available from the following link: http://www.casa-china.cn/uploads/soft/190419/6_1003488591.pdf The full roadmap in English is available from the following link: http://www.casa-china.cn/uploads/soft/190731/12_1010538821.pdf

 HIR HETEROGENEOUS INTEGRATION ROADMAP	The EPS Heterogeneous Integration Roadmap (HIR)
About the organization	The EPS Heterogeneous Integration Roadmap (HIR) is a sister IEEE roadmap.
Purpose of roadmap	HIR covers a wide range of electronic applications, including Integrated Power Electronics (IPE) as part of System-in-Package (SiP) solutions. The IPE is looking specifically at approaches that are mapping technologies for integrating power around the periphery or directly embedding power within a “component”. The “component” is a single package among other “components” within a SiP module, e.g. RF communications component, graphics processor component, or a photonic information processor. The HIR does not focus on discrete power electronics or standalone power conversion modules, whether outside or inside the SiP. Contrary to the ITRW and the above regional roadmaps, it does not address wide bandgap semiconductor power devices, but the activities of HIR are relevant to the activities of the ITRW packaging working group.
Accessibility	To maximize the efficiency and effectiveness of both road maps, the ITRW will work in cooperation with the HIR team by sharing knowledge and experience of power electronic packaging and integration. This will be facilitated by cross-representation within relevant groups of the ITRW and HIR. Further details of the HIR can be found at: https://eps.ieee.org/technology/heterogeneous-integration-roadmap.html

	NanoElectronics Roadmap for Europe Published in 2018
About the organization	The NEREID project has been funded in the framework of the call “H2020 ICT 25 – 2015: Generic micro- and nano-electronic technologies”, to develop a nano-electronic roadmap focused on European needs. The roadmap developed by NEREID combines an “application pull” approach, focusing on the requirements of European semiconductor and applications industry and main societal challenges, with the “technology push” coming from the advanced concepts developed by Research Centers and Universities in order to achieve an early identification of promising novel technologies. The NEREID roadmap focuses on the identification of medium and long-term requirements, gaps and recommendations. It covers the main technology sectors: More Moore (Innovative FETs and Memories), Connectivity, More than Moore (Smart Energy, including Energy Harvesting, and Smart Sensors), Beyond-CMOS, Heterogeneous Integration, System design, Materials, Equipment and Manufacturing Science.
Purpose of roadmap	NEREID takes into account the specificity of the European industrial and academic landscape, and will be very important to better coordinate academic and industrial research for equipment, semiconductors and application developments, as well as serving as the input for future research programs at European and National levels in order to join our efforts to overcome the main NanoElectronic challenges and put the EU at the forefront of future technological developments.
Accessibility	The document can be downloaded from the website https://www.nereid-h2020.eu/

Chapter 4: Wide Bandgap Power Electronics

4.1 Introduction

Semiconductor devices made from Silicon (Si) such as diodes, Insulated Gate Bipolar Transistors (IGBT), Thyristors, and Metal-Oxide Semiconductor Field-Effect Transistors (MOSFETs) have dominated the power electronics landscape becoming a stable and mature technology base for reliable power electronics systems. More recently, there has been the exciting shift to “wide bandgap” semiconductors such as Silicon Carbide (SiC) and Gallium-Nitride (GaN) which offer some significant advantage over their silicon counterparts.

What are wide bandgap power devices and what is driving the introduction of wide bandgap based power electronics? This chapter serves to give the reader a basic background and is divided into two parts; the first addresses the transition drivers, and the second gives a brief technology overview.

It is clear from the technology rush to wide band gap devices by the power electronics community, especially Silicon Carbide (SiC) and Gallium Nitride (GaN), such that the benefits of moving from Silicon to wide band gap technology (SiC or GaN) will have significant influence on the power electronics industry in the future.

Some key questions for the power electronics community to address when considering the adoptions of Wide bandgap (WBG) devices in applications are:

1. What are the decisive benefits for the adoption of WBG?
2. Can the investment costs be justified by the long-term benefits?
3. Which applications will benefit first from the adoption of WBG?
4. What is the threshold to move to WBG?

These questions are crucial in understanding how the transition to new technologies such as SiC can take place successfully. The choice of target application needs to be well matched to the characteristics of the new devices, to be able to take advantage of these characteristics.

4.2 Technology Drivers

We can classify a range of drivers to justify why we need a roadmap for wide bandgap power semiconductor based power electronics, and these can be broadly divided into top level drivers (market or application focused) and technical (technology focused). We can further refine these into specific drivers in each category as follows:

4.2.1 High Level Technology Drivers

The top-level drivers for power electronics include:

1. Reduced costs (bill of materials, manufacturing, O&M, recycling)
2. Increased efficiency (reduced losses through life)
3. Increased power density (reduced volume, reduced mass)
4. Ease of use (plug and go, modular solutions, higher voltage, higher current, load/source integration, simple maintenance)
5. Environmental tolerance (higher temperature, extreme temperature range, vibration and shock, radiation hard)
6. Reliability and robustness (longer application lifetimes, longer time between maintenance, fewer early failures, ability to survive fault events)

4.2.2 Technical Drivers

Specific technology drivers that contribute to the realization of the top-level drivers include:

1. Higher levels of structural and functional integration
2. Improved electrical and magnetic performance
3. Greater reliability and robustness
4. Extended operating range (voltage, current, temperature, and frequency)
5. Improved thermal performance

4.2.3 Using GaN Technology in Power Electronics Applications

The comparison suggests that replacing, for example, SJ MOSFETs with GaN HEMTs in a switching power supply would result in noticeable performance improvements. However, most of the early attempts [5] to demonstrate GaN HEMTs value proposition by simply replacing incumbent MOSFETs with GaN HEMTs have failed short of expectations either yielding incremental performance improvements or sometimes even causing performance degradation. The reason is that a GaN HEMT is not necessarily a drop-in replacement for Si or SiC MOSFET, but rather a complex switching device whose parameters affecting switching performance are quite different than those found in MOSFETs as detailed in Table 1.

Table 1. Application attributes of GaN HEMTs

Applications Attribute	Si SJ MOSFET, 650V	SiC MOSFET, 1200V	GaN HEMT, 600-650V	Notes
Gate Charge, Q_g	Highest	Slightly lower than SJ	Lowest	GaN HEMTs have the smallest die size and lateral architecture
Threshold Voltage, V_{th}	3V	3V	1.2-1.6V	GaN HEMTs may require negative V_{gs} for reliable turn OFF
Maximum V_{gs}	20V	20V	7-10V	GaN HEMTs are sensitive to gate overvoltage
Drain to Source leakage current in OFF state	Slightly higher than SiC	Lowest	Highest	GaN has impurities
Body Diode, V_f	~0.9V	~3.5V	No body diode	Intrinsic diode in SiC MOSFET is PN type and has high V_f
Reverse Recovery, Q_{rr}	Highest	Slightly higher than GaN	Lowest	GaN and SiC don't have reverse recovery losses, just Q_{oss}

3rd Quadrant Operation	Good	May need external diode	Need positive V_{gs}	
Avalanche Rating	High	High	None	GaN HEMTs don't have PN junction
Dynamic R_{ds_on}	None	None	Up to 30%	Charge trapping occurs in most GaN HEMTs
Thermals	OK	Best	Comparable to Si	SiC has the highest thermal conductivity
Most suitable for applications	Boost PFC, Two-transistor forward, flyback	Forward, half-bridge based topologies	Half-bridge based topologies	Overvoltages are clamped to the bus in half-bridge topologies

Therefore, to take full advantage of intrinsic capabilities of GaN HEMTs in switching power supply applications, it is necessary to employ a holistic approach that combines the two major elements:

1. Identification of new topologies and applications enabled by unique capabilities and performance attributes of GaN HEMTs
2. Ecosystem required to enable broad acceptance of GaN HEMTs

The first element encompasses road-mapping so-called “killer apps” or the applications either entirely enabled by GaN HEMTs or where GaN HEMTs offer dramatic performance improvements compared to other technologies and materials.

4.2.4 Using SiC for Power Electronics Applications

A long-term vision of the power electronics community is to develop highly integrated power electronic converters that are specialized for their specific applications. The motivation for a migration from silicon to silicon carbide power electronics is driven by the need for those power converters to have much greater power density, reliability, and overall system performance without an excessive cost in devices or design. The benefits of moving to silicon carbide are such that it is inevitable that there will be a transition from silicon to SiC technology for power electronics with a rapidly increasing trajectory

The *Silicon Carbide Application Working Group* (SiC-AWG) of the *International Technology Roadmap for Wide Band-gap Power Semiconductors* (ITRW) comprises the following four subgroups: utility; transportation; power supplies and drives; and medical and emerging applications. The goals of the SiC-AWG's are two-fold. First, it aims to identify how applications (e.g. traction drives) will experience the game-changing performance by adopting the appropriate SiC technology. Secondly, it will highlight the device criteria that manufacturers should target so the benefits of SiC components are fully realized in these design spaces. To accomplish this task, the team has developed a template to guide topic co-chairs while documenting the applications and establishing optimal characteristics for SiC components utilized in these areas.

Some key questions for the power electronics community to address when considering the adoptions of Silicon Carbide (SiC) devices in applications are:

1. What are the decisive benefits for the adoption of SiC?
2. Can the investment costs be justified by the long-term benefits?

3. Which applications will benefit first from the adoption of SiC?
4. What is the threshold to move to SiC?

These questions are crucial in understanding how the transition to new technologies such as SiC can take place successfully. The choice of target application needs to be well matched to the characteristics of the new devices, to be able to take advantage of these characteristics [1]. We can summarize some of the key characteristics in Table 1. From this table we can see some fundamental differences between the materials commonly used in power electronics, with the critical field highlighting the potential for higher breakdown voltages in wide band gap materials, the higher electron saturation velocity in SiC and GaN, and particularly the very high thermal conductivity in SiC. This particular combination of potential for high voltage, high speed and high temperature behavior intrinsic to SiC makes in extremely attractive for applications in power electronics that require these aspects of behavior.

Each subgroup has identified several application areas to focus on. The transportation subgroup is focusing on land vehicles, aviation hardware, naval systems, and rail-based systems. The applications for land vehicles include traction drives, battery chargers, and DC/DC converters. Aviation hardware and naval systems encompass generation equipment, solid-state circuit breakers, energy storage hardware, and propulsion systems. For rail-based systems, tractions drives and auxiliary power supplies are being examined. The utility subgroup is exploring high voltage DC (HVDC) transmission systems, flexible AC transmission systems (FACTS), solid-state transformers, energy storage systems, photovoltaic (PV) generation, and wind generation. For power supplies and drives, the areas include switch-mode power supplies for data centers and drives for refrigeration, air conditioning, and compressors. Finally, the medical and emerging applications subgroup is examining Computed Tomography (CT) scanners, power supply systems for fusion energy, wireless power transfer, and pulsed power supply systems.

The impact of SiC will be particularly immediate in fields where the cost of new devices is not a significant barrier or efficiency savings can be made (either through direct improvements in the efficiency of power converters, reduced weight and improved thermal performance or operation in harsh environments). Specific application areas where SiC technology is already being adopted and is growing rapidly are:

1. MRI – Medical Resonance Imaging: reduction in the size of power converters, noise and losses, improved resolution and dynamic performance.
2. Automotive: Reduced weight, possible high temperature operation, improved range, reduced charging times, and improved robustness.
3. Aerospace: Reduced weight, significantly reduced fuel costs, intrinsically radiation hardened, wide temperature range operation, and harsh environments.
4. Traction: Improved efficiency, high continuous power operation, energy utilization via improved regeneration at high power.
5. PV generation: Ultra-high efficiency inverters.

4.3 Fundamentals of Wide Bandgap Semiconductor Power Electronics

Power electronics are fundamental components in consumer electronics and clean energy technologies [1], [2]. For several decades, silicon (Si) has been the primary semiconductor choice for power electronic devices. Due to the many decades dedicated to the development and fabrication optimization of Si devices, as well as the large abundance of material, the manufacturing capability is high, and the costs are extremely low. However, Si is quickly approaching its limits in power conversion [3], [4]. Wide bandgap (WBG) semiconductors offer promise of improved efficiency, reduced size, and lower overall system cost [3].

Of the various types of WBG semiconductors, silicon carbide (SiC) and gallium nitride (GaN) have proven to be the most promising technologies [3], [4], with several devices already being sold commercially. GaN is favored for its high frequency switching capability, though its high costs and driving intricacies currently limit its commercial viability [5]. In particular, the lack of quality substrates hinders the development of the economically advantageous vertical structure [4], [5]. Vertical devices are desirable as the lateral arrangement limits the current density, and voltage

capabilities of the GaN technology [5]. Alternatively, SiC has shown tremendous high temperature capability, as well as aptitude for high voltage applications [3], [4]. Furthermore, the cost of SiC devices has decreased within the last decade, and the performance has proven superior to that of conventional Si devices.

Some of the potential application areas for these WBG devices include: transportation electrification, downhole drilling, and renewable energy [3]. Transportation electrification is a major trend today, especially in the aerospace and automotive industries. Further success of electric cars and more-electric aircrafts hinges on advances in power electronics. Within the oil and gas industry, high temperature electronics are in demand for downhole drilling in the exploration of petroleum. These deep oil wells can reach temperatures greater than 200 °C, which is far beyond the limit of traditional Si devices. Regarding renewable energy, in particular, WBG devices have been explored for replacement of Si insulated gate bipolar transistors (IGBTs) in photovoltaic inverters in order to improve efficiency. WBG semiconductors are well suited for these applications due to their high temperature capabilities, fast switching speeds, and low losses. The high temperature capability of these devices allow them to better withstand the harsh environmental conditions, and have relaxed cooling system requirements [3], [4]. In addition to this high temperature capability, the fast switching speed of these devices allows for higher frequency operation thereby resulting in the reduction of the passive components, which decreases the total size, weight, and cost of the system [3].

These benefits of WBG semiconductors, such as SiC and GaN, arise from their greater thermal conductivity, higher critical electric field for breakdown and saturation electron drift velocity, and lower intrinsic carrier concentration compared to Si [6]-[8]. The higher thermal conductivity of SiC makes it superior to Si in terms of heat transfer to the environment, which allows for increased power densities [4]-[7]. SiC has a breakdown electric field that is more than eight times higher than that of Si [6]. This means that, for a given blocking voltage, SiC can be more heavily doped than Si, thereby reducing the on-state resistance [6]. The higher saturation drift velocity makes SiC more suitable for high frequency applications [7]. Also, due to the wider bandgap of SiC, it has a lower intrinsic carrier concentration, which allows these devices to operate at higher temperatures without suffering from excessive leakage [6], [7].

Infineon Technologies and Cree Inc. introduced the first commercial SiC power devices in 2001 and 2002, which were SiC Schottky diodes [9], [10]. The wild success of the SiC Schottky diodes led to the further development, and eventual commercialization, of the vertical trench SiC junction field-effect transistor (JFET) by SemiSouth Laboratories in 2008. Although SemiSouth has dissolved, other companies such as Infineon and United Silicon Carbide, Inc. (USCi) have taken over the production of SiC JFETs, though choosing to focus on the depletion mode (normally-on) SiC JFET [11], [12].

Just three years after the release of the SiC JFET, Cree commercialized the first SiC metal-oxide-semiconductor field-effect transistor (MOSFET) [9]. Though the gate-oxide reliability proved to be a challenge, improved manufacturing has resulted in new generation SiC MOSFETs with superior oxide quality [13]. Today, more manufacturers are producing SiC MOSFETs, including General Electric (GE) [14], ROHM Semiconductor [15], Mitsubishi Electric [16], and Sumitomo Electric Industries [17].

Bipolar devices are also becoming available. In 2013, GeneSiC Semiconductor released its SiC super junction transistor (SJT) [18], and Fairchild Semiconductor was developing its SiC bipolar junction transistor (BJT) [19]. These SiC devices experience quite different characteristics as compared to their Si counterparts, such as high current gains and fast switching speeds, as will be seen in a later chapter. Figure 1 summarizes this progression of the commercialization of SiC power semiconductors.

The benefits of SiC arise from its material properties (Figure 1-1). Specifically, SiC has a greater thermal conductivity, lower intrinsic carrier concentration, higher saturation electron drift velocity and critical electric field for breakdown compared to silicon [45],[46],[47]. The higher thermal conductivity of SiC makes it superior to silicon in terms of heat transfer to the environment, which allows for increased power densities [9],[45],[46]. Additionally, due to the wider bandgap of SiC, it has a lower intrinsic carrier concentration, which allows these devices to operate at higher temperatures without suffering from excessive leakage, hence also allowing for higher power densities [45],[46]. The higher saturation drift velocity also makes SiC suitable for high-frequency applications [46].

The initial fabrication of SiC MOSFETs was stifled by two main issues: (1) the quality of the SiC/SiO₂ interface [4], and (2) the high electric field generation [6]. The poor quality of the boundary between the SiC surface and oxide resulted in low mobility in the inversion layer, while the high electric field generated within the SiC caused degradation of the gate oxide [6]. Significant improvement has been made to the manufacturing of SiC MOSFETs, and thus it has been demonstrated that these issues have been resolved [13], [15].

4.4 Bipolar Devices: BJT

The Bipolar Junction Transistor (BJT) was invented by William Schokley, John Bardeen, and Walter Brattain in 1949 at Bell Laboratories [64]. The BJT requires a junction of n-type and p-type doped semiconductor, which results in the bidirectional flow of two types of charge carriers: electrons and holes. This is contrary to unipolar devices, such as FETs, which only have one kind of charge carrier.

The BJT was eventually replaced by MOSFET technology for a number of reasons. First, the Si BJT featured low current gain; that is, the ratio of supplied base current (input) to the resulting collector current (output). The current gain would also reduce as the current rating of these Si BJTs was increased [65]. Second, the large dynamic resistance within the device resulted in high switching losses [65]. Third, the BJTs experienced numerous reliability issues. In forward-biased mode, high temperature filaments could arise with current crowding, thereby leading to device failure [65]. Reverse-biased breakdown could also occur due to voltage and current stress present during inductive load switching, which cause the electric field stress to extend beyond the drift region [65]. Consequently, strict limits were placed on the Reverse Safe Operation Area (RSOA) of the Si BJTs, and thus they had no short-circuit capability [65].

It is thought that the implementation of BJTs in SiC would revive this technology. The wider band gap of SiC allows for a higher current gain [65] and therefore lower driver losses and improved efficiency. The higher breakdown field of SiC also makes the BJT less susceptible to thermal runaway and thus increases the reliability [65]. In this chapter, the background of SiC BJTs will be discussed, and then the results of the high temperature static and dynamic characterization will be presented.

As mentioned in the previous section, SiC BJTs feature many improvements over their Si counterparts. SiC BJTs also possess qualities that make them good contenders to Si IGBTs. These benefits include wider RBSOA, lower stored charge allowing for faster switching, more modest reduction of the current gain at high current densities, smaller collector-emitter saturation voltage, and high temperature operation [66]. Of course, unlike Si IGBTs, the SiC BJT requires a continuous base driving current, which can complicate driving in some applications. However, as will be shown in this chapter, SiC BJTs have large current gains, thereby reducing the needed base current.

4.5 JFET

In the early stages of the SiC technology, issues with the gate oxide of the SiC MOSFET hindered its progression. Consequently, many companies focused their attention on the SiC JFET, and thus the technology quickly matured. Utilizing SiC instead of Si allows for practical enhancement and depletion mode JFETs. As in the case for SiC MOSFETs, SiC JFETs allow for fast unipolar switching that is seen by devices such as the Si SJ-MOSFETs, but at higher voltage ratings of 1.2 kV or more. Moreover, unlike MOSFETs, JFETs do not have a gate oxide [76], [77]. Thus, although advancements have been demonstrated in the gate-oxide reliability and threshold voltage instability of the SiC MOSFETs, the SiC JFETs still feature lower failure risks due to the lack of oxide processing [76], [77].

The enhancement mode JFETs are normally off devices, but require some dc current in the on state, similar to a BJT [80]. These devices also have low threshold voltages, which are determined by the built-in potential of the gate-source pn junction diode [78], [79]. Therefore, a negative off-state gate voltage is typically recommended [80]. Furthermore, enhancement mode JFETs also experience manufacturing complexity due to strict dimension requirements needed to achieve normally off behavior [78]. As a result, precise lithography and patterning techniques are needed [77]. Due to the vertical structure, these devices also do not possess internal body diodes [80].

Alternatively, the depletion mode JFETs, which conduct current when no bias is applied to the gate, are a mature technology that has comparably relaxed processing requirements. Of course, since these devices are normally on, added attention must be paid to the driving of these JFETs. Significant research has been done on the driving of these types of devices, with particular attention paid to the cascode configuration, where a low voltage Si MOSFET is connected in series with the JFET, and is always conducting [81]. This MOSFET allows the device to be driven as if it is normally off [81]. The addition of this Si MOSFET adds additional voltage drop, which contributes to the on-state losses [81]. However, [76] proposed a “direct-driven JFET” concept where a separate drive is used for the Si MOSFET. This method eliminates the switching losses due to the Si MOSFET, and reduces its contribution to the conduction losses [76]. Additionally, depending on the layout of the cascode structure, the MOSFET may limit the high temperature operation of the normally on JFET. For instance, if the JFET is mounted to the heat sink, while the MOSFET is soldered to the PCB, then there is minimal thermal coupling between the devices.

The common structure for the SiC normally on JFET has as quasi-vertical structure has a lateral channel, which can conduct bidirectional current, as well as an inherent body diode. The current flow is controlled by the p-type gate and n-type source pn junction. The lateral channel thus narrows when a negative bias is applied across the gate-source junction. Once the bias has surpassed the pinch-off voltage, the channel is pinched-off and thus no current flows and the device is off.

4.6 References

- [1] J. Chang, “Advancements and trends of power electronics for industrial applications,” *IEEE IECON*, vol. 3, pp. 3021-3022, Nov. 2003.
- [2] B. K. Bose, “Technology advancement and trends in power electronics,” *IEEE IECON*, vol. 3, pp. 3019-3020, Nov. 2003.
- [3] J. Rabkowski, D. Peftitsis, H. P. Nee, “Silicon carbide power transistors-A new era in power electronics is initiated,” in *IEEE IEM*, pp. 17-26, June. 2012.
- [4] J. Millan, P. Godignon, X. Perpina, A. Perez-Tomas, J. Rebollo, “A survey of wide bandgap power semiconductor devices,” *IEEE Trans. on Power Electron.*, vol. 29, no. 5, pp. 2155-2163, May 2014.
- [5] S. Chowdhury, U. K. Mishra, “Lateral and vertical transistors using the AlGaIn/GaN heterostructure,” *IEEE Trans. Electron. Devices*, vol. 60, no. 10, pp. 3060-3066, Oct. 2013.
- [6] J. Baliga, *Fundamentals of Power Semiconductor Devices*. New York, Springer, 2008.
- [7] W. Liu, “Electro-thermal simulations and measurements of silicon carbide power transistors,” Ph.D. dissertation, Dept. Microelectron. Inf. Technol., KTH Royal Inst. Technol., Stockholm, Sweden, 2004.
- [8] M. Hasanuzzaman, S. K. Islam, L. M. Tolbert, M. T. Alam, “Temperature dependency of MOSFET device characteristics in 4H- and 6H-silicon carbide (SiC),” *Solid State Electron.*, vol. 48, no. 10-11, pp. 1877-1881, Oct.-Nov. 2004.
- [9] <http://www.cree.com/About-Cree/History-and-Milestones/Milestones>
- [10] P. Friedrichs, T. Kimoto, L. Ley, G. Pensl, *Silicon Carbide: Power Devices and Sensors*, vol. 2, Wiley, 2011.
- [11] Infineon Technologies, “1200V CoolSiCTM & direct drive technology- The green revolution is taking place,” Product Brief, Nov. 2012.
- [12] <http://www.unitedsic.com/news.shtml#20140314>
- [13] D. A. Gajewski, S. H. Ryu, M. Das, B. Hull, J. Young, J. Palmour, “Reliability performance of 1200 V and 1700 V 4H-SiC DMOSFETs for next generation power conversion applications,” *Mater. Sci. Forum*, vol. 778-780, pp. 967-970, Feb. 2014.
- [14] J. S. Glaser, J. J. Nasadoski, P. A. Losee, A. S. Kashyap, K. S. Matocha, J. L. Garrett, L. D. Stevanovic, “Direct comparison of silicon and silicon carbide power transistors in high-frequency hard-switched applications,” *IEEE APEC*, pp. 1049-1056, March 2011.

- [15] ROHM Semiconductor, "SiC power devices and modules," Appl. Note Rev. 001, June 2013.
- [16] Y. Tarui, N. Kaguchi, T. Nakamura, "Silicon carbide semiconductor device," U.S. Patent US8525189 B2, Oct. 2011.
- [17] S. Yamada, "Method of manufacturing silicon carbide semiconductor device," U.S. Patent US8415241 B2, Jan. 2012.
- [18] R. Singh, S. Sundaresan, E. Lieser, M. Digangi, "1200 V SiC 'super' junction transistors operating at 250 °C with extremely low energy losses for power conversion applications," *IEEE APEC*, pp. 2516-2520, Feb. 2012.
- [19] Fairchild Semiconductor, "SiC BJT characteristics and switching," Appl. Note AN-9764, Rev. 1.0.1, Oct. 2012.
- [20] Z. Chen, Y. Yao, M. Danilovic, D. Boroyevich, "Performance evaluation of SiC power MOSFETs for high temperature applications," *IEEE ECCE*, pp. DS1a.8-1-DS1a.8-9, Sept. 2012.
- [21] Z. Chen, D. Boroyevich, R. Burgos, F. Wang, "Characterization and modeling of 1.2 kV, 20 A SiC MOSFETs," *IEEE ECCE*, pp. 1480-1487, Sept. 2009.
- [22] H. Sheng, Z. Chen, F. Wang, A. Millner, "Investigation of 1.2 kV SiC MOSFET for high frequency high power applications," *IEEE APEC*, pp. 1572-1577, Feb. 2010.
- [23] T. Zhao, J. Wang, A. Q. Huang, A. Agarwal, "Comparisons of SiC MOSFET and Si IGBT based motor drive systems," *IEEE IAS*, pp. 331-335, Sept. 2007.
- [24] L. D. Stevanovic, K. S. Matocha, P. A. Losee, J. S. Glaser, J. J. Nasadoski, S. D. Arthur, "Recent advances in silicon carbide MOSFET power devices," *IEEE APEC*, pp. 401-407, Feb. 2010.
- [25] J. McBryde, A. Kadavelugu, B. Compton, S. Bhattacharya, M. Das, A. Agarwal, "Performance comparison of 1200 V silicon and SiC devices for UPS application," *IEEE IECON*, pp. 2657-2662, Nov. 2010.
- [26] S. Piasecki, A. M. Cantarellas, J. Rabkowski, P. Rodriguez, "Design of AC-DC power converters with LCL + tuned trap line filter using Si IGBT and SiC MOSFET modules," *IEEE IECON*, pp. 5957-5962, Nov. 2013.
- [27] D. Han, J. Noppakunkajorn, B. Sarlioglu, "Analysis of a SiC three-phase voltage source inverter under various current and power factor operations," *IEEE IECON*, pp. 447-452, Nov. 2013.
- [28] A. Trentin, P. Zanchetta, P. Wheeler, J. Clare, "Performance evaluation of high-voltage 1.2 kV silicon carbide metal oxide semi-conductor field effect transistors for three-phase buck-type PWM rectifiers in aircraft applications," *IEEE Power Electron. IET*, vol. 5, no. 9, pp. 1877-1881, Nov. 2012.
- [29] A. Merket, T. Krone, A. Mertens, "Characterization and scalable modeling of power semiconductors for optimized design of traction inverters with Si- and SiC- devices," *IEEE Trans. Power Electron.*, vol. 29, no. 5, pp. 2238-2245, May 2014.
- [30] B. Ozpineci, L. M. Tolbert, S. K. Islam, M. Hasanuzzaman, "Effects of silicon carbide (SiC) power devices on HEV PWM inverter losses," *IEEE IECON*, vol. 2, pp. 1061-1066, Nov. 2001.
- [31] A. Blinov, A. Chub, D. Vinnikov, T. Rang, "Feasibility study of Si and SiC MOSFETs in high-gain DC/DC converter for renewable energy applications," *IEEE IECON*, pp. 5975-5978, Nov. 2013.
- [32] L. C. Yu, G. T. Dunne, K. S. Matocha, K. P. Cheung, J. S. Suehle, K. Sheng, "Reliability issues of SiC MOSFETs: A technology for high temperature environments," *IEEE Trans. Device Mater.*, vol. 10, no. 4, pp. 418-426, Dec. 2010.
- [33] W. T. Franke, "Comparison of six different SiC power switching devices in the 1200 V range," *PCIM*, 2012.
- [34] A. Lemmon, M. Mazzola, J. Gafford, K. M. Speer, "Comparative analysis of commercially available silicon carbide transistors," *IEEE APEC*, pp. 2509-2515, 2012.
- [35] B. Rubino, M. Macaudo, M. Nania, S. Buonomo, "Direct comparison among different technologies in silicon carbide," *PCIM*, 2012.
- [36] K. Haehre, M. Meisser, F. Denk, R. Kling, "Characterization and comparison of commercially available silicon carbide (SiC) power switches," *PEMD*, pp. 1-6, 2012.
- [37] S. H. Ryu, S. Krishnaswami, B. A. Hull, B. Heath, F. Husna, J. Richmond, A. Agarwal, J. Palmour, J. Scofield, "A

- comparison of high temperature performance of SiC DMOSFETs and JFETs,” *Mater. Sci. Forum*, vols. 556-557, pp. 775-778, Sept. 2007.
- [38] C. DiMarino, Z. Chen, M. Danilovic, D. Boroyevich, R. Burgos, P. Mattavelli, “High-temperature characterization and comparison of 1.2 kV SiC power MOSFETs,” *IEEE ECCE*, pp. 3235-3242, Sept. 2013.
 - [39] C. DiMarino, Z. Chen, D. Boroyevich, R. Burgos, P. Mattavelli, “Characterization and comparison of 1.2 kV SiC power semiconductor devices,” *IEEE EPE*, pp. 1-10, Sept. 2013.
 - [40] C. DiMarino, Z. Chen, D. Boroyevich, R. Burgos, P. Mattavelli, “High-temperature characterization and comparison of 1.2 kV SiC power semiconductor devices,” *J. Microelectron. Electron. Packaging*, vol. 10, no. 4, pp. 138, 2013.
 - [41] R. Callanan, “Application of silicon carbide MOSFETs,” *Power Mag.*, vol. 12, no. 3, pp. 39-40, 2011.
 - [42] A. Ong, J. Carr, J. Balda, A. Mantooth, “A comparison of silicon and silicon carbide MOSFET switching characteristics,” *IEEE Region 5 Tech. Conf.*, pp. 273-277, Apr. 2007.
 - [43] M. Mudholkar, S. Ahmed, M. N. Ericson, S. S. Frank, C. L. Britton, H. A. Mantooth, “Datasheet driven silicon carbide power MOSFET model,” *IEEE Trans. Power Electron.*, vol. 29, no. 5, pp. 2220-2228, May 2014.
 - [44] T. McNutt, A. Hefner, A. Mantooth, D. Berning, S. H. Ryu, “Silicon carbide power MOSFET model and parameter extraction sequence,” *IEEE Trans. Power Electron.*, vol. 22, no. 2, pp. 353-363, March 2007.
 - [45] Cree, “CMF20120D-Silicon carbide power MOSFET,” *CMF20120D* datasheet, Rev. C, Sept. 2012.
 - [46] Cree, “C2M0080120D- Silicon carbide power MOSFET,” *C2M0080120D* datasheet, 2013.
 - [47] General Electric, “General Electric SiC MOSFET: Ultrafast, low on-resistance, 1200V power switch,” *GE12N20L* datasheet, Rev. 1.0, May 2013.
 - [48] ROHM Semiconductor, “N-Channel SiC power MOSFET,” *SCT2080KE* datasheet, Rev. B, Sept. 2012.
 - [49] ROHM Semiconductor, “N-Channel SiC power MOSFET,” *SCT2160KE* datasheet, Rev. A, May 2013.
 - [50] Z. Chen, “Characterization and modeling of high-switching-speed behavior of SiC active devices,” M.S. thesis, Dept. Elect. Comput. Eng., Virginia Tech, Blacksburg, VA, 2009.
 - [51] R. Burgos, Z. Chen, D. Boroyevich, F. Wang, “Design considerations of a fast 0- Ω gate-drive circuit for 1.2 kV SiC JFET devices in phase-leg configuration,” *IEEE ECCE*, pp. 2293-2300, 2009.
 - [52] IXYS, “HiPerFETTM Power MOSFETs,” *IXFH10N100* datasheet, DS91531F, Jan. 2004.
 - [53] R. Ramovic, M. Jevtic, J. Hadzi-Vukovic, D. Randjelovic, “A novel analytical model of a SiC MOSFET,” *IEEE MIEL*, vol. 2, pp. 447-450, 2002.
 - [54] F. Wang, S. Kher, T. Fichtner, J. Aurich, “A new power MOSFET model and an easy to use characterization tool using device datasheet,” *IEEE COMPEL*, pp. 1-5, June 2013.
 - [55] Vishay Siliconix, “Power MOSFET basics: Understanding gate charge and using it to assess switching performance,” Appl. Note 608, Rev. Dec. 2004.
 - [56] Fairchild Semiconductor, “Power MOSFET switching waveforms: A new insight,” Appl. Note 7502, Oct. 1999, Rev. A1 2002.
 - [57] K. Nielsen, “Audio power amplifier techniques with energy efficient power conversion,” Ph.D. thesis, Dept. Appl. Electron., Tech. Univ. of Denmark, Kongens Lyngby, Denmark, 1998.
 - [58] Z. Chen, “Electrical integration of SiC power devices for high-power-density applications,” Ph.D. dissertation, Dept. Elect. Comput. Eng., Virginia Tech, Blacksburg, VA, 2013.
 - [59] <http://www.cree.com/Power/Tools-and-Support/MOSFET-model-request-form>
 - [60] <http://www.rohm.com/>

- [61] GeneSiC Semiconductor, "SPICE model parameters," *GA06JT12-247*, <http://www.genesicsemi.com/>, Aug. 2013.
- [62] <http://www.infineon.com/cms/en/product/power/sicarbide-sic/coolpic-tm-sic-jfet/channel.html?channel=db3a3043372d5cc801377a322ac7554c>
- [63] J. B. Witcher, "Methodology for switching characterization of power devices and modules," M.S. thesis, Dept. Elect. Comput. Eng., Virginia Tech, Blacksburg, VA, 2013.
- [64] W. Shockley, "The theory of p-n junctions in semiconductors and p-n junction transistors," *Bell System Tech. J.*, vol. 28, pp. 435-489, 1949
- [65] S. Araujo, L. Menezes, P. Zacharias, T. Hjort, "Resurrecting the BJT: New silicon carbide BJTs enable lower system cost in a photovoltaic inverters," <https://www.fairchildsemi.com.cn/Assets/zSystem/documents/collateral/technicalArticle/Resurrecting-the-BJT.pdf>, accessed: Apr. 2014.
- [66] M. Domeij, H. S. Lee, C. M. Zetterling, M. Ostling, "Analysis of the base current and saturation voltage in 4H-SiC power BJTs," *IEEE EPE*, pp. 1-7, Sept. 2007.
- [67] Fairchild Semiconductor, "57 mΩ silicon carbide NPN power transistor," *FSICBH057A120* datasheet, Rev. 0.0.7, Nov. 2012.
- [68] Fairchild Semiconductor, "17 mΩ silicon carbide NPN power transistor," *FSICBH017A120* datasheet, Rev. 0.0.7, Nov. 2012.
- [69] GeneSiC Semiconductor, "Normally-off silicon carbide super junction transistor," *GA06JT12-247* datasheet, Feb. 2013.
- [70] Y. Gao, "Analysis and optimization of 1200V silicon carbide bipolar junction transistor," Ph.D. dissertation, Dept. Elect. Eng., NC State Univ., Raleigh, NC, 2007.
- [71] Claudio, H. Wang, A. Q. Huang, A. K. Agarwal, "Static and dynamic characterization of silicon carbide bipolar junction transistor," *IEEE IECON*, vol. 2, pp. 1173-1178, Nov. 2003.
- [72] Buono, "Simulation and characterization of silicon carbide power bipolar junction transistors," Ph.D. dissertation, Integr. Dev. Circuits Dept., KTH Royal Inst. Technol., Stockholm, Sweden, 2012.
- [73] B. Buono, R. Ghandi, M. Domeij, B. G. Malm, C. M. Zetterling, M. Ostling, "Modeling and characterization of current gain versus temperature in 4H-SiC power BJTs," *IEEE Trans. Electron. Dev.*, vol. 57, no. 3, pp. 704-711, March 2010.
- [74] M. J. Kumar, P. Vinod, "Enhanced current gain in SiC power BJTs using surface accumulation layer transistor (SALTran) concept," *J. Microelectron. Eng.*, vol. 81, pp. 90-95, April 2005.
- [75] X. Li, Y. Luo, L. Fursin, J. H. Zhao, M. Pan, P. Alexandrov, M. Weiner, "On the temperature coefficient of 4H-SiC BJT current gain," *Solid-State Electron.*, vol. 47, pp. 233-239, April 2003.
- [76] R. Siemienieć, U. Kirchner, "The 1200V direct-driven SiC JFET power switch," *IEEE EPE*, pp. 1-10, Sept. 2011.
- [77] W. Berger, F. Bjoerk, D. Domes, G. Deboy, "Infineon's 1200V SiC JFET- The new way of efficient and reliable high voltages switching," www.infineon.com, accessed: Apr. 2014.
- [78] I. Sankin, D. C. Sheridan, W. Draper, V. Bondarenko, R. Kelley, M. S. Mazzola, J. B. Casady, "Normally-off SiC VJFETs for 800 V and 1200 V power switching applications," *IEEE ISPSD*, pp. 260-262, May 2008.
- [79] S. Basu, "On understanding and driving SiC power JFETs," *IEEE APEC*, pp. 1071-1075, March 2011.
- [80] SemiSouth Laboratories, "Silicon carbide enhancement-mode junction field effect transistor and recommendations for use," Appl. Note SS1.
- [81] R. Siemienieć, G. Nobauer, D. Domes, "Stability and performance analysis of a SiC-based cascade switch and an alternative solution," *Microelectron. Rel.*, vol. 52, no. 3, pp. 509-518, March 2012.
- [82] A. Maralani, "Design of operational amplifiers and utilizing SiC JFET for analog design," Ph.D. dissertation, Dept. Elec. Comput. Eng., MS State Univ., Starkville, MS, 2009.

- [83] Infineon, “SiC-JFET, CoolSiCTM,” *IJW120R100T1* datasheet, Rev. 2.0, Sept. 2013.
- [84] Infineon, “SiC-JFET, CoolSiCTM,” *IJW120R070T1* datasheet, Rev. 2.0, Sept. 2013.
- [85] SemiSouth, “Normally-off trench silicon carbide power JFET,” *SJEP120R100* datasheet, Rev. 3.0, 2012.
- [86] D. C. Sheridan, A. Ritenour, V. Bondarenko, P. Burks, J. B. Casady, “Record $2.8\text{m}\Omega\text{-cm}^2$ 1.9kV enhancement-mode SiC VJFETs,” *IEEE ISPSD*, pp. 335-338, June 2009.
- [87] M. Chinthavali, P. Ning, Y. Cui, L. M. Tolbert, “Investigation on the parallel operation of discrete SiC BJTs and JFETs,” *IEEE APEC*, pp. 1076-1083, March, 2011.
- [88] Y. Zhang, K. Sheng, M. Su, J. H. Zhao, P. Alexandrov, X. Li, L. Fursin, M. Weiner, “Development of 4H-SiC LJFET-based power IC,” *IEEE Trans. Electron. Devices*, vol. 55, no. 8, Aug. 2008.
- [89] D. C. Sheridan, A. Ritenour, R. Kelley, V. Bondarenko, J. B. Casady, “Advances in SiC VJFETs for renewable and high-efficiency power electronics applications,” *IEEE IPEC*, pp. 3254-3258, June 2010.
- [90] S. Round, M. Heldweign, J. Kolar, I. Hofsjager, P. Friedrichs, “A SiC JFET driver for a 5 kW, 150 kHz three-phase PWM converter,” *IEEE IAS*, vol. 1, pp. 410-416, Oct. 2005.

Chapter 5: Wide Bandgap Power Devices Adoption on the Rise^[1]

5.1 Introduction

As power electronics continues to extend its reach into the emerging markets of renewable energies, smart grids, smart homes, lighting, electric and hybrid electric vehicles (EV/HEVs), transportation electrification, and other developing industrial and medical applications, including Internet of Things (IoT) and 5G communications, more and more designers are attracted by the benefits of emerging wide bandgap (WBG) power devices, namely silicon carbide (SiC) and gallium nitride (GaN). With falling prices, improved reliability and commercial availability from multiple sources, the adoption of SiC and GaN power devices is on the rise in the emerging new applications, as well as existing ones like chargers for consumer devices, such as smartphones, laptops, and TVs, on-board chargers (OBCs) for xEVs and power supplies. According to PowerAmerica's annual report, the market for WBG devices is at a tipping point with unprecedented growth forecast over the next decade.

The latest report from market research firm Yole Développement SA suggests that total market for SiC power devices by 2023 will be over \$1.5 billion with a compound annual growth rate (CAGR) of 31% for next five years. Similarly, PowerAmerica's SiC power semiconductor market projection chart (Figure 1) shows that total market revenues for SiC devices will cross the \$3 billion mark by 2025 and reach around \$10 billion by 2027. Key drivers for this growth are EVs, HEVs and power supplies.

5.2 SiC Growth Predictions

Although initially the SiC market was primarily driven by diodes used in power factor correction (PFC) and photovoltaic (PV) applications, the Yole analyst expects transistors to be the key drivers for SiC device market going forward. This rise in adoption is attributed to improved transistor performance and reliability as compared to the first generation products.

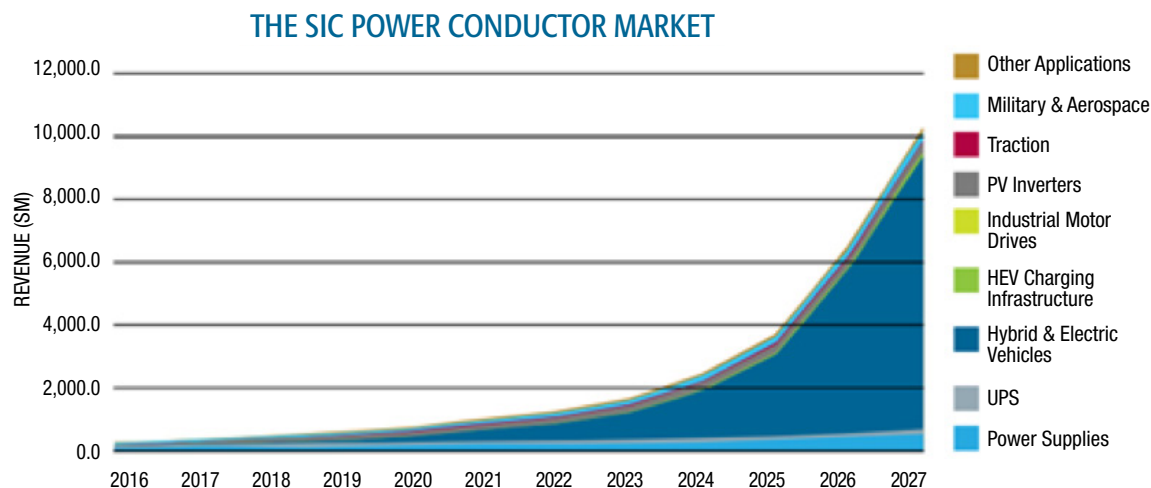


Figure 1: The market for silicon carbide power devices will experience an unprecedented growth over the next decade.

Source: PowerAmerica

Per Yole's research, SiC adoption will rise in automotive applications in the next 5-10 years. In fact, as of 2018, more than 20 automotive manufacturers have already used SiC Schottky barrier diodes (SBDs) or SiC MOSFET transistors in on-board chargers (OBCs) while adoption in the main inverter is on the growth path. With the xEV charging infrastructure market growing and rapidly adopting SiC devices, leading Chinese xEV supplier Build Your Dreams, better known as BYD, has confirmed its use of SiC devices in its on-board chargers. While leading American EV maker Tesla has adopted STMicroelectronics' SiC MOSFET based power module in the inverter section of Tesla

Model 3, which is in production. Similarly, in Europe, Micropower and ABB have unveiled automotive battery chargers based on SiC devices. According to Yole, all carmakers, including Chinese auto manufacturers, have plans to implement SiC devices in the main inverter in the coming years.

To meet emerging capacity demands, manufacturers have transitioned from 4-inch (100-mm) to 6-inch (150-mm) SiC wafers while concurrently expanding capacity. New players are joining the business to cater to this increasing demand. For example, II-VI Inc. has announced that it will supply 200-mm SiC substrates to Horizon 2020, a four-year program funded by the European Commission. The goal is to establish in Europe the world's first 200-mm pilot production facility with the aim of scaling the mass production of 200-mm substrates for SiC devices in power applications ranging from electric cars and renewable energy systems to smart grids. Similar efforts are also underway at PowerAmerica where the institute is readying a WBG manufacturing process based on 200-mm wafers to further drive the cost down. As a result, according to Victor Veliadis, PowerAmerica's deputy executive director and CTO, "WBG are expected to reach a 1.5-times cost parity with silicon within five years."

To ensure supply of wafers in the future, manufacturers like STMicroelectronics has inked a multi-year agreement with Cree to supply its 150-mm Wolfspeed SiC wafers to the company whose SiC power modules have been adopted in Tesla Model 3. Accelerated adoption of SiC devices has also prompted Infineon Technologies to sign a long-term SiC wafer supply agreement with Cree

These developments are also attracting more foundry service providers to enable fabless and fab-lite companies to launch SiC devices for emerging applications. A good example is the new 6-inch wafer foundry Clas-SiC Wafer Fab Limited established in 2017 in Scotland, U.K. In Lubbock, Texas, X-FAB plans to double its 6-inch SiC process capacity, and similar efforts are also underway at Taiwan based Episl Technologies and TSMC.

As standards are essential to drive the adoption of new technologies, JEDEC Solid State Technology Association has announced the formation of a new JEDEC committee: JC-70. JEDEC president John Kelly added, "The formation of the JC-70 committee is part of an ongoing effort within JEDEC to extend our standards setting expertise to new technologies to meet market demands. We welcome all interested companies to participate in the development of open industry standards within JEDEC."

5.3 More SiC Suppliers

Meanwhile, the number of SiC MOSFET suppliers has doubled in the past several years, and it is expected to increase as new fabless suppliers of SiC power devices enter the marketplace. For easier entry into the SiC supply chain, researchers from North Carolina State University (NCSU) have developed a new manufacturing process and chip design for SiC power devices. Process Engineered for Manufacturing SiC Electronics-Devices (PRESiCE) was developed with support from the PowerAmerica Institute funded by the Department of Energy. Details of the process were presented at the International Conference on Silicon Carbide and Related Materials 2017 (ICSCRM 2017), Washington, DC [3].

To implement this process, PRESiCE team is using the X-Fab Texas foundry, located in Lubbock. Currently, the process is using 6-inch substrates. Researchers have already licensed the PRESiCE process and chip design to one U.S. based semiconductor company, and several others are in discussion.

In addition, Germany based Danfoss Silicon Power GmbH has collaborated with General Electric (GE) to build SiC modules in Marcy, New York based facility called Computer Chip Commercialization Center (Quad-C). GE will supply SiC chips for these modules aimed at PV inverters, EVs and other applications. With plans to begin production by late 2018, New York State is investing more than US\$200 million for the construction, tools and equipment for the factory and Danfoss is leasing the space and equipment to produce SiC modules for the U.S. market.

Likewise, component supplier Littelfuse acquired technology start-up Monolith Semiconductor to develop high voltage SiC diodes and MOSFETs for a wide range of applications, while key SiC power device suppliers like Wolfspeed, ROHM Semiconductor, STMicroelectronics, Infineon, Microsemi Corporation, GeneSiC Semiconductor, and UnitedSiC, among others, have begun expanding the production of their respective SiC MOSFETs, junction FETs (JFETs) and transistors, and diodes, including modules, to meet market demands.

Concurrently, as a new member of PowerAmerica, semiconductor supplier Microsemi Corporation was awarded a contract to accelerate the adoption of SiC devices in products and systems. Under a US\$70 million contract, Microsemi will develop 1.7 kV and 3.3 kV SiC MOSFETs and Schottky diodes for a wide range of industrial, aerospace and defense markets, as well as solar inverters and motor drives.

A significant development further accelerating the adoption of SiC MOSFETs is the development of gate driver ICs and boards. They are available from STMicroelectronics, Infineon, Power Integrations, Texas Instruments, Microsemi, Prodrive Technologies and ROHM. To simplify the use of gate drivers, start-up Agileswitch of Philadelphia, Penn., has readied software-configurable gate driver boards, optimized for use with SiC MOSFETs and SiC modules.

5.4 Growing GaN-on-Silicon Market

With rising GaN adoption in a variety of applications, Yole is projecting the GaN device business to reach around \$423 million by 2023 with a CAGR of 55%. While the near-term drivers for this market are power supplies, UPS, PV inverters and other applications, long-term applications include power supplies, EVs and HEVs, military & aerospace, industrial motor drives and other applications like light detection and ranging (LiDAR), envelope tracking and wireless power. As a result, according to PowerAmerica's 2018 report, the total GaN power semiconductor revenues could reach \$1.8 billion by 2027 (Figure 2). It expects this market to be over \$800 million by 2025.

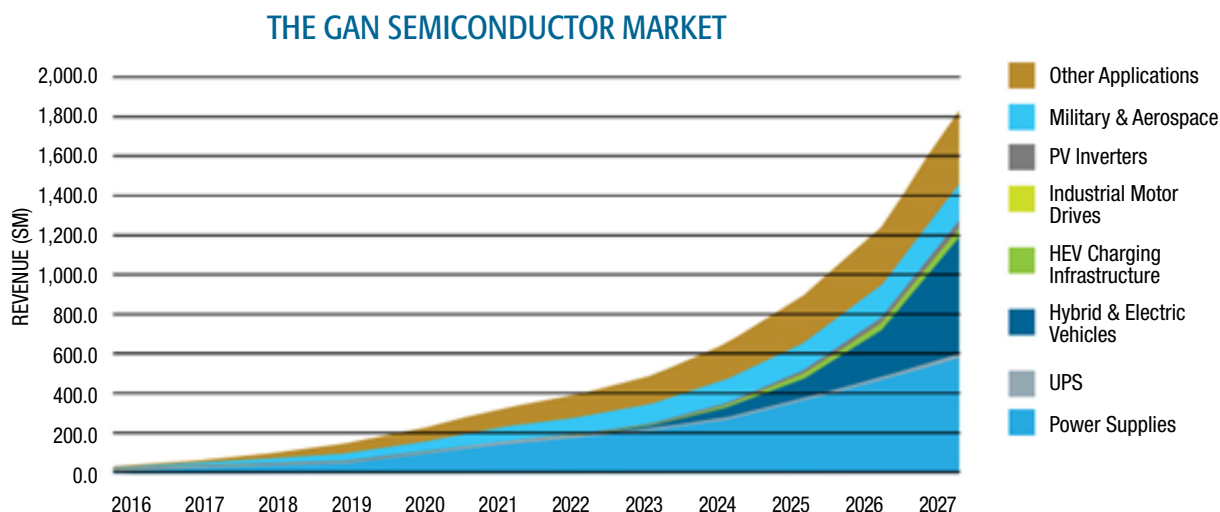


Figure 2: The adoption of GaN-on-Silicon power transistors, including modules and ICs, is on the rise. It is predicted to reach around \$1.8 billion by 2027.

Source: PowerAmerica

Consequently, the production of GaN-on-Silicon power transistors is on the rise, including GaN power modules, and GaN power integrated circuits (ICs). Lately, GaN power ICs have emerged to further drive the system cost down and accelerate the adoption of GaN power devices in a variety of applications. Recently, Navitas Semiconductor demonstrated the commercial feasibility of monolithically integrating GaN FET, GaN driver and GaN logic to set the stage for GaN power ICs. Others driving GaN integration include Dialog Semiconductor, Efficient Power Conversion Corp., GaN Systems and Texas Instruments Inc. to name a few.

As demand rises, more and more start-ups are entering the fray and partnering with foundries to boost production capacities. Toward that goal, French start-up Exagan, in partnership with X-FAB Silicon Foundries, demonstrated the capability to manufacture high-voltage GaN power devices on 200-mm (8-inch) GaN-on-silicon wafers using X-FAB's standard CMOS production facility in Dresden, Germany. The maker affirmed that this will significantly lower the cost of mass-producing GaN-on-silicon devices, thereby accelerating its adoption in EV chargers, and power supplies for servers and industrial systems. The company has announced the availability of 650 V and 1200 V G-FETs for converters

in solar, automotive and data center applications. Likewise, to boost production, start-up Navitas Semiconductor announced a partnership with TSMC and Amkor Technology. At the same time, German automaker BMW's venture capital arm BMWi Ventures has announced its strategic investment in GaN Systems of Ottawa, Canada.

Other major investors attracted to GaN-on-Silicon technology include power supply giant Delta Electronics and the Japanese mechatronics maker Yaskawa Electric Corporation. Like BMWi, Delta has also joined GaN Systems as a strategic investor. Although details were not available, it is obvious that the leading power supply manufacturer is planning to tap the benefits of GaN devices for server and data center power supplies, laptop/tablet/smartphone chargers, and motor drives for industrial applications, to name a few.

Satisfied with the impact of high voltage GaN on its servo motor drive, Yaskawa Electric Corporation announced a US\$15 million investment in GaN manufacturer Transphorm Inc., located in Goleta, California. Implementing Transphorm's JEDEC and AEC-Q101 qualified 650 V GaN-on-silicon FETs in TO-220 package; Yaskawa Electric has released an ac servo motor with a built-in amplifier (model Σ -7 F) with unprecedented performance and power density. Higher performance from a smaller form factor has prompted the mechatronics leader to extend the benefits of GaN FETs to the entire Σ -7 F product line, which ranges from 100 W to 400 W.

Other manufacturers adopting Transphorm's 650 V GaN FETs include Telcodium, Bel Power Solutions, Delta and Tata Power. While Telcodium has introduced a high efficiency redundant power supply in 30% smaller footprint using GaN FETs, Bel Power has released an ac-dc frontend using GaN based bridgeless totem-pole power factor correction (PFC) topology with >96% efficiency to achieve 80 Plus Titanium efficiency and to earn a CE Mark per EC's low voltage directive. Similarly, Delta has exploited the benefits of high-voltage GaN to build an 80Plus Platinum 800 W power supply unit (PSU) with a built-in lithium-ion battery. The use of GaN enabled the maker to reduce size by 25%, allowing for battery inclusion within a desired form factor. For the Indian market, Tata Power has launched a 1.5 kW PV inverter, which is fully qualified for production.

Concurrently, Transphorm has also started sampling its 900 V GaN FET in standard TO-220 package with on-resistance of 170 m Ω and 50 m Ω , respectively. The 900 V platform was developed with supplemental funding from U.S DoE's Advanced Research Projects Agency-Energy (ARPA-E). To enable U.S. manufacturers to jump start the adoption of 900 V GaN devices, Transphorm has developed a 3 kW PV inverter reference design in collaboration with PowerAmerica, according to Transphorm's co-founder and COO Primit Parikh.

In short, SiC and GaN devices have currently reached a level of maturity that is further accelerating their adoption in existing and emerging applications ranging from chargers for consumer appliances to power trains for EVs & HEVs inverters/converters for renewable energy sources. The growth projections for both SiC and GaN power devices look encouraging as manufacturers of WBG devices to continue to improve performance, cost and reliability.

5.6 References

- [1] A. Bindra, "Wide-Bandgap Power Devices: Adoption Gathers Momentum," in *IEEE Power Electronics Magazine*, vol. 5, no. 1, pp. 22-27, March 2018. doi: 10.1109/MPEL.2017.2782404
- [2] Silicon Carbide and Gallium Nitride Power Semiconductors-2016, IHS Markit Technology
- [3] SiC technologies adoption is going to accelerate with a tipping point in 2019, Yole Développement SA, August 2017 http://www.yole.fr/STATUS_SIC_MARKET_TECHNOLOGY_TRENDS.aspx#.Wj1jh1WnGuk
- [4] B.J. Baliga, K. Han, J. Harmon, A. Tucker, S. Syed, and W. Sung "PRESiCE: Process Engineered for manufacturing SiC Electronic-devices," 17- 22 September 2017, International Conference on Silicon Carbide and Related Materials, Washington, D.C.
- [5] Anna Wennberg, "Micropower charges with SiC," *Elektronik Tidningen*, November 2017 <http://etn.se/index.php/reportage/64078-micropower-laddar-med-kiselkarbid>
- [6] J. Plummer, "Power Semiconductor Devices-Silicon vs New Materials, IEEE Control and Modeling for Power Electronics (COMPEL) Workshop, Stanford, CA, July 10, 2017

Chapter 6: Roadmap for WBG and UWBG Materials and Devices

6.1 Introduction

The purpose of the International Technology Roadmap for Wide-Bandgap Power Semiconductors (ITRW) Materials and Devices Working Group, which considers the materials science of Wide- and Ultra-Wide-Band-Gap (WBG and UWBG) semiconductors, in addition to device design, fabrication, and evaluation, is to formulate a long-term, international roadmap for WBG and UWBG materials and devices, consistent with the packaging and applications working groups of ITRW. The working group is co-chaired by Victor Veliadis (primarily representing silicon carbide (SiC) and related materials) and Robert Kaplar (primarily representing gallium nitride (GaN) and related materials, as well as emerging ultra-WBGs) and is split into four sub-working-groups, which are:

1. SiC materials and devices (co-chairs Jon Zhang and Mietek Bakowski)
2. Lateral GaN materials and devices (co-chairs Sameh Khalil and Peter Moens)
3. Vertical GaN materials and devices (co-chairs TBD)
4. Emerging UWBG materials and devices (co-chairs Mark Hollis)

The first two subgroups represent technology that is far more mature than that of the latter two, and devices are available as commercial products in power applications. The primary focus of this article will be on developments in subgroups 1 and 2, with only brief descriptions of the latter two sub-groups, including future activities as they mature technologically.

6.2 SiC Materials and Devices

The SiC Materials and Devices sub-Working-Group (SiC-MDWG) of the ITRW was formed in 2018 and is composed of experts from industry and academia with specialization in SiC materials and devices. The team is documenting SiC power devices at different development stages. This includes desired device types (e.g. MOSFET, JFET, IGBT, GTO, etc.) for specific applications, operating conditions, and required voltage and current ratings in the short-term (~1-5 years), and long-term (~6-10 years). The cost target in \$/A is key for SiC devices to be widely adopted in power systems, and will be emphasized. In 2018, a technical questionnaire for SiC materials and devices was created. The questionnaire was distributed internationally (US, Asia, and Europe) to collect feedback from SiC experts in academia and industry. The roadmap on SiC materials and devices incorporates feedback from this questionnaire and from other communications with SiC experts. The roadmap outlines key markets and application areas for SiC power electronics, the performance targets SiC technologies are expected to meet over time, technical barriers to achieving those targets, and activities needed to overcome those barriers.

With respect to SiC substrates and epitaxy, 150 mm wafers are commercially available, and the majority of commercial devices are produced on wafers of this area. 200 mm wafers have been demonstrated at trade shows, and their market introduction in the near future is expected to lead to cost reductions and to improved competitiveness with Si. 650 to 1700 V SiC planar metal-oxide-semiconductor field-effect transistor (MOSFETs) are commercially available from several vendors, and certain voltage ratings also have trench MOSFET offerings. 3.3 kV MOSFETs are rigorously investigated for near term product release. 6.5 and 10 kV MOSFETs have been demonstrated and are next in line for commercialization. The issue of MOSFET degradation with body diode current conduction is being addressed by multiple vendors, and this will accelerate 3.3 to 10 kV MOSFET product releases. The SiC IGBT is the device of choice for applications requiring +15 kV power conditioning. Work in tailoring the ambipolar lifetime to optimize performance is underway, and large-scale commercialization of these devices is about 10 years away.

6.3 Market Forecast of SiC Power Devices and Applications

Prices for SiC devices have been declining rapidly in the past few years, helping fuel the recent market growth. SiC MOSFET prices, for example, dropped 50% between 2012 and 2015 according to IHS Markit [1]. Although prices rose in 2017 due to wafer supply shortages, a growing number of wafer suppliers and improved wafer quality should

allow prices to stabilize in 2019 and then continue to decline in the foreseeable future. This improvement in cost competitiveness is helping SiC dislodge Si in certain applications (e.g. hybrid and electric vehicles). In addition, leading manufacturers now have trillions of hours of field device experience to assuage any reliability concerns that might dampen growth. A recent study by IHS Markit projects annual SiC revenues will reach \$10 billion by 2027, with hybrid and electric vehicles making up the vast majority of sales. Across these applications, discrete power devices will account for most of the growth as they are expected to take off faster than power modules and integrated circuits. A separate forecast by Cree Inc. also predicts that EVs will present a tremendous growth opportunity for WBG materials, particularly SiC [2]. To date, automakers have announced plans to spend \$150 billion in the EV market. Cree estimates that even modest EV adoption – approximately 10% of total vehicles sales by 2027 – could result in SiC revenues of \$6 billion. The same forecast places the total SiC PE market at over \$2 billion by 2022, largely driven not only by EV adoption but also by industrial and telecom applications. SiC technology has already proven its technical advantages over Si, and recent decreasing prices are fueling adoption.

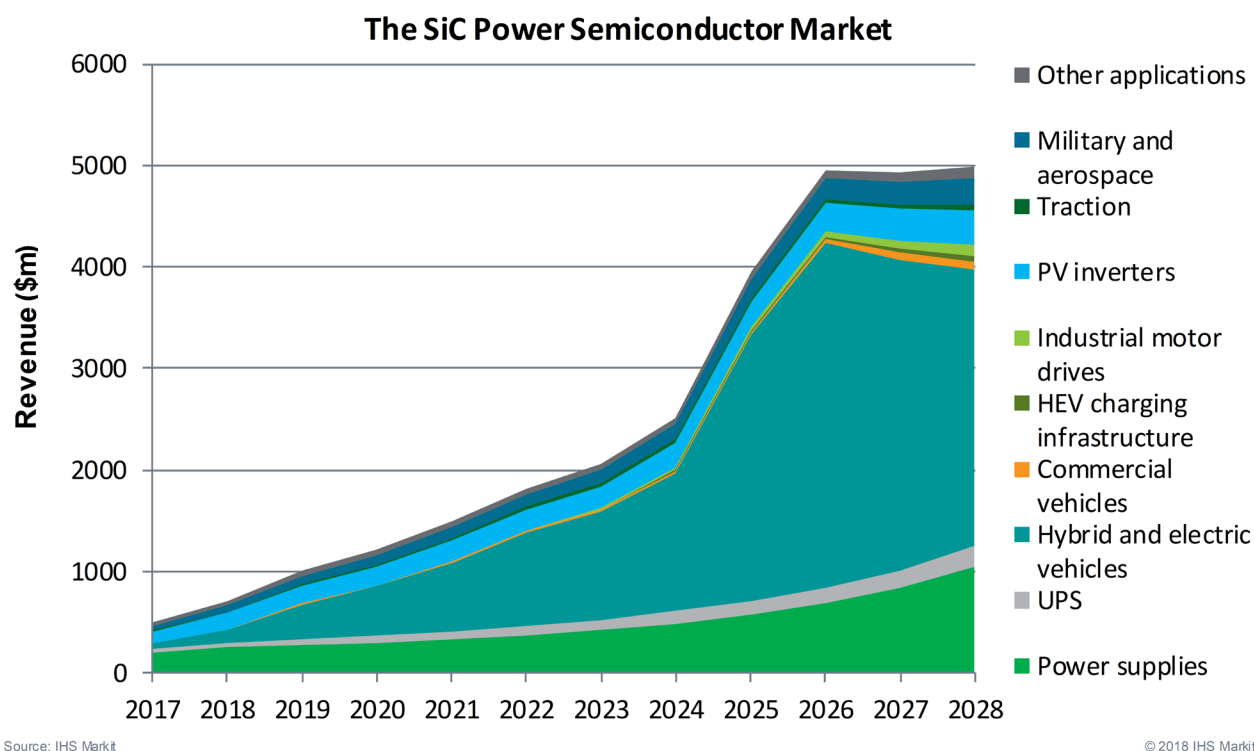


Figure 1: The world market for SiC power devices. Source: IHS Markit

Figure 1 shows various SiC device applications and their potential market size over time [3]. The key near-term application with large market penetration size is EV/HEV inverters, and its revenue could reach \$4 billion in 2026. According to the Electric Vehicles Initiative, over 20 million electric vehicles may be on the road in 2020, climbing to over 200 million by 2030 in the most ambitious scenario. Motor drive inverters with high power density and efficiency at elevated temperatures are essential to EVs. SiC semiconductors are ideal for such applications because of their advantages over Si in high-temperature and -frequency operation. Lower system costs and vehicle design simplifications will improve PE device integration with vehicles. The cost competitiveness of SiC semiconductors is the main hurdle in EV/HEV inverter adoption. UPS, power supply, PV, and traction are currently big markets for SiC power devices as well, and their market penetration is expected to further increase in the coming years.

6.4 Performance Targets for SiC Power Devices in Various Applications

To accelerate SiC technology adoption, the PE community must focus on applications that are most likely to deliver immediate (within 5 years) improvements in efficiency, reliability, and total cost of ownership. These in turn will

encourage further market growth and industry demand. For long-term applications, SiC power devices normally have higher voltage ratings than their near-term counterparts. Fabricating high-voltage devices at high yields in large volumes will remain a challenge for at least the next few years. Table 1 shows SiC MOSFET performance targets for some of the near term and long-term applications.

Table 1. Performance targets for SiC applications [4].

Performance Target	Year 1-5	Year 6-10
Electric Vehicle Inverters		
Rated Current for MOSFET [A]	> 50	> 100
Cost [\$/A]	< 0.3	Cost parity to Si IGBT
PV Inverters		
Rated Current for MOSFET [A]	> 50	> 100
Cost [\$/A]	< 0.3	< 0.2
Data Center (SiC, 650V)		
Rated Current for MOSFET [A]	> 50	> 100
Cost [\$/A]	~ 0.1	~ 0.1
Grid-tied energy storage (SiC, 3.3-10kV)		
Rated Current for MOSFET [A]	> 10	> 50
Cost [\$/A]	< 5	< 2
Heavy-duty vehicle (SiC, 1.7-3.3kV)		
Rated Current for MOSFET [A]	> 50	> 100
Cost [\$/A]	< 2	< 1

6.5 SiC Power Devices Pricing Comparison

Figure 2 compares the cost of various Si and SiC power devices. The data were obtained in December of 2017 from the Digikey distribution price list. SiC diode and MOSFET devices listed in PowerAmerica's Device Bank were also included in the comparison, although the number of comparable devices was limited. In general, SiC power devices become increasingly more expensive than their Si counterparts as voltage rating increases. For example, at 1200 V, the price of SiC diodes is 2.5-3x higher than that of their Si counterparts. At 10 kV, SiC diodes can cost as much as \$20/Amp. For 600 V SiC switches, there is approximate cost parity with Si. Above 900 V, SiC device prices are 2-4x higher than those of Si IGBTs. The price differential is primarily attributed to the higher material cost of the SiC substrate/epitaxy and the higher cost of SiC devices with larger chip area. Devices with larger chip areas and thicker epitaxy experience lower yields due to the presence of material and processing defects, which reduce yield thereby increasing cost.

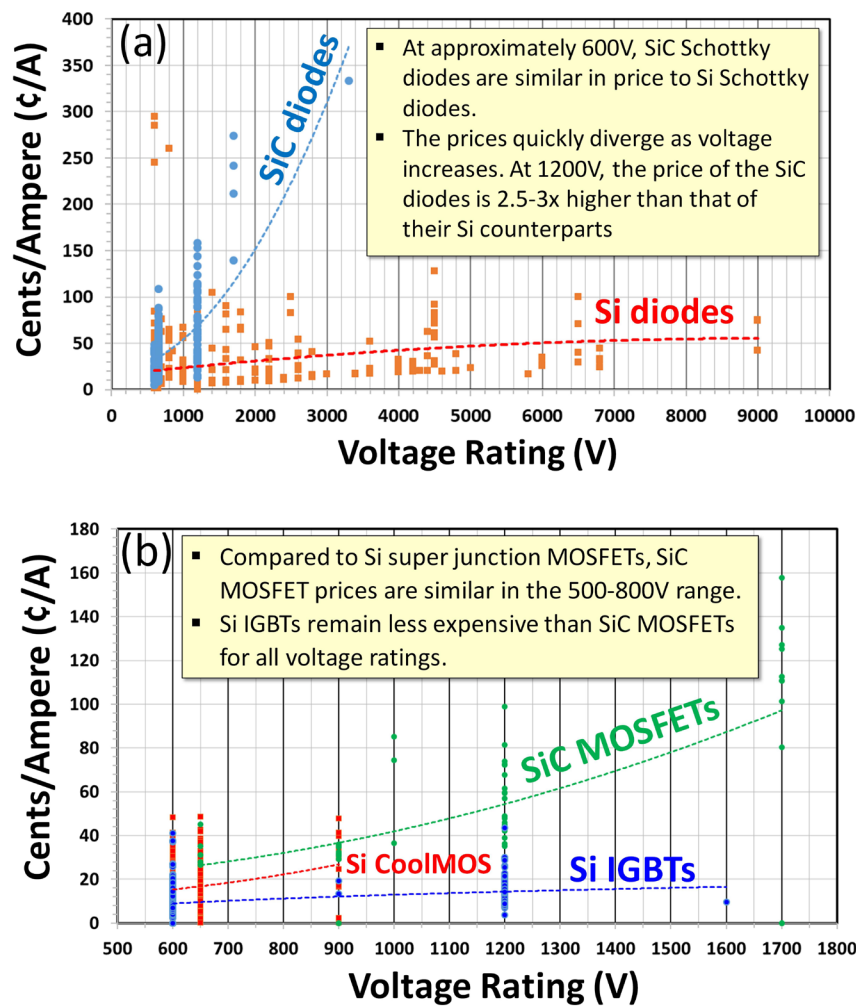


Figure 2: Cost comparison between SiC and Si power devices (a) diodes, and (b) switches [4].

In addition to higher quality SiC wafer material to increase yield and reliability, other areas of improvement include wafers with higher planarity to accommodate processing in high volume foundries, higher quality gate oxides to improve threshold voltage stability and mobility, and improvements in short-circuit performance and ruggedness to rival those of Si devices. Better understanding of degradation/failure mechanisms under harsh conditions (i.e. high voltages and/or high temperatures) will drive device design optimization. Development of SiC-specific high voltage/frequency/temperature reliability standards will facilitate system insertion. Finally, to increase fabrication throughput and yields, innovations in SiC doping will focus on room temperature implantation with minimal generation of killer defects.

6.6 Lateral GaN Materials and Devices: Current State-of-the-Art

The adoption of normally off lateral GaN power devices in the commercial market has so far been limited to p-GaN gate devices and d-mode GaN HEMTs in cascode configuration with an e-mode Si MOSFET. Panasonic's gate injection transistor (GIT) is one such example of a p-GaN gate device, with a demonstrated drain current of 200 mA/mm, on-state resistance of $2.6 \text{ m}\Omega\text{-cm}^2$, and an off-state breakdown of 800 V [5]. To solve the issue of current collapse, the GIT structure was modified to include a hybrid drain (HD-GIT), shown in Figure 3, which is used to compensate for hole emission in the buffer and has been demonstrated to eliminate current collapse phenomena up to 980 V [6,7]. More work has been done on the HD-GIT structure to reduce process variation associated with gate-recess, and VT standard deviation has been reduced significantly as a result of a new process named "through recess and regrowth gate" [6].

In the cascode configuration, a low-voltage normally off Si MOSFET is paired with a normally on high-voltage GaN HEMT to produce an effective e-mode power device (Figure 4 (a)). This cascode approach offers the advantage of improved switching speed and reduced switching loss due to the mitigation of the Miller effect. Recent work has shown progress for monolithically integrated cascode structures, including an all-GaN integrated structure (shown in Figure 4 (b)) [8]. This not only reduces package parasitics between the Si and GaN devices of the conventional cascode but also reaps the benefits of the native e-mode GaN device and the reduced circuit instability caused by device mismatch.

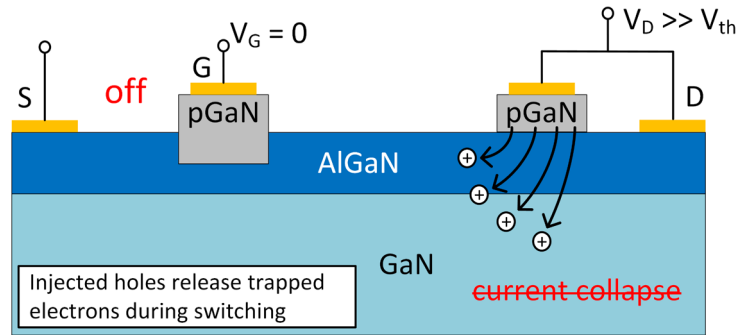


Figure 3: Example of HD-GIT structure showing injection of holes in the off-state to eliminate current collapse.

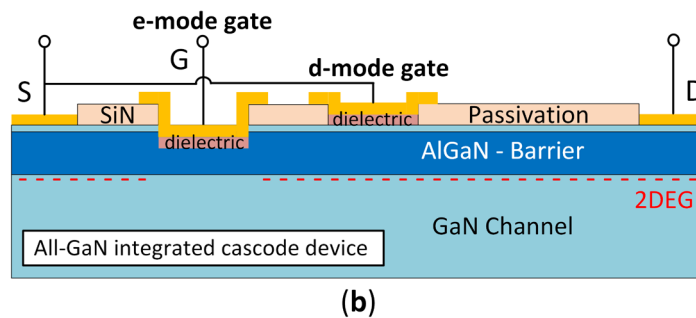
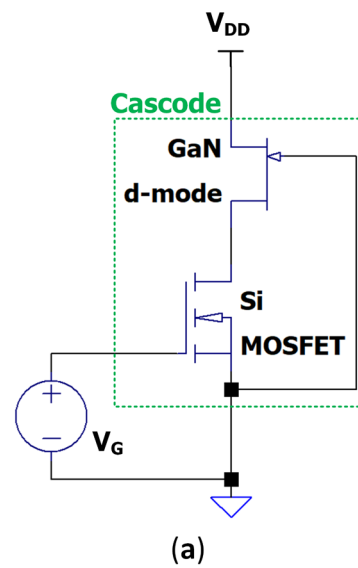


Figure 4: (a) Conventional GaN+Si cascode circuit diagram, and (b) Example of an all-GaN monolithically integrated cascode device.

To enable growth of GaN on large-diameter Si wafers, it is necessary to reduce bowing and growth stress. The improvement of crystallinity in epitaxial films and precise management of growth stress are critical to the success

of large-diameter GaN processing, which allows for cost reductions. Recent work with superlattice structures has demonstrated crack-free 2 μm GaN layers grown on 200 mm Si using AlN/GaN superlattice layers [9]. With a superlattice buffer structure, compressive and tensile stresses can be precisely adjusted resulting in controllable wafer bow (Figure 5). The caveat is that compared to a graded buffer structure, a superlattice structure (SLS) suffers from poorer thermal performance. From this perspective, it is desirable to further improve the thermal performance of superlattice structures or to reduce inherent growth stress and defect density in graded buffer structures.

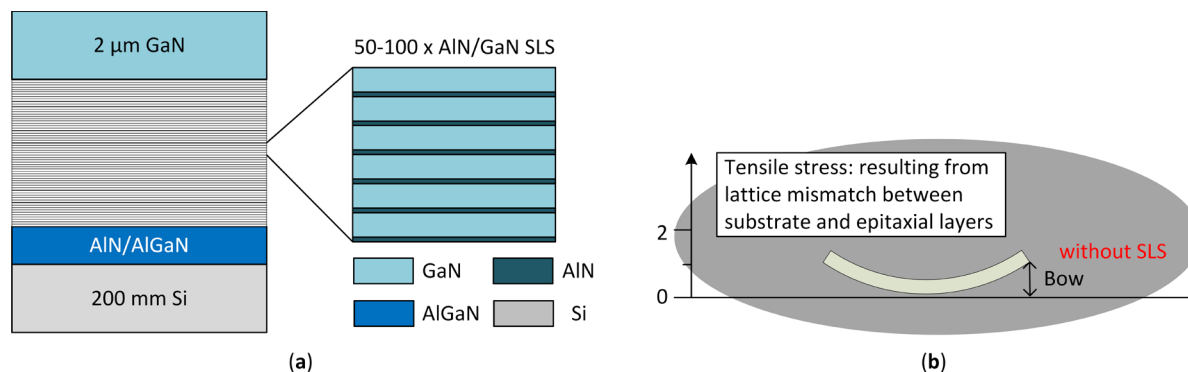


Figure 5: (a) Examples of a superlattice buffer structure used to mitigate growth stress, and (b) tensile stress that occurs in the absence of an SLS due to lattice mismatch.

Current collapse is still a critical issue for commercial GaN power devices, although progress has been made recently in mitigating this issue. As mentioned, Panasonic's HD-GIT structure is known to eliminate current collapse by compensating for hole emission in the buffer. Another method to suppress current collapse is to compensate for surface interface traps resulting from dangling bonds and nitrogen vacancies. Recent work by Tang et al. [10] demonstrates that a charge-polarized AlN passivation layer can suppress current collapse so long as the polarization charge density is higher than the interface trap density. In addition, the use of a nitrated interfacial layer via in-situ plasma nitridation prior to deposition of the gate dielectric or passivation has proven successful for compensating nitrogen vacancies and passivating Ga dangling bonds, creating a high-quality interface and reducing interface traps [11].

6.7 Lateral GaN Foundational Keystones

The foundational requirements for GaN devices and materials fall mainly into three categories: improved heterostructure growth, robust normally off device designs, and refined reliability characteristics. As GaN power HEMTs are transitioning in commercial production from 6" to 8" Si wafers, minimizing wafer bow and the associated growth stress becomes increasingly critical to creating a high-yield and uniform process. Revisions to the heterostructure design and growth processes are necessary not only to reduce stress but also to decrease material defects and to improve thermal performance. GaN technology enables a significant improvement in power density and with an associated reduction in die size. This introduces thermal management challenges like effectively extracting heat from the system. These challenges must be addressed both at the materials and at packaging levels.

Recent advances in normally off device designs, including the HD-GIT, have produced more robust outcomes, but more progress must be made in the areas of both process uniformity and reliability. Recently, device reliability has been studied extensively, resulting in significant improvements in this area. However, additional work must be done to fully understand degradation mechanisms and further improve reliability. Two areas of note include studying circuit and mixed-mode reliability, as well as expanding device ruggedness and safe operating area to rival those of Si. Circuit designers need to be confident in the long-term reliability of GaN power devices – otherwise adoption will be limited and slow.

6.8 Lateral GaN Concluding Remarks

The Lateral GaN Materials and Devices sub-Working-Group (L-GaN-MDWG) of the ITRW covers a similar scope as the SiC-MDWG described earlier in this document, but with a focus on Lateral GaN power devices (AlGaIn/GaN High Electron Mobility Transistors, HEMTs) as outlined in the previous section. Such devices are now beginning to play an important role in power electronics due to their superior intrinsic material properties as compared to those of present silicon-based power devices. This is based on several key technical advances, listed below along with representative seminal publications on each topic:

- Epitaxial growth of GaN layers on Si substrates up to 8" diameter [12]
- Device processing in CMOS-compatible fabrication lines [13]
- Achievement of normally-off operation using approaches such as p-gates or a cascode configuration [14]
- Extension of breakdown voltage using edge termination techniques such as field plates [15]
- Mitigation of reliability concerns such as dynamic on-resistance, threshold voltage shift, and time-dependent dielectric breakdown [16]

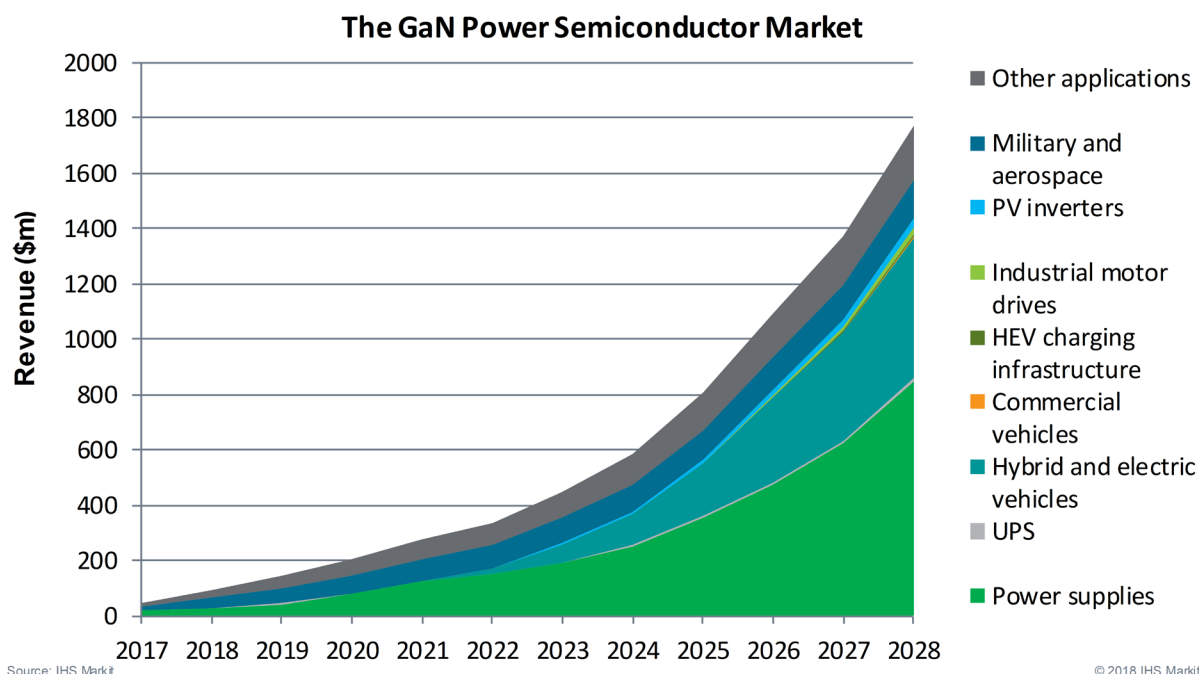


Figure 6: GaN Market Forecast [3]. Source: IHS Markit

GaN-based Radio-Frequency (RF) devices have already demonstrated commercial success valued at more than \$100M, the key near-term applications with large market penetration size are power supplies and hybrid/electric vehicles as shown in Figure 6. Although GaN power device R&D efforts have nearly reached their 15-year mark, revenue forecasts have been lowered over the last few years [17], signaling market penetration slower than originally projected. However, the fundamentals of the value proposition remain strong, and several key industry-wide efforts are underway to advance the technology including ITRW as well as JEDEC and JETTA that seek to standardize reliability testing, qualification, parametric testing, and datasheets. These efforts will contribute to the widespread adoption of GaN power electronics technologies. In parallel with these efforts, much research remains to be done to address issues such as device robustness (since GaN power HEMTs do not exhibit avalanche breakdown), reliability concerns (such as hard-switching reliability), and integration with gate drive circuitry and other components (reactive and/or thermal management) at the package/module and chip levels.

6.9 Vertical GaN Development

Availability of native 2- and 4-inch GaN substrates [18, 19] are enabling breakthroughs in vertical GaN devices. Compared to their lateral counterparts, vertical GaN devices can deliver higher breakdown voltage and higher power density. In addition, vertical GaN promises higher reliability as a result of shifting the peak electric field from the surface to the bulk, which also serves to minimize trapping effects and to reduce dynamic on-resistance. Another key benefit is that unlike lateral GaN, vertical GaN devices can survive and operate in the avalanche region; avalanche energy capability has been demonstrated at 1000 mJ [20]. Fabricating vertical devices on CTE- and lattice-matched bulk GaN is the best way to maximize the potential of GaN so that device performance fully exploits the excellent intrinsic material properties.

There has been recent progress both in vertical GaN diodes and transistors grown on native substrates. High breakdown voltages have been reported, with p-n diode breakdown voltage demonstrated at over 4 kV [21], and a reported blocking voltage of 1.6 kV for vertical GaN trench MOSFETs [22]. However, much work remains to be done to reach a higher breakdown target. One such challenge is the design of effective junction termination extension (JTE) structures in GaN. These structures are necessary to alleviate electric field crowding at the edge of the junction, which causes premature breakdown. In vertical Si and SiC power devices, this can be done by implanting a p-region at the edge of the junction that is charge balanced to the drift region. However, this single zone JTE design is sensitive to interface charge, dose, and process variations. If the single-zone JTE is not effectively charge-balanced then its efficiency can be severely impacted. More complex structures can be designed to help compensate for this, however, even in Si and SiC this poses a challenge. In GaN, the problem becomes more severe due to the present limitations in selective-area p-type doping. Activation of an implanted p-type dopant in GaN requires an annealing temperature of over 1300 °C, which causes decomposition of GaN at atmospheric pressure [19, 23]. As a result, Mg implantation suffers from low activation efficiency, and a limited effective range of implantation energy and dose [19, 23].

Vertical GaN power devices show promise for high voltage, high current operation, in which they can potentially outperform lateral GaN as well as vertical SiC devices. Many challenges in vertical GaN still need to be overcome at the materials and processing levels, as this technology is far less mature than lateral GaN and SiC.

6.10 Ultra-Wide-Bandgap Semiconductors

Beyond vertical GaN, a new class of materials known as ultra-wide-bandgap (UWBG) semiconductors is presently being investigated [24]. The leading materials in this class are Aluminum Nitride / Aluminum Gallium Nitride, Gallium Oxide, and Diamond, although others are also being investigated. The potential advantage of these materials, as with the wide-bandgap materials, lies in their ability to withstand higher electric fields without going into avalanche breakdown. Just like SiC and GaN have an advantage over Si in this respect, UWBG semiconductors may have an advantage over SiC and GaN. However, work on these materials is still in an early research phase, and fundamental challenges such as the growth of large-area substrates, the ability to effectively dope the materials, the ability to form low-resistance ohmic contacts, the ability to integrate the semiconductors with other materials such as dielectrics, and the understanding of the intrinsic breakdown mechanisms largely remain to be addressed. Nevertheless, work on these materials has ramped up significantly over the past couple of years, and working device prototypes have been demonstrated that suggest that the materials may indeed one day out-perform SiC and GaN.

6.11 Concluding Remarks

This IEEE ITRW effort addresses the SiC and GaN HEMT device roadmaps in close collaboration with the module / packaging working group as well as the SiC and GaN applications working groups. Additionally, key recent publications in the literature have been consulted, for example, the recent review article by H. Amano et al. [25]. The article takes a top-down approach where key power electronics applications and high-value markets are identified and prioritized. SiC devices are expected to target high-value applications with voltage ratings above 900 V, and lateral GaN devices are expected to address high-value applications at 650 V and below. However, some overlap in the 650 to 900 V range is likely to occur. In the long term, vertical GaN devices, and even ultra-WBG devices composed

of materials such as gallium oxide, aluminum nitride, and diamond may compete with SiC, although such devices are still in the research phase. Metrics defining the value proposition for high-value applications will drive device design, packages and modules, and circuit topologies. Example device parameters are shown in Table 2. The device specifications and reliability requirements are driven by market needs: such as in consumer, industrial, or automotive markets. Other performance metrics may influence adoption as well. For example, GaN may be favored earlier in some niche applications, such as in satellite power systems, due to its intrinsic resilience to space irradiation. Charting a roadmap with short-, medium-, and long-term projections for GaN power devices should be informed by the targeted performance, application space, reliability, cost per Amp, efficiency, frequency, and power density.

Table 2. Key device metrics for benchmarking power devices [26].

	Silicon	SiC	GaN
Concept	Super junction	Planar MOSFET	e-mode HEMT
Blocking voltage	600V	900V	600V
On-state resistance (typ.)	56 (Ω)	65 (Ω)	55 (Ω)
Reverse recovery charge	6000 nC	130 nC	0 nC
Energy stored in C_{oss} @ 400V	8.1 μ J	8.8 μ J	6.4 μ J
Charge stored in C_{oss} @ 400V	420 nC	70 nC	40 nC
Turn-off loss @ 10A / 400V	15 μ J	10 μ J	10 μ J

6.12 Path Forward

The ITRW materials and devices working group convenes periodically at major industry events, such as the Applied Power Electronics Conference (APEC) and the Workshop on Wide-bandgap Power Devices and Applications (WiPDA) to present its strategy for defining the SiC, GaN, and emerging power device roadmaps. This initial version of the roadmap will be continuously updated via these meetings, coupled with additional communications within and between the different working groups.

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6.14 Acknowledgement

The work at Sandia was supported by the U.S. DOE Office of Electricity's Energy Storage Program. Sandia National Laboratories is a multitechnology laboratory managed and operated by National Technology & Engineering Solutions of Sandia, LLC, a wholly owned subsidiary of Honeywell International Inc., for the U.S. Department of Energy's National Nuclear Security Administration under contract DE-NA0003525. The views expressed in this article do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

6.15 References

- [1] IHS Markit, "Market for GaN and SiC Power Semiconductors to Top \$1 Billion in 2020," <https://news.ihsmarket.com/press-release/technology/market-gan-and-sic-power-semiconductors-top-1-billion-2020> (March 9, 2016)
- [2] Cengiz Balkas, "Wolfspeed," <https://www.sec.gov/Archives/edgar/data/895419/000089541918000019/analystdayfebruary262018.htm> (February 26, 2018).
- [3] IHS Markit, Technology Group, SiC & GaN Power Semiconductors Report (2019).
- [4] PowerAmerica Technology Roadmap 4.0 (2019)
- [5] Y. Uemoto et al., "Gate Injection Transistor (GIT)—A Normally-Off AlGaIn/GaN Power Transistor Using Conductivity Modulation," *IEEE Trans. Elec. Dev.*, vol. 54, no. 12, pp. 3393–3399, 2007.
- [6] H. Okita et al., "Through Recess and Regrowth Gate Technology for Realizing Process Stability of GaN-Based Gate Injection Transistors," *IEEE Trans. Elec. Dev.*, vol. 64, no. 3, pp. 1026–1031, 2017.
- [7] S. Kaneko et al., "Current-collapse-free operations up to 850 V by GaN-GIT utilizing hole injection from drain," in *IEEE 27th International Symposium on Power Semiconductor Devices ICs (ISPSD)*, pp. 41–44, 2015.
- [8] S. Jiang et al., "All-GaN-Integrated Cascode Heterojunction Field Effect Transistors," *IEEE Trans. on Power Elec.*, vol. 32, no. 11, pp. 8743–8750, 2017.
- [9] J. Su, E. A. Armour, B. Krishnan, S. M. Lee, and G. D. Papasouliotis, "Stress Engineering with AlN/GaN Superlattices for Epitaxial GaN on 200 mm Silicon Substrates Using a Single Wafer Rotating Disk MOCVD Reactor," *J. Mater. Res.*, vol. 30, no. 19, pp. 2846–2858, 2015.
- [10] Z. Tang, S. Huang, Q. Jiang, S. Liu, C. Liu, and K. J. Chen, "High-Voltage (600-V) Low-Leakage Low-Current-Collapse AlGaIn/GaN HEMTs with AlN/SiN_x Passivation," *IEEE Elec. Dev. Lett.*, vol. 34, no. 3, pp. 366–368, Mar. 2013.
- [11] S. Yang et al., "High-Quality Interface in Al₂O₃/GaN/AlGaIn/GaN MIS Structures with In-Situ Pre-Gate Plasma Nitridation," *IEEE Elec. Dev. Lett.*, vol. 34, no. 12, 2013.
- [12] M. Ishida, T. Ueda, T. Tanaka, and D. Ueda, "GaN on Si Technologies for Power Switching Devices," *IEEE Trans. Elec. Dev.*, vol. 60, no. 10, p. 3053, 2013.
- [13] D. Marcon et al., "Manufacturing Challenges of GaN-on-Si HEMTs in a 200 mm CMOS Fab," *IEEE Trans. Semicond. Manuf.*, vol. 26, no. 3, p. 361, 2013.
- [14] K. J. Chen et al., "GaN-on-Si power technology: Devices and applications," *IEEE Trans. Elect. Dev.*, vol. 64, no. 3, pp. 779–795, 2017.
- [15] S. Karmalkar and U. K. Mishra, "Enhancement of Breakdown Voltage in AlGaIn/GaN High Electron Mobility Transistors Using a Field Plate," *IEEE Trans. Elec. Dev.*, vol. 48, no. 8, p. 1515, 2001.
- [16] G. Meneghesso et al., "Reliability of GaN High Electron Mobility Transistors: State of the Art and Perspectives," *IEEE Trans. Dev. Mat. Rel.*, vol. 8, no. 2, p. 332, 2008.
- [17] Yole Développement Annual Market Report, 2016.
- [18] I. C. Kizilyalli, "Vertical GaN Power Devices," in *ISPSD*, 2018.

- [19] J. Hu, Y. Zhang, M. Sun, D. Piedra, N. Chowdhury, and T. Palacios, “Materials and processing issues in vertical GaN power electronics,” *Mater. Sci. Semicond. Process.*, vol. 78, pp. 75–84, May 2018.
- [20] I. C. Kizilyalli, A. P. Edwards, H. Nie, S. Member, D. Disney, and D. Bour, “High Voltage Vertical GaN p-n Diodes With Avalanche Capability,” *IEEE Trans. Electron Devices*, vol. 60, no. 10, 2013.
- [21] H. Ohta et al., “Vertical GaN p-n Junction Diodes With High Breakdown Voltages Over 4 kV,” *IEEE Electron Device Lett.*, vol. 36, no. 11, 2015.
- [22] D. Shibata et al., “Vertical GaN-based power devices on bulk GaN substrates for future power switching systems,” in *Gallium Nitride Materials and Devices XIII*, 2018, vol. 10532, p. 36.
- [23] M. J. Tadjer et al., “Selective p-type Doping of GaN:Si by Mg Ion Implantation and Multicycle Rapid Thermal Annealing,” *ECS J. Solid State Sci. Technol.*, vol. 5, no. 2, pp. P124–P127, Dec. 2016.
- [24] J. Y. Tsao et al., “Ultra-Wide-Bandgap Semiconductors: Research Opportunities and Challenges,” *Advanced Electronic Materials* 4, 1600501 (2018).
- [25] H. Amano et al., “The 2018 GaN Power Electronics Roadmap,” *J. Phys. D*, vol. 51, p. 163001, 2018.
- [26] G. Deboy et al., “Si, SiC, and GaN Power Devices: An Un-Biased View on Key Performance Indications,” in *Proc. IEDM 2016*, pp. 20.2.1–20.2.4, 2016.

Chapter 7: Roadmap for GaN Applications

7.1 Introduction

The International Technology Roadmap for Wide bandgap Power Semiconductors (ITRW) has a specific working group for the area of GaN applications. This chapter summarizes the progress of the working group to date. The team has a range of members drawn from industry and academia to give the short, medium and long-term perspectives required for a roadmap.

7.2 Suitable Figure of Merit

A fundamental benefit of the GaN high-electron-mobility transistor (HEMT) is that it has a substantially lower switching figure of merit (FOM [NJ1]) than comparably rated silicon super junction (SJ) and SiC MOSFETs. The switching figure of merit is the product of the gate charge and the resistance between the drain and source terminals in the ON state as shown in Equation (1).

$$FOM = Q_g \times R_{ds_on} \quad (1)$$

The FOM is a proxy for both switching and conduction losses of a transistor operating in most switching power supply applications. As it can be seen from Figure 1, the FOM of GaN HEMTs is 8 to 25 times lower than SJ and SiC MOSFETs [1], mainly because die sizes of GaN HEMTs are substantially smaller than those of MOSFETs with the same R_{ds_on} ratings. This means that GaN HEMTs are capable of switching considerably faster or at higher frequencies than both SJ and SiC MOSFETs, potentially enabling more efficient, smaller and/or lighter power converters [NJ2] [2].

One issue that is often overlooked is that while the “raw” speed of devices may be better, often in power electronics this simply moves a problem to another part of the design. For example, gate drives circuits need to be redesigned to cope with much higher frequencies of operation and the resulting EMC behavior needs to be managed effectively. Therefore, it is important to note that this FOM is only one aspect of comparing the ultimate performance of power electronics modules and systems.

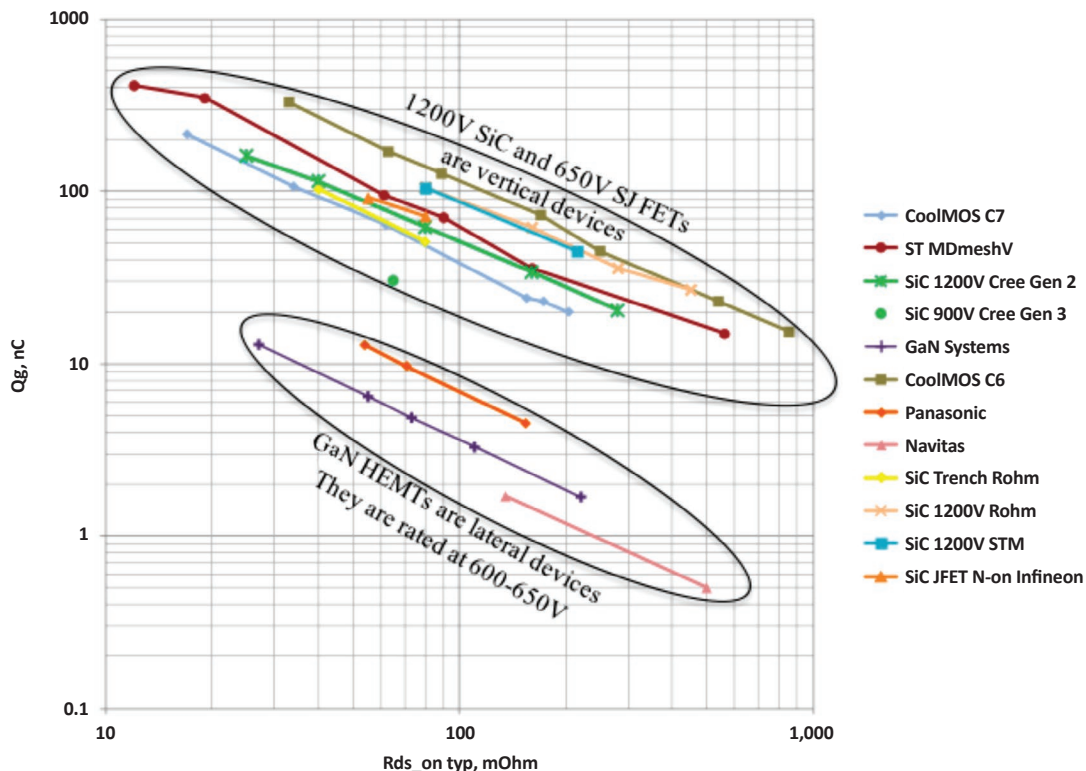


Figure 1: Switching Figure of Merit of Publicly Disclosed GaN HEMTs, SiC, and Si SJ MOSFETs

7.3 Using GaN Technology in Power Electronics Applications

The comparison suggests that replacing, for example, SJ MOSFETs with GaN HEMTs in a switching power supply would result in noticeable performance improvements. However, most of early attempts [5] to demonstrate GaN HEMTs value proposition by simply replacing incumbent MOSFETs with GaN HEMTs have failed short of expectations either yielding incremental performance improvements or sometimes even causing performance degradation. The reason is that a GaN HEMT is not necessarily a drop-in replacement for Si or SiC MOSFET, but rather a complex switching device whose parameters affecting switching performance are quite different from those found in MOSFETs as detailed in Table 1.

Table 1. Application attributes of GaN HEMTs

Applications Attribute	Si SJ MOSFET, 650V	SiC MOSFET, 1200V	GaN HEMT, 600-650V	Notes
Gate Charge, Q_G	Highest	Lower than SJ	Lowest	GaN HEMTs have the smallest die size and lateral architecture
Threshold Voltage, V_{th}	3V	3V	1.2-1.6V	GaN HEMTs may require negative V_{gs} for reliable turn OFF
Maximum V_{gs}	20V	20V	7-10V	GaN HEMTs are sensitive to gate overvoltage
Drain to Source leakage current in OFF state	Slightly higher than SiC	Lowest	Highest	GaN has impurities
Body Diode, V_f	~0.9V	~3.5V	No body diode	Intrinsic diode in SiC MOSFET is PN type and has high V_f [NJ3]
Reverse Recovery, Q_{rr}	Highest	Slightly higher than GaN	Lowest [NJ4]	GaN and SiC do not have reverse recovery losses, just Q_{oss}
3 rd Quadrant Operation	Good	May need external diode	Need positive V_{gs} [NJ5]	
Avalanche Rating	High	High	None	GaN HEMTs do not have PN junction
Dynamic R_{ds_on}	None	None	Up to 30%	Charge trapping occurs in most GaN HEMTs

Thermals	OK	Best	Comparable to Si	SiC has the highest thermal conductivity
Most suitable for applications	Boost PFC, Two-transistor forward, flyback	Forward, half-bridge based topologies	Half-bridge based topologies	

Therefore, to take full advantage of intrinsic capabilities of GaN HEMTs in switching power supply applications, it is necessary to employ a holistic approach that combines the two major elements:

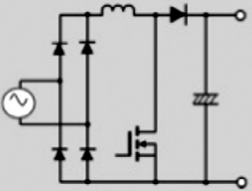
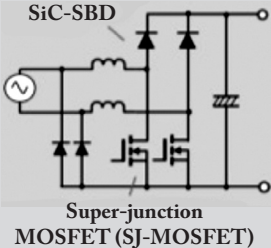
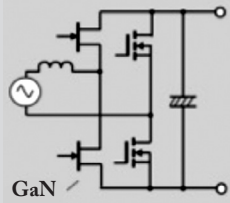
1. Identification of new topologies and applications enabled by unique capabilities and performance attributes of GaN HEMTs
2. Ecosystem required to enable broad acceptance of GaN HEMTs

The first element encompasses road-mapping so-called “killer apps” or the applications either entirely enabled by GaN HEMTs or where GaN HEMTs offer dramatic performance improvements compared to other technologies and materials.

7.4 Examples of GaN Technology Applied to Power Electronics

An example of such application is the totem pole bridgeless power factor correction circuit used in AC/DC power supplies [6, 7]. The totem pole circuit is only feasible with GaN HEMTs that have $Q_{rr} \approx 0$ as shown in Table 2.

Table 2. Roadmap to the totem pole PFC enabled by GaN HEMTs with no reverse recovery (Panasonic)

	Classic PFC	Interleaved Bridgeless PFC	Totem Pole Bridgeless PFC
Schematic			
Efficiency	95-97%	97-98%	98-99%
Number of Components	7 – 4 x Si rectifiers, SJ MOSFET, SiC SBD, Inductor	8 – 2 x SJ MOSFETs, 2 x SiC SBDs, 2 x Si rectifiers, 2 x Inductors	5 – 2 x GaN HEMTs, 2 x Si MOSFETs, Inductor
Rectification Losses	2 diode drops	1 diode drop	None

Other examples include switching power supply topologies employing bidirectional switches; high efficiency power converters [4]; full ZVS/ZCS topologies to eliminate source of EMI and take advantage of high frequency capabilities of GaN HEMTs [3], and high current high frequency synchronous rectifiers [8] where both full load and quiescent power losses are significantly reduced thanks to very low Q_g and the practically non-existent Q_{rr} of GaN HEMTs.

7.5 Future Applications of GaN Technology in Power Electronics

Existing and emerging applications where GaN HEMTs offer dramatic performance improvements include wireless power transfer systems, plasma generators, RF power supplies, and envelope tracking converters. As GaN HEMTs get

closer to their cost and demonstrated reliability targets, more applications in high performance AC/DC networking and server power supplies, photovoltaic inverters, electrical vehicles, aerospace, and consumer electronics will take advantage of higher efficiency, smaller size, and lower weight enabled by GaN HEMTs.

A significant area of interest with GaN technology is the potential ability to have highly integrated 3D modules. Given the potential for extremely high efficiency, the reduced thermal constraints compared to silicon devices lead to some very interesting possibilities for packaging with GaN devices, as shown in Figure 2. There will be a direct link with the ITRW modules and packaging technical working group on this topic as the roadmap develops in the future.

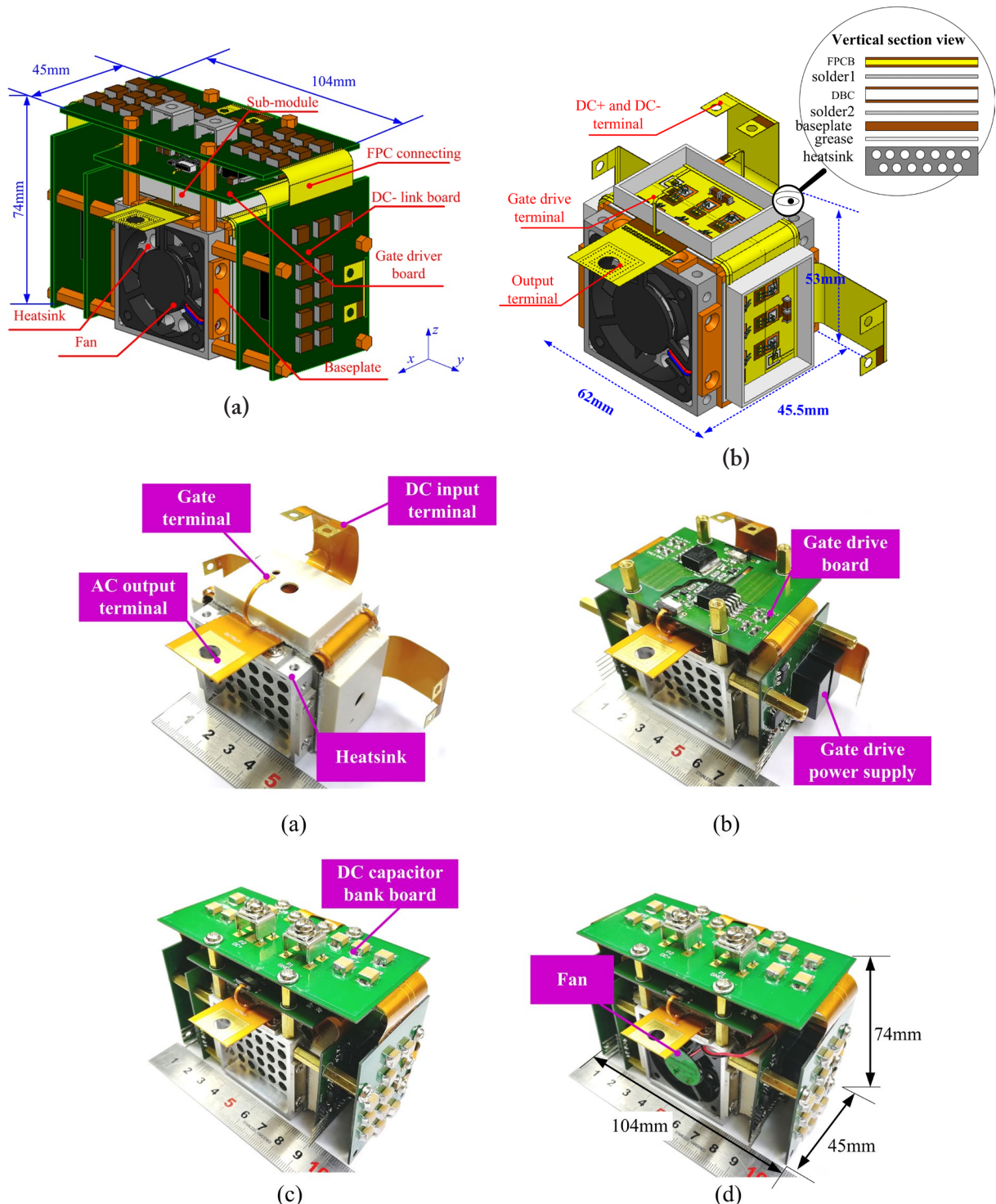


Figure 2: 3D GaN High Density Power Module (Courtesy Fang Luo, University of Arkansas)

7.6 Designing Power Electronics with GaN

The second element pertains to an ecosystem required to achieve broad acceptance of GaN HEMTs comprising multiple components from different domains, including:

- Reference designs
- Simulation models
- Device drivers
- Integration
- Magnetics and other passive components
- Device compatibility and interoperability
- Engineering education

7.7 Reference Designs

Reference designs address the fact that GaN HEMTs are not direct replacements for existing switching devices. Power supply topology [9], device drive requirements, PCB layout, EMI design, and even aspects of thermal and mechanical designs are different in applications employing GaN HEMTs. In order to make it easy for customers to use GaN HEMTs it is necessary to create a roadmap for reference designs addressing specific system integration issues. As the development of GaN power designs migrate from the academic labs (where designs and techniques are published in conferences and journals) into industry, there will be an increasing library of white papers and reference designs available for designers to access.

7.8 Simulation Models

From a simulation perspective, capturing the effects of reactive parasitic interaction becomes more critical. Even the parasitic elements of the standard TO packages have become limiting factors for high frequency operation. True impedance modeling is needed to properly predict effects such as voltage overshoot of a propagating wavefront traveling down a transmission line. New modeling techniques and models that combine the efficiency of piecewise linear simulators (e.g. SIMPLIS), along with the real-world transient effects of the true linear simulators (e.g. SPICE) are needed to address the challenges of the GaN HEMT-based switching power supplies [10].

7.9 Integration

The topic of device drivers is closely related to device integration and packaging roadmaps. With the EMI signature being addressed in the topology selection, integration becomes a significant enabler of high switching frequency operation.

7.10 Device Drivers

GaN HEMTs' relatively low range of V_{gs} , sensitivity to gate overvoltage, and the need for negative V_{gs} in some applications require coordinated efforts in development of device drivers [11]. Driving a GaN HEMT by a conventional MOSFET driver connected to a low voltage supply would either fail to extract true value of the GaN device or result in overstressing the gate and compromising long-term reliability of the GaN device.

7.11 Magnetics and Other Passive Components

In order to take advantage of the higher switching speeds of GaN HEMTs, the passive components must also support the higher switching frequencies. Currently, magnetics are a key limiting element causing significant efficiency degradation at frequencies above several hundred kilohertz [14][15]. New materials and assembly techniques including 3D printed magnetics that are explored as a potential solution to this problem need to be mapped.

An established supply chain is a vital component of a product ecosystem. Incompatibility of GaN HEMTs from different device manufacturers stifles their adoption in the market that is used to dealing with multiple functionally and mechanically equivalent sources for every component. [NJ6]

7.12 Device Compatibility and Interoperability

Moreover, lack of standardization in GaN HEMTs' specification, testing, and reliability qualification makes it difficult for customers to compare performance and reliability of various GaN HEMT devices. While this is being addressed by JEDEC that formed the new JC-70 Wide Bandgap Power Electronic Conversion Semiconductors Committee, the roadmap to true pin-to-pin compatibility of GaN HEMTs from different manufacturers is heavily dependent on business models and go-to-market strategies of the manufacturers.

7.13 Engineering Education

One of the largest barriers to the adoption of GaN HEMTs is lack of engineering experience with the devices. Power supply engineers who are not comfortable with GaN HEMTs are reluctant to use them in their designs. This issue will need to be addressed through road mapping both engineering education and the customer support model that is evolving into a more complete model requiring significant application engineering resources.

7.14 Applications to Benefit from and be Enabled by GaNs

Table 2 lists the projected timeline for the applications that most likely will benefit from or be enabled by GaN.

Table 3. Project timelines for potential GaN applications

Current or near term applications (< 3 years)	Mid term applications (3 to 8 years)	Long term applications (>8 years)
• LIDAR**** • data center*** • consumer power electronics (e.g. charger)** • drones*	• Motor Controls ***** • EV auxiliary power supplies ***** • PV inverter (rooftop) *** • Wireless power transfer ** to **** • EV onboard charger ** to **** • Aerospace **	• Medical devices ** • Integration (monolithic) *****

Note: the number of * indicates the degree of difficulty with 10 being the highest and 1 being the lowest

The major roadblocks to realize the potential applications include:

- Cost
- Reliability
- Supply chains
- Packaging challenges, especially at higher power
- Thermal management (higher heat flux)

The technology development needs to happen include:

- Better model
- Better and cheaper instrumentation (probes, scopes, sensors)
- Better controllers
- Better magnetics
- Understanding and characterization under radiation/cosmic ray.

Some of the issues have been discussed in sections 7.8 to 7.13.

7.15 Summary of Road Mapping Progress to Date

In order to develop a roadmap for the GaN application related elements described above, the GaN application working group is taking the following steps:

1. Identify systems integration issues and barriers to broad acceptance of GaN HEMTs
2. Translate the issues and barriers into specific attributes, components, and services
3. Develop performance metric(s) for each attribute, component, and service
4. Map evolution of the performance metrics over time

The team's findings on items 1 and 2 have been described in this article. In the next step, the team is going to focus on items 3 and 4. In order to successfully develop a comprehensive set of performance metrics, the team will collaborate with GaN HEMTs manufacturers, industry groups such as PSMA, and with academia and national laboratories.

The outcome of the team's activities is a complete set of roadmaps for application attributes, components, and services that are required to ensure wide adoption of GaN HEMTs in power electronics applications.

7.16 Conclusion

There is huge promise in the potential for GaN technology being applied to power electronics. However, there are many significant challenges to be addressed before this potential can be fully realized. The roadmap team for GaN applications within ITRW is focused on identifying these challenges and providing vision as to which priority areas need to be addressed to realize the potential of GaN in power electronics.

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7.18 References

- [1] J. Lautner and B. Piepenbreier, "Performance comparison of cascode GaN HEMT and Si MOSFET based inverter for motor drive applications," 2017 IEEE 12th International Conference on Power Electronics and Drive Systems (PEDS), Honolulu, HI, 2017, pp. 81-87.
- [2] K. Li, P. Evans and M. Johnson, "SiC/GaN power semiconductor devices: a theoretical comparison and experimental evaluation under different switching conditions," in IET Electrical Systems in Transportation, vol. 8, no. 1, pp. 3-11, 3 2018.
- [3] W. Zhang, F. Wang, L. M. Tolbert, B. J. Blalock and D. Costinett, "Investigation of soft-switching behavior of 600 V cascode GaN HEMT," 2014 IEEE Energy Conversion Congress and Exposition (ECCE), Pittsburgh, PA, 2014, pp. 2865-2872.
- [4] Y. Wu, M. Jacob-Mitos, M. L. Moore and S. Heikman, "A 97.8% Efficient GaN HEMT Boost Converter With 300-W Output Power at 1 MHz," in IEEE Electron Device Letters, vol. 29, no. 8, pp. 824-826, Aug. 2008.
- [5] X. Huang, Z. Liu, Q. Li and F. C. Lee, "Evaluation and Application of 600 V GaN HEMT in Cascode Structure," in IEEE Transactions on Power Electronics, vol. 29, no. 5, pp. 2453-2461, May 2014.
- [6] L. Zhou, Y-F. Wu, U. Mishra, "True Bridgeless Totem-pole PFC based on GaN HEMTs", PCIM Europe 2013, pp. 1017-1022, May 2013.
- [7] Z. Liu, X. Huang, M. Mu, Y. Yang, F. C. Lee and Q. Li, "Design and evaluation of GaN-based dual-phase interleaved MHz critical mode PFC converter," 2014 IEEE Energy Conversion Congress and Exposition (ECCE), Pittsburgh, PA, 2014, pp. 611-616.
- [8] S. Aldhaher, D. C. Yates and P. D. Mitcheson, "Load-Independent Class E/EF Inverters and Rectifiers for MHz-Switching Applications," in IEEE Transactions on Power Electronics, vol. PP, no. 99, pp. 1-1.
- [9] L. Xue and J. Zhang, "Active clamp flyback using GaN power IC for power adapter applications," 2017 IEEE Applied Power Electronics Conference and Exposition (APEC), Tampa, FL, 2017, pp. 2441-2448.
- [10] L. Wu and M. Saeedifard, "A Simple Behavioral Electro-Thermal Model of GaN FETs for SPICE Circuit Simulation," in IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 4, no. 3, pp. 730-737, 2016
- [11] P. Jacqmaer, J. Everts, R. Gelagaev, P. Tant and J. Driesen, "Fast robust gate-drivers with easily adjustable voltage ranges for driving normally-on wide-bandgap power transistors," Proceedings of 14th International Power Electronics and Motion Control Conference EPE-PEMC 2010, Ohrid, 2010, pp. T2-44-T2-51.
- [12] K.J. Chen et al., "GaN-on-Si Power Technology: Devices and Applications," in IEEE Transactions on Electron Devices, vol. 64, no. 3, pp. 779-795, March 2017.
- [13] JEDEC Committee: JC-70 Wide Bandgap Power Electronic Conversion Semiconductors, <https://www.jedec.org/committees/jc-70> (last accessed 26th March 2018)
- [14] P. A. Kyaw, A. L. F. Stein and C. R. Sullivan, "Fundamental Examination of Multiple Potential Passive Component Technologies for Future Power Electronics," in IEEE Transactions on Power Electronics, vol. PP, no. 99, pp. 1-1, 2018
- [15] C. R. Sullivan, "Prospects for advances in power magnetics," CIPS 2016; 9th International Conference on Integrated Power Electronics Systems, Nuremberg, Germany, 2016, pp. 1-9.

Chapter 8: Roadmap for SiC Applications

8.1 Introduction

A long-term vision of the power electronics community is to develop highly integrated power electronic converters that are specialized for their particular applications. This requires power converters to have much greater power density, higher reliability, and better overall system performance without costly devices or designs. These needs motivate the migration from silicon (Si) to silicon carbide (SiC) based power electronics. A rapid transition from Si to SiC technology is inevitable due to SiC's superior atomic properties.

The Silicon Carbide Application Working Group (SiC-AWG) of the International Technology Roadmap for Wide-Bandgap Power Semiconductors (ITRW) is comprised of four subgroups: utility, transportation, power supplies and drives, and medical and emerging applications. The goals of the SiC-AWG are twofold. First, to identify how applications (e.g., traction drives) experience game-changing performance by adopting the appropriate SiC technology. Second, to establish device specifications that guide manufacturers during product development so the benefits of SiC components are fully realized in these design spaces.

To accomplish these tasks, the team has developed a template to assist subgroup co-chairs with documenting application parameters and establishing optimal characteristics for SiC components in a specified role. Through discussions held at the workshops this past year, the working group identified the following applications that either would be the earlier adopters of SiC or have large market potentials:

- Traction drive for hybrid and pure electric vehicles;
- Fast charging stations for electric vehicles;
- String inverters for photovoltaic based power productions;
- Integrated drives for generators and motors onboard aircrafts;
- Traction drives and auxiliary power supplies for high speed and commuter trains;
- Commercial and industrial motor drives; and
- Solid state transformer for smart power stations.

Industry and academia members within the groups have filled in templates for these applications. The working groups is in the process of reviewing the gathered data and seeking broader feedback on its implications. The group aims to present a summary of the findings at the Seventh IEEE Workshop on Wide Bandgap Power Devices and Applications (WiPDA) in November 2019.

8.2 Key Questions

The key questions for the power electronics community to address when considering to adopt SiC devices in their applications include:

- What are the significant benefits of adopting SiC?
- Can the investment costs be justified by the long-term benefits?
- Which applications will benefit first from the adoption of SiC?
- What is the motivation to use SiC technology?

These questions are crucial to understand how to successfully transition to new technologies such as SiC. The choice of a target application needs to be well suited to the characteristics of the new devices so that it takes advantages of the technology's characteristics [1], some of which are summarized in Table 1. There are some fundamental differences between Si and its wide bandgap competitors. These include the higher critical electric field strength, which enables higher breakdown voltages in wide-bandgap materials, the faster electron saturation velocity in SiC and GaN, and the very high thermal conductivity in SiC. The combination of the high-voltage, high-speed, and high-temperature characteristics of SiC, make it extremely attractive for certain power electronics applications.

Table 1. Key characteristics of Si, SiC, and GaN.

Property	Si	SiC	GaN
Bandgap (eV)	1.1	3.2	3.4
Critical Field (10^6 V/cm)	0.3	3.0	3.5
Electron mobility (cm^2/Vs)	1,450	900	2,000
Electron saturation velocity (10^6 cm/s)	10	22	25
Thermal conductivity ($\text{W}/\text{cm}^2 \text{ K}$)	1.5	5	13

It is expected that SiC will have large market shares in automotive, aviation, railway, industrial drives, renewable energy and utility-based power electronics applications such as solid-state transformers. For these applications, SiC will increase system efficiency, which lowers operation costs. For electric vehicles, the increased efficiency of traction systems will also reduce the size of batteries onboard the vehicles or it can extend driving distances between charges. In the long run, the aviation industry will move from more electric aircrafts (MEA) to hybrid electric propulsion. For hybrid electric propulsion systems, SiC based motor drives with power levels ranging from hundreds of kilowatts to megawatts will become popular. Cost reductions of SiC devices will gradually happen with more field implementations, but it will be the potential system level cost savings, improved efficiency and higher power density that will make SiC more appealing than its Si counterparts.

At this time, the major concerns and technological challenges are application specific qualifications, reliability, short circuit capability, and high temperature packaging, which is of great interest. While high dV/dt and dI/dt are currently challenges for power converter and electric machine designs, these obstacles most likely be overcome gradually. Also, some requirements for current Si based systems may need to be modified to reflect the developments in fault detection and protection methodologies. For example, the existing short circuit capability requirements for IGBTs are based on past experience from motor drives. With the recent development of fast and reliable detection and protection methods, the required short circuit withstand time for SiC devices may need to be revisited.

8.3 Application Areas

Each subgroup has identified several application areas in need of information.

- The transportation subgroup is focusing on land vehicles, aviation hardware, naval systems, and rail-based systems. The applications for land vehicles include traction drives, battery chargers, and dc-dc converters, while aviation hardware and naval systems encompass generation equipment, solid-state circuit breakers, energy-storage hardware, and propulsion systems. For rail-based systems, tractions drives and auxiliary power supplies are being investigated.
- The utility subgroup is exploring high-voltage dc transmission systems, flexible ac transmission systems, solid-state transformers, energy storage systems, photovoltaic (PV) generation, and wind generation.

In recent years, because of the potential size and weight reductions, solid-state transformers (SST), also called power electronics transformers (PET), have been studied extensively by both rail and electric power utility industries. Thus, the transportation subgroup and utility subgroup will both study potential benefits and roadmap of SiC based SST/ (PET).

- The power supplies and drives subgroup is focusing on switch-mode power supplies for data centers and drives for refrigeration, air conditioning, and compressors.
- Finally, the medical and emerging applications subgroup is examining computed tomography scanners, power supply systems for fusion energy, wireless power transfer, and pulsed power supply systems.

SiC technology is being adopted and is growing rapidly in specific application areas with varied results. The impact of SiC will be immediate, particularly in applications where the cost of new devices is not a significant barrier. For instance, aerospace applications are expected to benefit from SiC based power converters because of reduced weight, improved fuel saving, better intrinsic radiation hardness, and wider temperature operation range. For more cost sensitive applications such as photovoltaic string inverter, the main benefit from the adaptation of SiC devices will be higher efficiency and higher energy throughput.

8.4 SiC Applications

Some key advantages of SiC technology are its ability to support higher voltages, switch at faster speeds, and operate at higher temperatures. These strengths require special attention during integration, packaging and circuit design to ensure that SiC's advantages are harnessed in a final product. The a 250-kW SiC three-phase inverter shown in Fig. 1 showcases a successful design based on these considerations [2]. It uses Wolfspeed's CAS325M12HM2 power modules, which were developed with the key advantages of SiC in mind. This all-SiC power module is rated at 1,200 V and 445 A, with a 3.6-m Ω on-resistance and a maximum operating temperature of 175 °C [3], [4]. Detailed electromagnetic and thermal analyses were conducted to design the internal bus structure and gate drive circuit, resulting in the compact layout shown in Fig. 1.

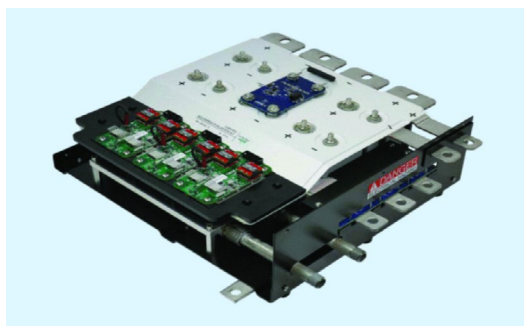


Figure 1: A compact 250-kW inverter using SiC MOSFETs.

8.5 SiC Application Examples

8.5.1 SiC-Based Commuter-Train Traction Inverters

3.3-kV, 1.5-kA rated SiC MOSFET power modules have been implemented in 1.5-kV traction drive inverters for Japan's commuter trains. Because of the higher efficiency that is enabled by SiC MOSFET, compared to Si IGBT based traction drive, 30% energy consumption reduction has been realized. Fig. 2 shows a typical traction drive for commuter trains.



Figure 2: SiC based traction drive system for commuter train (information and picture provided as a courtesy from Mitsubishi Electric).

Compared to its Si IGBT counterpart, the unit also has a 65% reduction in both weight and size. Fig. 3 shows the roadmap of SiC devices for rail applications in Japan.

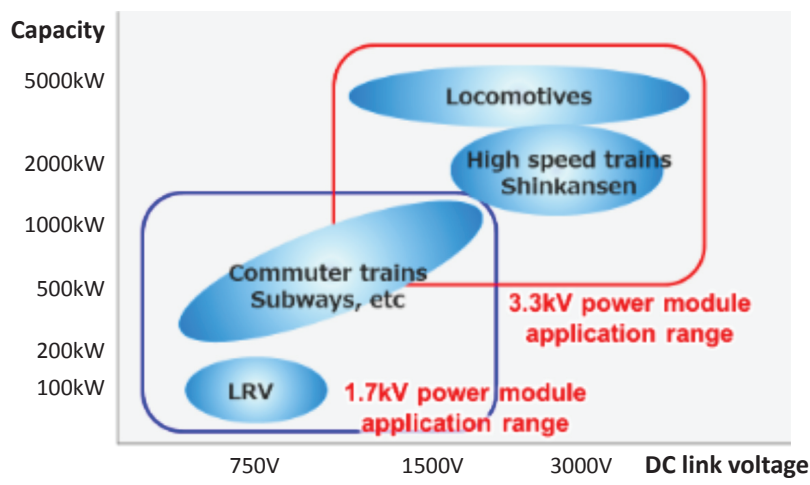


Figure 3: Road map of SiC power modules in Japan's rail systems (information and picture provided as a courtesy from Mitsubishi Electric).

8.5.2 SiC-Based Medium-Voltage Modular Multiple Converter

The performance capabilities of SiC technology were recently demonstrated by a 1-MVA 7-kV SiC-based modular multilevel converter (MMC) for motor drives [5]. The MMC consists of 36 submodules built with 1.7-kV-rated SiC MOSFETs. Fig. 4 shows a single submodule and the full assembly of the 1-MVA MMC. Due to SiC's high efficiency, the size and power of the thermal management system for each submodule is minimized. The low switching losses of SiC MOSFETs also enable high-frequency circulating current to be injected into the system. This mitigates the double line-frequency ripple voltage that occurs in the MMC arms and enables size of the dc link capacitors in the submodules to be reduced. Because of these advantages, the MMC in Fig. 4(b) achieved an unprecedented inverse volumetric power density of 1.19 m³/MW and an ultrahigh efficiency of 99.6% at full load.

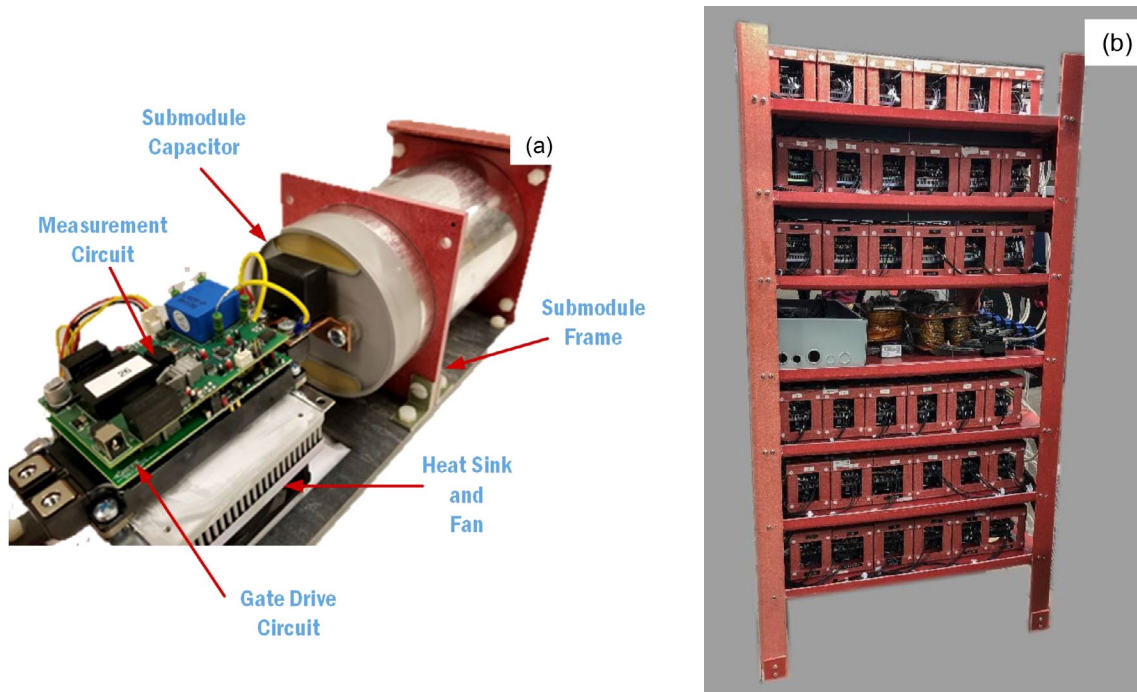


Figure 4: The SiC-based 1-MVA, 7-kV MMC for the motor drive.
(a) The submodule of the MMC and (b) the full prototype.

8.5.3 SiC-Based 100 kW Photovoltaic Inverter

One objective of using SiC technology is to reduce the size of bulky power inductors by increasing the switching frequency. However, as frequency increases, the specific power loss of conventional power inductors may increase significantly. This can offset the benefit of using smaller inductance and make it difficult to reduce the cost of the inductors. Recently, a 100-kW filter-less three phase two-stage SiC PV inverter has been presented in Fig. 5 [11]. By using an intercell transformer (ICT) based multilevel technology, the inverter can be connected to a utility grid without any power filter [12]. The weight of ICT is 0.8 kg. The total weight of inverter is 20 kg. Experimental results have showed a 2.41 % current THD measured in a grid with 3.53 % voltage THD. The ground leakage current is 205 mA [13]. The measured CEC efficiency under 720 V input voltage is 99.2 %, and the peak efficiency is 99.5 %. With a lightweight EMI filter that weights 3.8 kg the PV inverter meets 47 CFR 15.107 standard, i.e., averaged conducted emission ≤ 66 dB μ V at 150 kHz – 500 kHz, and ≤ 60 dB μ V at 500 kHz – 30 MHz.

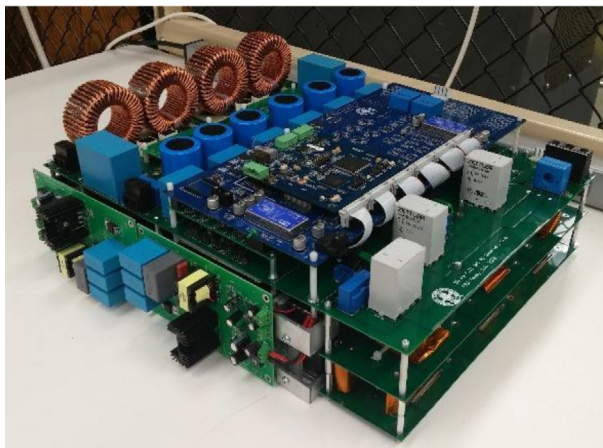


Figure 5: A 100-kW filter-less three phase two-stage SiC PV inverter including dc-dc boost stage and dc-ac inverter stage.

8.6 Important Issues for SiC

Key for the wide adoption of SiC by power electronics engineers is the availability of detailed device models, packaging and circuit design tools, reference designs, standards, and test data. As the power electronics community transitions from Si to SiC devices, it is necessary to have accurate simulation models to ensure that the behavior of the overall design is consistent with the empirical data. The type of model needed depends on the purpose of the simulation; structural models for device analysis and higher-level models for circuit design. SPICE (compact) models are commonly used to simulate the behavior of SiC devices in circuit design [6]. It is also vital that models have enough scalability to be used in larger simulations [7]. Since one of the key advantages of SiC is that it can operate at higher frequencies, models must also have accurate representations of the device's high frequency behavior to be useful [8, 9, 10].

8.7 Information Gathering

The SiC-AWG is using questionnaires to gather information from industrial and academic experts on the requirements to insert SiC technology in power electronics. To collect data in a standardized format, a template has been created that focuses on the specific power electronics application requirements. For example, the questionnaire collects high-level information relating to: the eventual application; device types (e.g., MOSFETs, IGBTs, diodes, and so on); environmental conditions; thermal constraints; packaging; expected lifetime; desired efficiency; power density; target cost (i.e., US\$/kW), and voltage and current ratings. In each case, the data has relevance in different timescales that have been defined in this context as short term (five years), medium term (ten years), and long term (15 years).

With the applications fully defined, the template turns its attention to the power device specifications. Devices are defined by a large number of parameters. One goal of this subgroup is to gather device information from power

application engineers to inform device manufacturers on the relevant device characteristics and tradeoffs specific to each application.

The purpose is to collect data on the following important device characteristics for each application:

1. maximum gate-source voltage
2. threshold voltage
3. output capacitance
4. miller capacitance
5. switching energy.

Device data on figures of merit (e.g., R_{DS_ON} versus Q_G) and fault tolerance capabilities (e.g., short-circuit protection time) are also collected in this section.

At the end of the template, a short questionnaire is used to gather additional specific application information. Suggestions and comments on design parameters and constraints, such as dV/dt and dI/dt , voltage profiles, power cycling profiles, thermal cycling profiles, regular overload conditions, surge capability, health and environmental reporting requirements, preferred packaging methods, and tolerable failures in time rate, are among the information collected. In addition to this, the report will estimate points in time when SiC-based systems are expected to reach cost parity with their Si-based counterparts.

8.8 Conclusions

The insertion of SiC in high-voltage, high-speed power electronics over the last five years has been dramatic, and the technology has rapidly moved from the research lab to mainstream power applications. The progress of SiC power electronics designs has been slower. More compact designs with smaller passives, enabled by the higher frequency operation, are being implemented. SiC has excellent thermal properties and radiation hardness, making it ideal for applications in extreme environments. Issues remain with gate drivers and the speed of operation, and further advancements in packaging are key for high volume adoption of SiC.

8.9 References

1. J. Millan, P. Godignon, X. Perpina, A. Perez-Tomas, J. Rebollo, "A survey of wide-bandgap power semiconductor devices", *IEEE Trans. Power Electron.*, vol. 28, pp. 899-907, Feb. 2013.
2. A. H. Wijenayake, K. J. Olejniczak, D. Simco, S. Minden, M. Feurtado, B. Passmore, T. McNutt, A. Lostetter, D. Martin, "Design of a 250 kW 1200 V SiC MOSFET-based three-phase inverter by considering a subsystem level design optimization approach", Proc. 2017 IEEE Energy Conversion Congr. Exposition (ECCE), pp. 939-946.
3. B. McPherson, P. D. Killeen, A. Lostetter, R. Shaw, B. Passmore, J. Hornberger, T. M. Berry, Low-profile highly configurable current-sharing paralleled wide-bandgap power device power module, Aug. 2016.
4. Data sheet; CAS325M12HM2 1200V 3.6mW Dual power module, Apr. 2018, [online] Available: www.wolfspeed.com.
5. H. Li, K. Potty, Z. Ke, J. Pan, Y. Chen, F. Zhang, M. Sabbagh, W. Perdikakis, G. Li, X. Ye, R. Na, J. Zhang, L. Xu, and J. Wang "Hardware Design of a 1.7 kV SiC MOSFET Based MMC for Medium Voltage Motor Drives", in Proc. 2018 IEEE Applied Power Electronics Conference and Exposition (APEC), San Antonio, TX, pp. 1649-1655, Mar. 2018.
6. Y. Zhou, Y. Li, B. Wang, "Spice modeling of 4H-SiC MOSFET-based on the advanced mobility model", Proc. 2016 IEEE 4th Workshop Wide-Bandgap Power Devices and Applications (WiPDA), pp. 1-7.
7. A. Merkert, T. Krone, A. Mertens, "Characterization and scalable modeling of power semiconductors for optimized design of traction inverters with Si- and SiC-devices", *IEEE Trans. Power Electron.*, vol. 29, no. 5, pp. 2238-2245, May 2014.

8. X. Li, J. Jiang, A. Huang, S. Guo, X. Deng, B. Zhang, X. She, "A SiC power MOSFET loss model suitable for high-frequency applications", *IEEE Trans. Ind. Electron.*, vol. 64, no. 10, pp. 8268-8276, Oct. 2017.
9. H. A. Mantooth, K. Peng, E. Santi, J. L. Hudgins, "Modeling of wide-bandgap power semiconductor devices—Part I", *IEEE Trans. Electron. Devices*, vol. 62, no. 2, pp. 423-433, Feb. 2015.
10. E. Santi, K. Peng, H. A. Mantooth, J. L. Hudgins, "Modeling of wide-bandgap power semiconductor devices—Part II", *IEEE Trans. Electron. Devices*, vol. 62, no. 2, pp. 434-442, Feb. 2015.
11. Y. Shi, L. Wang, R. Xie, and H. Li, "Design and implementation of a 100 kW SiC filter-less PV inverter with 5 kW/kg power density and 99.2% CEC efficiency," *Proc. IEEE APEC*, Mar. 2018, pp. 393-398.
12. Y. Shi, L. Wang, H. Li, "Stability Analysis and Grid Disturbance Rejection for a 60kW SiC based Filter-less Grid-connected PV Inverter", *IEEE Transactions on Industrial Applications*, vol. 54, no. 5, pp. 5025-5038, Sept.-Oct. 2018.
13. L. Wang, Y. Shi, Y. Shi, R. Xie and H. Li, "Ground Leakage Current Analysis and Suppression in a 60-kW 5-Level T-Type Transformerless SiC PV Inverter," *IEEE Transactions on Power Electronics*, vol. 33, no. 2, pp. 1271-1283, Feb. 2018.

Chapter 9: Roadmap for WBG Packaging and Integration

9.1 Introduction

New approaches to packaging and integration will be needed to unlock the full potential of wide bandgap devices. Compared to silicon power device technology, their greatly increased switching speeds allow much higher switching frequencies to be achieved, leading, for example, to smaller passive component requirements, more power dense converters and a reduced bill of materials (BoM). However, such desirable features can only be achieved if circuit parasitics and associated electromagnetic interference can be reduced to unprecedentedly low levels. This demands radically new approaches to integration, moving from assemblies of discrete components, each of which is designed and packaged separately, to fully integrated assemblies comprising power devices, gate drives, filters, sensing, and control functions.

The progression to integrated functional blocks will occur over differing timescales according to the power level and application. Many low-power (less than a few hundred watts) dc-dc converters are already fabricated as single-package, integrated assemblies, whereas the norm for power levels greater than a few kilowatts typically combines surface-mount components on a multi-layer printed circuit board (PCB) with larger discrete passive components and power semiconductor modules. Where high switching frequency is paramount, for example in GaN dc-dc converters, there will be an early move to smaller commutation cells, favoring increased use of smaller, surface-mount components, embedded component technologies and 3D stacked structures. Consequently, it is likely that higher power converters will be realized as several lower-power modular blocks, combined in parallel or series, so that the benefits of higher frequency switching can be maintained. For applications where switching frequency is less important, for example SiC inverters and high-voltage conversion, end-user familiarity, low-cost and established capability will favour conventional modules with adaptations for lower inductance and improved thermal management.

9.2 Focus and Scope

The primary technology focus of the Packaging and Integration working group is on high-speed switching, identifying the materials, design, and manufacturing approaches that will support the thermal, electromagnetic, and through-life management of power electronic converters at power levels from hundreds of watts to hundreds of kilowatts and beyond. A secondary focus is on technologies enabling operation at higher temperatures and other harsh environments. The roadmap targets technological developments over three distinct time horizons: short-term (commercial realization within 5 years), medium-term (commercial realization within 5-15 years), long-term (commercial realization at 15 years and beyond).

9.2.1 Top-level Drivers

The top-level drivers for power electronics that apply to packaging and integration include:

- Reduced costs (bill of materials, manufacturing, O&M, recycling)
- Increased efficiency (reduced losses through life)
- Increased power density (reduced volume, reduced mass)
- Ease of use (plug and go, modular solutions, higher voltage, higher current, load/source integration, simple maintenance)
- Environmental tolerance (higher temperature, extreme temperature range, vibration and shock, radiation hard)
- Reliability and robustness (longer application lifetimes, longer time between maintenance, fewer early failures, ability to survive fault events)

9.2.2 Technical Drivers

The overarching challenge for the Packaging and Integration working group is the realization of cost-effective solutions that can unlock the full potential of WBG semiconductors. Specific technology drivers for packaging and modules that contribute to the realization of the top-level drivers include:

- Higher levels of structural and functional integration
- Improved electrical and magnetic performance
- Improved reliability and robustness
- Extended operating range (voltage, current, temperature, and frequency)
- Improved thermal performance

9.2.3 Mapping Packaging and Integration Technology Drivers onto Power Electronic Drivers

Table 1. Summary of the primary influences of the packaging and modules technology drivers on the top-level drivers and their target metrics.

Power Electronics High-Level Drivers	High-Level Targets	Packaging & Integration Technology Drivers				
		Higher levels of structural & functional integration	Improved electrical & magnetic performance	Improved reliability & robustness	Extended operating range (V, I, T, f)	Improved thermal performance
Cost	Reduced BoM Low-cost manufacturing Easier recycling Reduced O&M cost Reduced recycling cost	+	0-	+	+	+0-
Efficiency	Lower through-life energy losses	0	+	0	0-	+
Power Density	Smaller Lighter	+	+0-	0-	+	+
Ease of Use	Plug and go Simplified thermal management Modular & reconfigurable Minimised EMI Load/source integration	+	+	+	+	+
Environmental Tolerance	Higher temperature Extreme vibration & shock	+0	0	+	+	+0

Table 1 summarizes the primary influences of the packaging and modules technology drivers on the top-level drivers and their target metrics. For example, top-level driver “cost” is influenced positively (+) by “higher levels of structural and functional integration” (reduced BoM, low cost assembly), “improved reliability & robustness” (reduced O&M costs, reduced recycling), and “higher temperature operation” (reduced BoM). Conversely, the need for “improved electromagnetic management” and “improved thermal management” may be cost neutral (0) or add (-) to the BoM, assembly costs, etc.

9.2.4 Technology Interventions

Each of the technology drivers can be associated with specific technology interventions, which in turn define the overall technology scope for the roadmap. The interventions reflect the major areas for innovation and development and are separated into four broad intervention categories:

- Architecture (e.g. package type, layout, level of integration)
- Materials (e.g. substrate materials, interconnect materials)
- Processes (e.g. assembly, manufacturing)
- Tools (e.g. design, modelling, characterization, standards)

9.2.4.1 Interventions: Higher Levels of Structural & Functional Integration

- **Materials:** Passive component materials; Substrate materials; Laminate materials; Thermal management materials; Dielectric isolation & encapsulation materials; Interconnection & termination materials; Die and substrate metallisation materials
- **Processes:** Assembly and manufacturing processes; Electro & electroless-plating; Additive manufacturing; Additive layer manufacturing; Subtractive manufacturing; Interconnection & termination processes; Rapid prototyping; Joining technologies; Passive component manufacturing; Die and substrate metallisation
- **Tools:** Multi-domain virtual prototyping; Multi-scale simulation methods
- **Architecture:** Interconnection & termination architecture; Integrated gate drives; Component embedding; 2.5D and 3D packaging; Integrated thermal management; Thermal management architecture; Integrated sensors; Integration with loads and sources; Integrated passive components; Integrated controls

9.2.4.2 Interventions: Improved Electrical & Magnetic Performance

- **Materials:** Dielectric isolation & encapsulation materials; Interconnection & termination materials
- **Processes:** Interconnection & termination processes; Assembly and manufacturing processes; Additive manufacturing; Additive layer manufacturing; Subtractive manufacturing; Passive component manufacturing
- **Tools:** Electromagnetic modelling; Standards & test; Multi-domain virtual prototyping
- **Architecture:** Controlled parasitics; Interconnection & termination architecture; Layout and geometry optimisation; Embedded screening; Embedded shielding; Integrated filters; Creepage and clearance mitigation; Integrated gate drives; Integrated control

9.2.4.3 Interventions: Improved Reliability & Robustness

- **Materials:** CTE engineered materials; Thermal management materials; Dielectric isolation & encapsulation materials; Substrate materials; Laminate materials; Thermal interface materials; Interconnection & termination materials; Die and substrate metallisation materials
- **Processes:** Interconnection & termination processes; Die attach; Wire/foil bond; Assembly and manufacturing processes; Die and substrate metallisation
- **Tools:** Reliability modelling; Prognostics & health management; Reliability characterisation; Standards and test; Physics of failure models; Condition monitoring; Life-consumption estimation; Diagnostics; Multi-domain virtual prototyping
- **Architecture:** Fault tolerant architecture; Redundancy; Interconnection & termination architecture

9.2.4.4 Interventions: Extended Operating Range (V, I, T, f)

- **Materials:** Thermal management materials; Dielectric isolation & encapsulation materials; Housing materials; Interconnection & termination materials; Substrate materials; Die and substrate metallisation materials

- **Processes:** Die attach; Wire/foil bond; Interconnection & termination processes; Joining technologies; Assembly and manufacturing processes; Die and substrate metallisation
- **Tools:** Standards and test; Thermal modelling; Multi-domain virtual prototyping
- **Architecture:** Thermal management architecture; Interconnection & termination architecture

9.2.4.5 Interventions: Improved Thermal Management

- **Materials:** Thermal management materials; Thermal interface materials; Interconnection & termination materials; Substrate materials; Laminate materials
- **Processes:** Assembly and manufacturing processes; Interconnection & termination processes; Die attach
- **Tools:** Standards and test; Thermal modelling; Multi-domain virtual prototyping
- **Architecture:** Integrated thermal management; Thermal management architecture; 2-phase cooling; Cooling media; Layout and geometry optimisation

9.3 The Roadmap

Presentation of the Roadmap:

The roadmap for this working group is presented in two parts:

- The evolution of a range of selected interventions in materials, processes and tools
- The evolution of selected package architectures, making reference to interventions in architecture, materials, processes and tools

The interventions in materials processes and tools have been simplified to include to following categories: interconnects and vias; die attach and mount-down; substrates; encapsulation; baseplate and thermal management; housing and termination; metallization; passive components and sensors; simulation and virtual prototyping; testing and qualification.

9.3.1 Definition of Package Architectures

The package types have been divided into three broad classes, representing both state-of-the-art and emerging solutions: 1) discrete or single functional switches, including through-hole, surface mount, and chip direct packages; 2) modules, including conventional mono-planar, hermetic power hybrid, press-pack, and conventional multi-planar; 3) integrated power modules, including substrate-assembled and additive-embedded variants. These may be based on state-of-the-art packages or be inspired by other influences e.g. microelectronics. Definitions of the levels of integration referred to in the roadmap are given in Table 2.

9.3.1.1 Through-hole and Surface-mount

Conventional through-hole (e.g. TO-247) & surface mount (e.g. D-Pak, PQFN) packages for a single functional switch (and exceptionally multiple dies). Characterized by lead-frame/DBC plus transfer-moulded construction, typically epoxy encapsulation. Typical state-of-art for SiC packaging when used as direct replacement for discrete Si technology. Low-cost, flexible layout, limited reliability, possible poor thermal performance, can be difficult to achieve low-inductance layout.

9.3.1.2 Chip Direct

Intended for chip-on-substrate mounting. characterized by no lead frame (i.e. Chip pads directly attached to substrate). May or may not have added encapsulation & housing. Examples include IR Direct FET, EPC solder-bar dies and GaN Systems packages. Typical of state-of-art GaN packaging for chip-on-board applications. Low inductance layout (vertical commutation loop) possible with multi-layer substrate.

9.3.1.3 Conventional Mono-planar

Conventional plastic-encapsulated power module. Typified by one or more ceramic substrates in DBC, DBA or AMB technologies, wire-bonded interconnect, bus-bar & screw terminals or solder pin/spring connections, soft encapsulation over dies and plastic housing or alternatively hard epoxy encapsulation e.g. by transfer moulding. May or may not have a separate base-plate and/or integrated cooler. Industry standard for high current & high voltage modules in many circuit topology variants. Adopted in packaging for SiC. Moderate cost, moderate to high reliability, good thermal performance.

9.3.1.4 Hermetic Power Hybrid

Multi-chip power or power hybrid module. Typified by one or more ceramic substrates (DBC, AMB, thick-film or LTCC), wire-bonded interconnect, hermetic housing and lead-out through glass-metal seals. May or may not have a separate base-plate and/or integrated cooler. May also be implemented as an IPM with a range of passive components and active control functions. Used in demanding environments e.g. space/aerospace. High cost, high reliability, moderate to good thermal performance.

9.3.1.5 Press-Pack

Originally developed for high power diodes/transistors/thyristors. Typified by single basic unit semiconductor in circular format, pressure contacts to one side or both sides, Mo interface washers and Cu pole pieces inside a hermetically-sealed ceramic housing. Developed into multi-die IGBT package with dies in carriers but with single pole piece. Alternative press-pack developed by ABB (Stakpack) which has individual dies mounted down onto a base-plate and individual spring contacts reacted by a top pole piece. Some research reports but no commercial uptake for WBG at present. High cost, high reliability, good thermal performance but complex mounting arrangements and relatively high commutation loop inductance.

9.3.1.6 Conventional Multi-planar

Range of research and some commercial modules typified by planar interconnect and possibly planar termination structures. Usually based on DBC/AMB substrate for die mount-down but with a range of planar interconnect technologies e.g. polyimide flex, DBC/AMB ceramic sandwich etc. May be double-side cooled with appropriate choice of interconnect e.g. DBC/AMB. Research examples and limited commercial application, mainly for Si-based IGBT.

9.3.1.7 Substrate-assembled IPM

IPM characterized by one or more separate substrates, possibly of differing composition e.g. DBC/IMS/FR4, each populated with a range of components fulfilling specific functions e.g. power devices & commutation loop, gate drives, bulk decoupling. Populated substrates assembled into an IPM, typically stacked vertically, with electrical and possibly thermal interconnections (e.g. solder pins, spring pins). Encapsulated post assembly (e.g. with silicone or epoxy) or during assembly as the layers are added. Alternatively, may be housed in a hermetic package. Cooling arrangements vary but will include some form of integrated cooler/heatsink for high power applications. Wide range of power levels and applications possible. Existing commercial examples range from low-power dc-dc converters (1 W-100s W) through to high power inverters (e.g. Semikron SKAI 10s – 100s kW). Moderate cost, easy to use, high reliability, potential to optimize dynamic and EMI performance within a single package.

9.3.1.8 Additive-embedded IPM

IPM characterized by sequential construction starting from a single substrate. Components, conductors and insulators added and encapsulated as part of a sequential build-up process, leaving the componentry embedded within an encapsulating medium. Substrate types may vary within a single build-up but are typically based on readily available PCB laminates. Additional processes may include electroplating to form interconnects and vias, etching, drilling, milling and pressure lamination. Future routes to manufacture could include a range of additive techniques

to supplement/replace existing processes. Low to moderate cost, easy to use, moderate to high reliability, potential to optimize dynamic and EMI performance within a single package.

Table 2. Definition of the levels of integration referred to in the roadmap.

Integration Level	Description	Applies to Package Types
Intelligent switch	Power semiconductors plus active low-level controls and sensors e.g. gate drive, temperature sensors, current sensors	Discrete, Module and Integrated Power Module
Commutation cell	Power semiconductors plus one or more passive components and possible EM screening within a single package. Multiple topologies possible, one or more commutation cells within a single package. Simple example would be a half bridge plus decoupling.	Module & Integrated Power Module
Switching cell	One or more commutation cells plus active low-level control elements, in particular sensors, gate drives and protection. May also include additional passive components and screening to provide filtering functions and EMI control.	Integrated Power Module
Converter cell	One or more switching cells plus functional controls, in particular switch timing (e.g. PWM), current/voltage control etc. Typically will also include passive components and screening to provide filtering functions and EMI control as appropriate to converter function	Integrated Power Module

Draft roadmaps for the interventions in materials, processes and tools and for the various package architectures are shown in Figure 1 and Figure 2, respectively.



Figure 1: Roadmap for supporting technology in packaging and integration for WBG power electronics

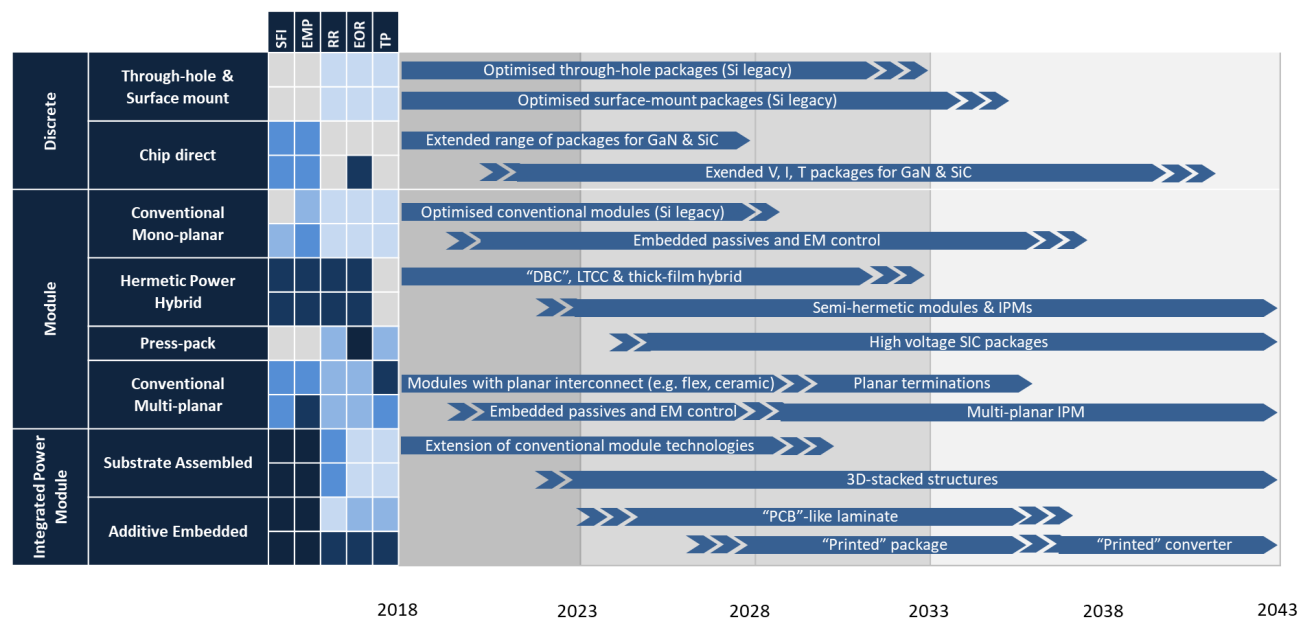


Figure 2: Roadmap for a range of package types applicable to WBG power electronics

Each technology bar shows the period of mass-market exploitation with chevrons indicating a degree of uncertainty; one chevron indicates some uncertainty around timing of mass market adoption or phase out, while two chevrons indicates considerable uncertainty around timing of mass market adoption or phase out. The relevance of each of the technology bars to the packaging and integration technology drivers are shown using color-coded bars as shown in Figure 3:



Figure 3: Relevance to technology driver color coding

Each bar has a color-coded key indicating the strength of impact on each of the drivers, with five separate columns are labeled as follows:

- SFI: Higher levels of structural and functional integration
- EMP: Improved electrical and magnetic performance
- RR: Improved reliability & robustness
- (V, I, T, f) EOR: Extended operating range
- TP: Improved thermal performance

9.3.2 Interventions in Materials, Processes and Tools

9.3.2.1 Interconnects and Vias

Ultrasonic Wire/Ribbon Bonding

Ultrasonically bonded Al wires/ribbons are the industry standard and will remain in use for the foreseeable future. There will be increased use of alternatives such as Cu for high reliability and high current density and Au or Pd wires for high temperature. In the longer term, enhanced techniques such as laser-assisted wire/ribbon bonding may be adopted.

Planar Interconnects (Bonded & Pressure)

Planar interconnects and double side bonded solutions based on ceramic, flex or FR4-based substrates will become increasingly important for conventional and substrate-assembled IPMs. Range of bonding technologies including solder, sinter, TLP, thermo-compression. Direct on-die bonding of copper/other metal foils will become increasingly important in power discretes as an alternative to ultrasonically bonded wire. Pressure contacts are an alternative. Bumps (solid, hollow) and pins in Cu, Cu/Mo, Mo etc. may be needed to provide standoff isolation and vias.

Planar & 3-D Interconnects (Deposited)

Electroplated, printed or ink/thick-film/paste-based interconnects and vias as part of layered build-up will become the norm for additive embedded IPMs. Solid/hollow pins may be more effective for extended vias.

9.3.2.2 Die Attach and Mount-down

Reflow Soldering

Reflow soldering with a mixture of alloys will remain the norm for many die attach and substrate mount-down applications. Sn-Bi, Sn-Ag, Sn-Sb commonly used with Pb-Sn and Au containing alloys reserved for high temperature. Evolution of alternative high temperature alloys e.g. Zn-based. Use of in-forms to improve reliability of substrate mount-down and other large-area bonds.

Sintering

Use of pressure-assisted sintering will become more widespread as demands for high reliability/high temperature increase. Pressure-less sintering will displace pressure-assisted in many applications, where the simplified process can help to reduce costs.

Epoxy, Glass and Cermet-based Die Attach

Improved epoxy-based systems (e.g. semi-sintered Ag) may offer an attractive route for additive embedded systems. Ag glass and Au cermets will find application in high-temperature applications and require high process temperature.

9.3.2.3 Substrate

Ceramic and Advanced Composites

Ceramic-based substrates will continue to dominate high power and high temperature modules. Evolution of new composite materials e.g. diamond/SiC with improved combination of e.g. cost, dielectric, thermal & mechanical performance. Multi-layer formats to support screening & shielding.

IMS/FR4/Polyimide

Technologies of choice for low power modules and auxiliary substrates in substrate-assembled IPMs. Evolution to more complex assemblies for additive-embedded IPMs where combinations of materials (metals, insulators etc.) are combined in 3-D structures.

Printed/Multi-functional

Evolution of printed, multi-functional substrates to form complex 3-D structures with regions of insulator, conductor, magnetic materials etc. Supports ultimate evolution to printed converter.

9.3.2.4 Encapsulation

Soft Encapsulation

Silicone will remain the dominant soft encapsulation material. Development of extended temperature range capability (-60 to >250C) likely to meet demands of automotive and aerospace sectors.

Hard Encapsulation

Epoxy-based materials will remain the dominant hard encapsulation/transfer moulding material. Continued development of improved composites with better matched CTE, higher thermal conductivity and extended temperature range will support demands of WBG applications.

Semi-hermetic Coatings

Semi-hermetic coating materials such as Parylene may displace the use of Silicone where moisture resistance and high temperature capability (>250C) are required.

9.3.2.5 Base-plate and Thermal Management

Cu and AlSiC will continue to dominate base-plates in conventional power modules. In the longer term, improved composites, based on low-CTE materials such as Carbon/graphite/SiC/diamond, will likely become established for high reliability applications.

9.3.2.6 Cooling Technologies

Pin fin, micro-channel and impingement integrated coolers will continue to evolve for single and double-side cooled conventional modules. Vapour chambers and heat pipes offer improved heat transfer performance compared to solid conductors and will become more commonplace in high power density applications or where the cooling geometry challenges conventional approaches. Additive processes will be introduced to realise complex geometries.

Cooling of complex 3-D assemblies in IPMs will evolve to provide targeted cooling and/or heat transfer via conductors, vapour chambers, heat pipes and other phase-change processes. Distributed cooling will replace the concept of a single cooled surface in pursuit of reduced weight/materials/cost. Use of additive manufacturing becoming more widespread supporting printed converter.

9.3.2.7 Housing and Termination

In the short term housings will be dominated by continuing use of the incumbent housing technologies according to package type: conventional = polymer (injection & transfer moulding), hermetic = metal/metal-ceramic, press-pack = metal-ceramic. In the longer term use of additive layer and printed manufacturing will displace the need for separate housings in IPMs and potentially in conventional planar and multi-planar modules. Mixture of materials according to desired function (thermal, EMI suppression).

Terminations will benefit from the development of low-profile screw terminal, solder pin and spring pin terminations to minimise inductance. Evolution to co-planar and coaxial terminations to further reduce connection inductances. In the longer term, a move toward integrated commutation cells will result in less stringent requirements for terminations. Terminations will likely be formed by additive processes.

9.3.2.8 Metallization

A greater range of die presentations will become available to match with selected die attach and interconnect

technologies e.g. Cu for Cu wire and electroplating, Ag for sintering, Ni-Au & Ni-Ag for soldering. Application of suitable diffusion barriers for extended temperature use e.g. W. Substrate metallization's for a range of die attach technologies will evolve to provide high reliability and high temperature compatibility. Use of alternative diffusion barrier metals e.g. W to replace Ni. Additive processes will replace plated layers in IPMs.

9.3.2.9 Passive Components and Sensors

Passive components (e.g. capacitors and inductors) suitable for incorporation into modules will require co-development alongside the wide band gap semiconductors to ensure realization of the potential benefits of improvements in higher energy density, higher temperature and new magnetic and dielectric materials. Next generation materials for passive components would provide a step change in performance. Potential new materials include carbon nanotube enhanced windings for inductors and improved magnetic and dielectric materials, which permit higher energy storage densities to be achieved.

Embedded sensors need to be able to tolerate higher temperatures, generate lower losses whilst maintaining high accuracy. Application to integrated WBG power modules will demand smaller physical sensors with extended high frequency capability. Wireless solutions will become more common as technology advances.

9.3.2.10 Modelling Tools

In the short term, there will be increased application of EMC modelling and physics-based reliability modelling. There will be development of quick and accessible simulation tools for design and optimization together with access to cloud-based solutions via tablets and phones. Simulation and validation over multiple timescales and prospective mission profiles. In the longer term, there will be routine application of multi-domain models: magnetics, electrical, thermal, mechanical & reliability for virtual prototyping and design optimization. There will be increasing reliance on models that inform design through material choice to end of life with further extension to virtual testing and qualification.

9.3.2.11 Testing and Qualification

In the short term, there will be extension of testing and standards to cover EMC for modules and IPMs together with the development of accelerated test methods for extended operating range and high-reliability products. There will be routine application of non-destructive techniques for reliability characterization and the development of combined stress test methods to shorten testing times. In the longer term, improved multi-domain models and application of virtual prototyping will reduce the need for physical testing and qualification. Virtual testing and qualification can be expected to replace routine physical testing for established manufacturing processes and materials.

9.3.3 Evolution of Selected Package Architectures

9.3.3.1 Through-hole and Surface-mount

End-user familiarity, low-cost, established capability and layout flexibility will mean that these conventional packages will endure well into the future even though they provide a sub-optimal solution for many WBG applications. Expect applications to be dominated by SiC switching at frequencies up to 100kHz. There are likely to be incremental improvements in package performance e.g. temperature capability as new materials are introduced and changes to lead-out configuration e.g. Kelvin gate connections, broad foils rather than pins. Surface mount offers greater opportunities for step change possibly tending towards chip-scale and chip-on-board packages.

9.3.3.2 Chip Direct

Chip-scale packages, suitable for chip-on-substrate integration and embedding, will become more popular for high frequency applications (>100kHz to 10MHz). Flip-chip and double side versions with a variety of metallisations suitable

for soldering, sintering, epoxy bonding and plated interconnect/die attach will become available. Although initially seen as a low-voltage (<1000V) GaN technology, application to higher voltage, current and temperatures is likely, driven by requirements for higher levels of integration at increased power levels and niche applications. Early application in integrated power modules but eventually superseded by use of bare die as die embedding technologies mature.

9.3.3.3 Conventional Mono-planar

End-user familiarity, low-cost and established capability will mean that these conventional modules will endure well into the future. There will be evolution towards lower inductance internal layout and terminations but ultimately will be restricted to lower frequency, higher voltage applications. Both SiC and GaN variants will emerge but SiC dominant. There are likely to be incremental improvements in package performance e.g. temperature capability, reliability as new materials and assembly processes are introduced. Passive components (R, L, C) embedded within module will provide functions such as decoupling, output filtering and EM control, together with the addition of EM screens to suppress conducted & radiated emissions e.g. CM current through base-plate. Gate drives and sensing functions may be introduced but the complexity will be limited by mono-planar substrate layout.

9.3.3.4 Hermetic Power Hybrid

Established capability for demanding environments will mean that these hermetic hybrid modules will endure well into the future. Use of LTCC & thick-film technologies may become established for integrated power module applications at low to medium power in both SiC and GaN technologies. Evolution towards semi-hermetic encapsulation is likely in the longer term, based on e.g. parylene coatings, to reduce cost for high temperature and high reliability applications. This will be accompanied by a move away from glass-metal seals towards alternative termination technologies. Greater levels of integration using a mixture of “DBC” and thick-film type approaches may emerge to create high-temperature, high-reliability IPMs. There will be incremental improvements in package performance e.g. temperature capability, reliability as new materials and assembly processes are introduced.

9.3.3.5 Press-pack

Established capability for high voltage Si devices may be transferred to high-voltage SiC MOSFET/IGBTs. Advantages include high reliability, good thermal and HV isolation performance plus potential fail to short-circuit behaviour. Suited to high-voltage stacks and specialist pulse-power applications but limited to low frequency because of extended commutation loops.

9.3.3.6 Conventional Multi-planar

In the short-term there will be application of a range of alternative planar interconnect technologies for reduced inductance and improved reliability. Vertical commutation loops and co-planar bus structures will evolve for termination, yielding improved high frequency performance for SiC and GaN applications. Potential for double sided cooling will be exploited in some package variants.

9.3.3.7 Substrate-assembled IPM

In the short-term, developments based on conventional planar and hermetic hybrid modules will be used to realise commutation cells, switching cells and converter cells. Selected combinations of technologies will be employed to realise low, medium and high power applications. There will be an application emphasis on efficiency and high frequency switching for both SiC and GaN. In the longer term, there will be use of multiple substrates, potentially in differing technologies, to realise 3D layouts for improved electrical, magnetic and thermal performance. The emphasis will change towards converter cell implementations for high power density and high reliability and end-user ease of use will be a priority. This approach may be combined with aspects of additive embedded modules as detailed below.

9.3.3.8 Additive-embedded IPM

In the medium term, there will be development and application of techniques for PCB laminates, including 3D interconnects, active & passive embedded components and EMI management structures to realize converter cells with improved power density and reduced cost. Compatibility with high-volume manufacturing will be a key feature and applications will favor high switching frequency compact designs in both SiC and GaN. A variety of thermal management options will be developed according to the power level. This approach will seek to benefit from improved materials e.g. low-CTE laminates. In the long term, direct additive processes (e.g. 3D-printed) will be applied to realize all package layers and interconnects (electrical insulators and conductors). Initially, these IPMs will employ embedded discrete components/dies but there is the prospect of printed dielectrics, magnetics and semiconductors and ultimately, to print custom materials with controllable properties. Additive embedded techniques may be combined with substrate-assembled methods to generate a range of hybrid approaches.

9.4 Examples of Packaging for Wide Band Gap Power Electronics

9.4.1 Case study 1: Integrated Switching Cells for Modular Wide Band-gap Conversion

The fast switching speeds and low specific conduction losses of wide band-gap semiconductors allow the realisation of high-frequency, high power-density switching converters with dramatically reduced passive component requirements compared to Silicon technology. However, careful attention must be paid to switching cell design to mitigate the effects of circuit parasitics and fast voltage transitions, which would otherwise limit the attainable switching speed and lead to increased levels of EMI. A modular, power-cell solution is proposed in [1] which allows the creation of any two-level topology converter. The cell structure enables fast switching of wide bandgap semiconductor devices while allowing high power converters to be fabricated using multiple, smaller commutation cells. Close integration of semiconductor dies, decoupling capacitors, gate drives and an output filter, using a single ceramic substrate to act as the thermal path, allows dramatic increases in power density without compromising converter performance. Figure 4 illustrates the assembled modular inverter and output waveforms.

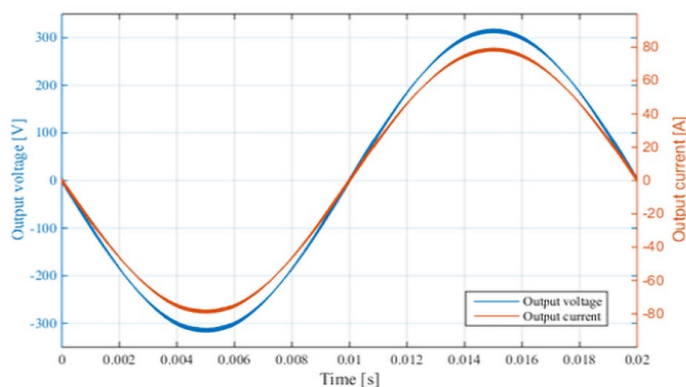
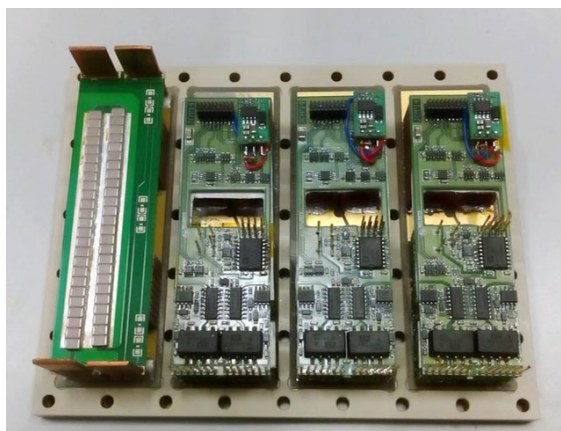


Figure 4: a) Modular 3-phase inverter based on half-bridge switching cells incorporating gate-drives, sensors, low-level controls and input and output filters; b) output V and I waveforms with a resistive load, demonstrating low levels of switching noise and ripple

9.4.2 Case study 2: High-voltage Packages for Silicon Carbide MOSFETs

High-voltage SiC devices offer an attractive combination of fast switching and low losses, giving application users unprecedented levels of flexibility in choice of topology and control strategy for medium- and high-voltage power conversion. However, the realisation of power modules that are optimised for the high electric fields, whilst maintaining compact commutation loops and effective thermal management, presents significant challenges including: 1) the mitigation of high electric fields in the substrates, interconnects and lead-out connections; 2) suppression of switching voltage overshoot and common-mode interference; 3) high-performance thermal management. The

assembled module described in [2] incorporates stacked substrates with vias to mitigate internal electric fields and allow realisation of vertical commutation loops. Embedded decoupling capacitors and an interlayer screen help to suppress voltage overshoot and heatsink-coupled common-mode interference. Finally application of a direct-substrate liquid impingement cooler permits exceptional thermal performance to be achieved within a small physical envelope. Figure 5 illustrates a cross-sectional schematic and photograph of the assembled half-bridge module.

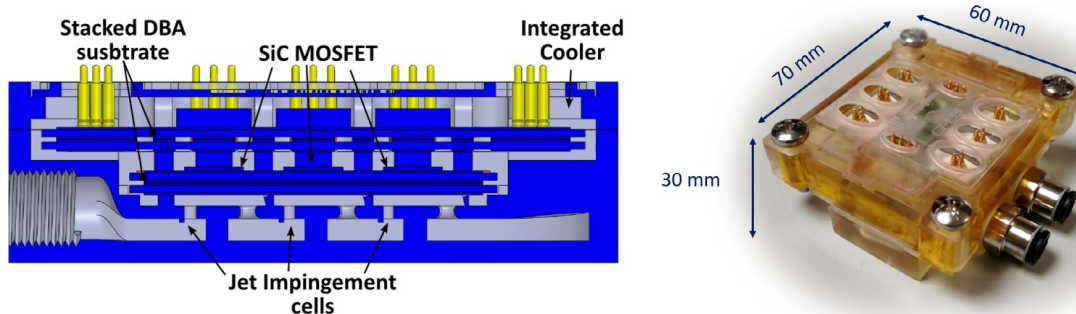


Figure 5: a) Cross-section through module and cooler, showing stacked substrates, interconnect posts and termination pins; b) physical realisation of the prototype 10 kV SiC MOSFET module.

9.4.3 Case study 3: Integrated Full-Bridge Power Module Based on GaN eHEMT Devices

The authors of [3] have proposed a power module design combining the inherently low interconnection inductance of a vertical commutation loop, combining a multilayer PCB and low inductance HEMT package, with the good thermal performance offered by DBC substrates. Figure 6 illustrates a schematic cross-section through the module and an assembled full-bridge. A pre-manufactured PCB is populated with components including, DC-link capacitors, gate drivers, voltage regulators and GaN eHEMT devices, which are mounted onto the rear side of the PCB. The exposed heat pads on the backside of the GaN eHEMT devices are then soldered to separate pads on an aluminium oxide (Al₂O₃) DBC. Press fit pins are used to interface the power module with DC voltage supply, gate driver signals, shunt resistor voltage signal and switching output. The assembly is enclosed by a plastic housing which may be gel-filled to improve electrical isolation within the package.

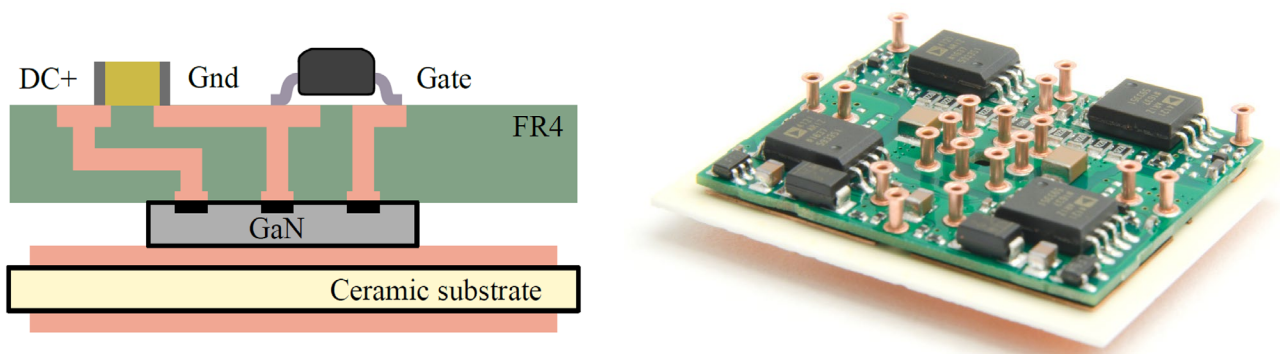


Figure 6: a) Schematic cross-section of the module; b) photograph of the assembled full bridge module.

9.5 Conclusions and Future Activities

The key challenge for the practical realization of wide band gap devices in power electronics applications will be effective packaging that addresses the challenges of high-speed operation and thermal management for a wide range of operating environments and applications. Although good progress is being made, there is much further work to be done in packaging for wide band gap devices in the longer term.

A future task for the working group will be to identify the research, development, and industrialization activities that are needed to support the roadmap priorities on a short (0-5 year), medium (5-15 year), and long-term (15-30 year) horizon. The broad categories defined by the working group are:

- Curiosity-inspired research 10+ years out from mass-market exploitation
- Industry-led development 5-10 years out from mass-market exploitation
- Industrialization initiatives 0-5 years out from mass-market exploitation

Realization of roadmap priorities will generally require all three stages in succession.

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9.7 References

- [1] Jordi Espina, Behzad Ahmadi, Lee Empringham, Liliana De Lillo, Mark Johnson, “Highly-integrated power cell for high-power wide band-gap power converters”, 2017 IEEE 3rd International Future Energy Electronics Conference and ECCE Asia (IFEEC 2017 - ECCE Asia). 2017. pp. 146-150
- [2] Bassem Mouawad, Robert Skuriat, Jianfeng Li, C. Mark Johnson, Christina Dimarino, “Development of a Highly Integrated 10 kV SiC MOSFET Power Module with a Direct Jet Impingement Cooling System”, 2018 IEEE 30th International Symposium on Power Semiconductor Devices and ICs (ISPSD)
- [3] Asger Bjørn Jørgensen, Szymon Beczkowski, Christian Uhrenfeldt, Niels Høgholt Petersen, Søren Jørgensen and Stig Munk Nielsen, “A Fast-Switching Integrated Full-Bridge Power Module Based on GaN eHEMT Devices”, *IEEE Transactions on Power Electronics*, vol. 34, no. 3, pp. 2494-2504, March 2019

Chapter 10: Design Automation for Wide Bandgap Power Electronics ^[1]

10.1 Introduction

From component modeling and schematic capture to circuit simulation and practical implementation, design tools have played a crucial role in the design and development of power electronics converters and motor drive systems for a long time. As the switching speeds rise and power densities increase, the pressure to improve automation in these design tools is gathering momentum. Modern integrated power systems are becoming multi-physics problems, need shorter design time, and require multivariable optimization to obtain higher efficiencies and more compact designs. As a result, design automation is bringing more integration in design tools for power electronics.

Although power electronics design tools, such as circuit simulation, thermal management, electromagnetic interference (EMI) compatibility, test and manufacturing, have improved significantly over time, they have remained separate and standalone for most of the time. Consequently, when engineers convert specifications to electrical circuits, and then implement them on circuit boards for manufacturing, they employ multiple tools to ensure that the final product is performing satisfactorily with high reliability and low cost.

For example, a designer may use Matlab/Simulink for simulating a converter circuit/schematic with control. Once the schematic is finalized and components selected, the engineer must then use tools for proper layout of components and traces on the printed circuit board (PCB) to minimize parasitics. Next, the designer will use another set of tools to perform thermal analysis of the circuit board to ensure that heat is spread properly and that there is no overheating in any section of the board. Likewise, the designer must perform EMI compatibility to ensure that there is no interference from switching at high frequencies or layout and interconnect parasitics. Testing is another task before sending the product to production. In short, a designer must use multiple disparate design tools before finalizing a power system design for manufacturing.

However, in the last decade or so, significant work has been done to integrate these varied tools into one integrated process flow for design and realization of power systems. As a result, automation in design tools for power electronics circuits continues to make progress in the design, simulation, and implementation of power electronics circuits and systems.

10.2 Heterogeneous Integration

To see how far tools for design automation have progressed, both academically and industrially, a workshop was held last September in Portland, Oregon called “Design Automation for Power Electronics – DAPE”, it was jointly organized by IEEE Power Electronics Society (PELS) and Council on Electronic Design Automation (CEDA) and co-located with IEEE Energy Conversion Congress and Exposition 2018 conference in Portland. The purpose of this first workshop was to understand the problems of design automation in power electronics, identify methodologies that have been used so far by academia and industry and identify the tools that have been developed to resolve the issues during design. The focus of the workshop was to bring together the experts in both power electronics and design automation and have them present their perspectives on the emerging needs driven by wide bandgap (WBG) technologies.

The workshop goals were as follows:

- identifying the common ground and synergy between power electronics and design automation
- understanding the limits of the current design automation technology and tools
- recognizing the areas for future work

Because designers are driven by the need for smaller size, higher efficiency, lesser weight, and lower cost, they have migrated to higher switching frequencies for smaller passives and higher power densities. This trend has continued for decades now, said the keynoter. While the benefits of switching at higher frequencies are plenty, it also comes with some penalties, such as EMI and parasitics. Furthermore, the complexity of modern power systems demand a more integral approach to design and realization. The design tools need to facilitate the design and analysis of a heterogeneously integrated system, which involves combining device models and modeling of the module, layout

optimization, thermal and EMI analysis, electrical performance measurement, and so forth. Consequently, a model based engineering design flow suitable for high-density power electronics is required, such as that developed at the University of Arkansas (Figure 1) [4].

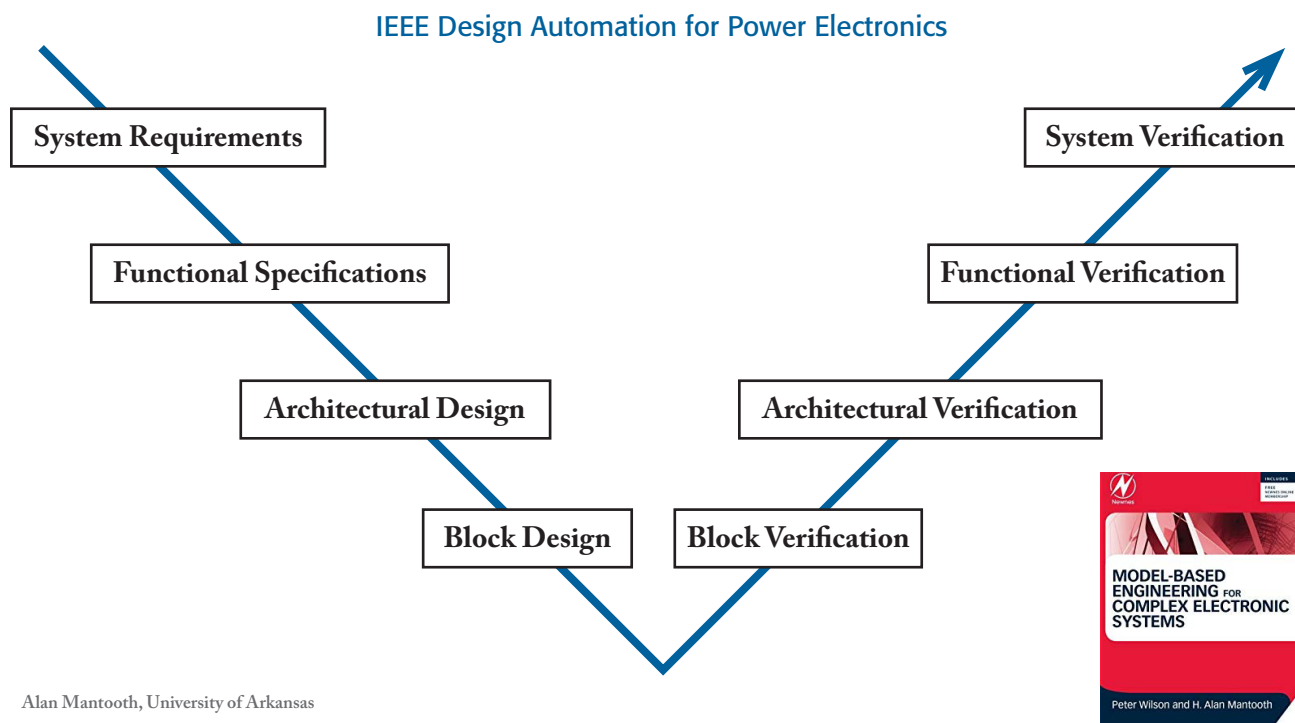


Figure 1: Model based engineering design flow for a complex power system.
(Figure courtesy of DAPE)

Traditionally the design flow progresses from specifications to circuit schematic by using tools, e.g., MATLAB/Simulink, PLECS, LTSpice, and Saber or something similar. Control is then finalized using tools such as MATLAB, hardware-in-the-loop (HIL) or power HIL, followed by component selection, electro-thermal circuit analysis, board layout, design rule checking/layout versus schematic/parasitic extraction, thermal, electrical, EMI analysis, cabinet design with a separate electro-thermal analysis, manufacturing and testing. Taking these factors into consideration, Prof. Mantooth's team at UARK has developed a software tool for the design and layout of multi-chip integrated modules called PowerSynth (Figure 2) [2]. By enabling the design and layout of integrated multi-chip power modules (MCPM), tools such as PowerSynth bridge the gap between circuit design and packaging engineers. It uses fast thermal and electrical models to gauge power module performance quickly, and offers multi-objective optimization capability to allow for many trade-off design solutions to be considered. When compared with conventional simulation runtime and workflow, tools such as PowerSynth can take considerably less computation and user time to produce several candidate layout solutions from which a designer may easily balance selected optimization targets.

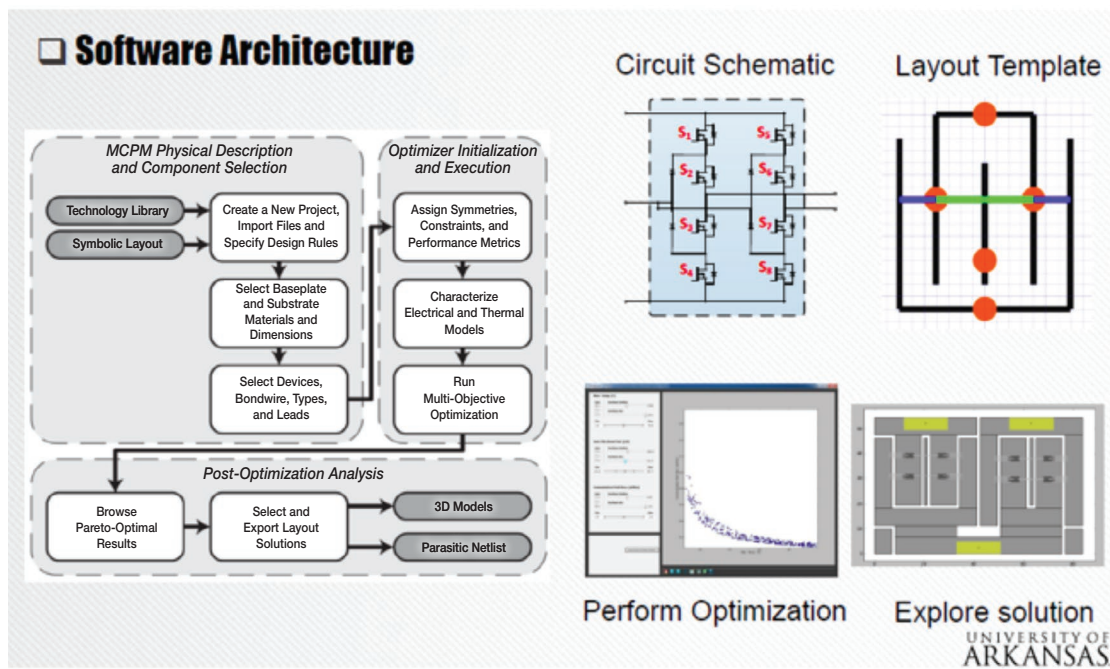


Figure 2: PowerSynth is a software tool for the design and layout of multi-chip integrated power modules. (Figure courtesy of UARK)

10.3 More R&D Work

In addition, design solutions generated can be easily exported to tools like ANSYS Q3D and SolidWorks for further analysis and manufacturing. However, accounting for EMI in layout is still under development. Ongoing research efforts include extending the layout representation to include heterogeneous component integration and 3D layout optimization along with and improved fidelity of the electrical and thermal models [3]. The work further shows that continued development of the electrical models will allow for better optimization for balanced branch-currents among paralleled devices and conducted EMI. This model can then be used to optimize the thickness of the substrate, attachment, and other related material layers to produce more reliable modules. Finally, techniques that will allow complete synthesis of the layout from a circuit netlist—further narrowing the technical divide between systems design and converter manufacturing are under investigation.

Recent work by Professor Johan Kolar indicates that multi-objective design faces challenges like counteracting the effects of key design parameters and mutual coupling of performance indices and trade-offs, which are dependent on applications such as consumer, industrial and aerospace. In essence, he suggests that multi-domain modeling/simulation/optimization ensures optimal mapping of the design space into the “performance space and identifies absolute performance limits, labeled as η - ρ -Pareto Front, or efficiency-density Pareto Front. Key benefits include excellent design insight, understanding of components, reduced design time and time-to-market. Future experimental work includes convergence measurement and simulation, resulting in augmented reality oscilloscope. As a result, measured signals and simulated inner voltages/currents/temperature could be displayed simultaneously, noted the speaker. Thus, automatic tuning of simulation parameters offer best fit of simulated and measured waveforms, concluded the talk. Concurrently, the talk also suggested that the concept of digital twins continues to evolve as designers convert information on paper into digital format for virtualization.

Some other researchers advancing the art of design automation include Prof. Frede Blaabjerg, president-elect of PELS and a professor in the Center of Reliable Power Electronics (CORPE), Department of Energy Technology, Aalborg University, Denmark, and Prof. Dushan Boroyevich of Center for Power Electronics Systems (CPES), Virginia Tech. Prof. Blaabjerg, with co-author Prof. Huai Wang of the same university, presented current status and outlook for mission-profile based reliability design tool developed at CORPE. Based on model-based platform, the

researchers described a design for reliability and robustness (DfR²) tool with its graphical user interface (GUI) and interfaces to other software tools from makers like Synopsys and ANSYS (Figure 3). Offering a modular model-based reliability analysis for power components and systems, DfR² performs parameter sensitivity and risk analysis on the main models (mission profile, electro-thermal, lifetime, etc.) used in the reliability assessment procedure. Plus, it offers multi-objective optimization procedure for power electronic systems aimed at variety of applications including PV systems, wind power and motor drives.

Design for Reliability and Robustness (DfR²)

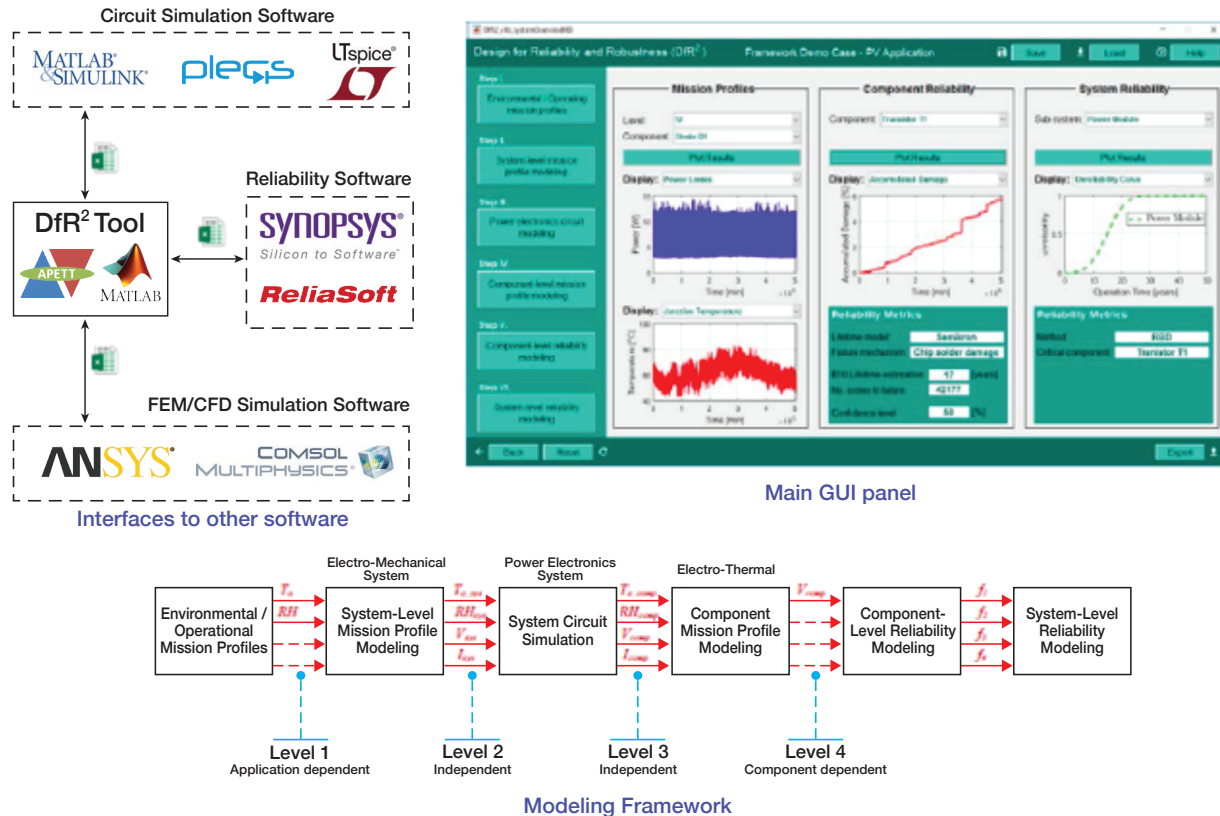


Figure 3: Design for reliability and robustness tool developed at CORPE, Aalborg University. (Figure courtesy of Center of Reliable Power Electronics, Aalborg University, Denmark)

According to the researchers, concurrent efforts to incorporate multitime scale modeling and multistressor modeling, which includes electrohygrothermal modeling and thermal coupling (e.g., system level, component level, and sub-component level) are underway. However, a major challenge that researchers foresee is the lack of models for new generation components, especially WBG power devices.

Prof. Boroyevich proposes a multidisciplinary design methodology (MDO) for emerging power systems in transportation and remote locations. Using model parameters and analysis data exchange, the MDO concurrently takes input from blocks like manufacturing, physics of failure, structural mechanics, thermo-dynamics, solid body geometry, electro-magnetics, device physics, systems & control, and circuits & topologies and feeds it into a design optimization block. The data extracted from all the functions described is used to create a model database for the design process/control/optimization block. Using this methodology, Prof. Boroyevich's team, comprising Rolando Burgos, Igor Cvetkovic, Bo Wen, Qian Li, Qiong Wang, Niloofar Rashidi, has created a multidisciplinary hierarchical optimization design flowchart depicted in Figure 4. A high performance isolated three-phase ac-dc converter example is then used to demonstrate the design and optimization methodology proposed. The results indicate that the converter meets an optimization objective with high efficiency and power density. Another example is used to demonstrate its use in minimizing system weight of aircraft power with high-peak pulsed-power loads.

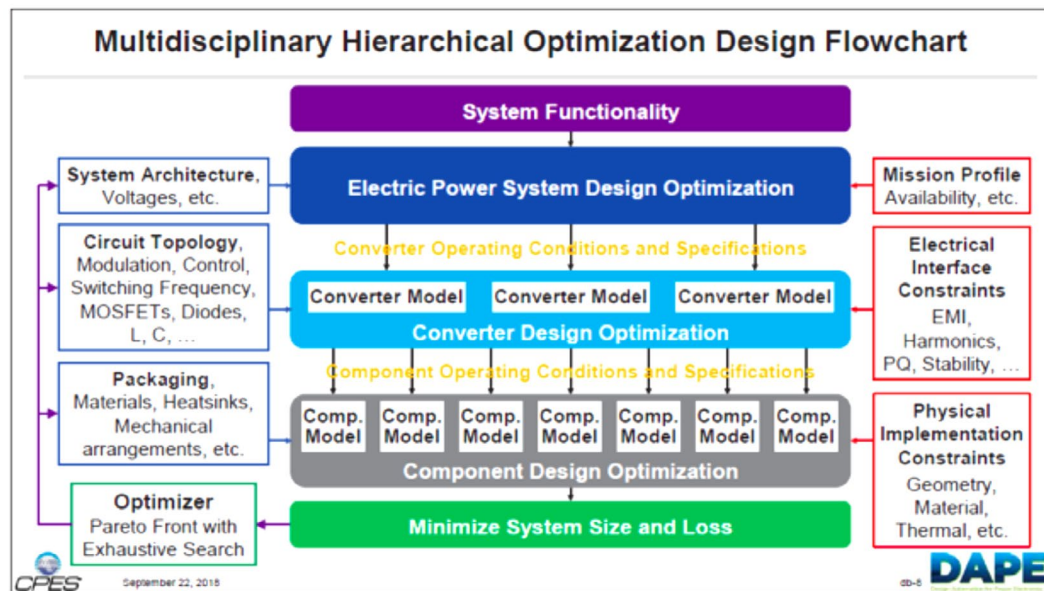


Figure 4: A multidisciplinary hierarchical optimization design flowchart proposed by Prof. Dushan Boroyevich, CPES. (Figure courtesy of CPES)

Prof. Boroyevich also proposed a modified design flow with parametric uncertainty quantification (UQ). The talk concludes that system optimization is enhanced if more interface standards are used as design variables instead of design constraints. The design models are preferably algebraic or databases (response surfaces) and time-domain and frequency-domain simulations are used to verify the designs before the results are passed between the levels. Simulation models should have different levels of abstraction at different design levels. Much of the design automation work in the power electronics community will remain inconsequential unless power electronics discipline develops a systematic verification and validation and a UQ methodology, according to Boroyevich.

10.4 Advances in Commercial Tools

Commercial design tools suppliers who presented their perspective on developments in design automation for power electronics include Mathworks, Mentor Graphics, ANSYS, Synopsys, and Typhoon HIL. From the industry user standpoint, General Motors (GM) and Boeing shared their current practices in automotive and aerospace applications, respectively. For developing custom design automation tools for such a complex multi-physics electrification problem, the EV maker is utilizing a detailed design analysis (DDA) philosophy with some pre-defined goals. Based on DDA philosophy, a custom software toolbox developed internally for design automation and application. Employing this optimization tool, the designers were able to minimize stray inductance for the dc capacitor and automotive power modules, respectively, thus enabling the designers to tap the benefits of SiC transistors in automotive power modules.

There are challenges with EV powertrain design. These include obtaining realistic models for early trade studies, selecting topology & architectures, selecting & sizing of components, optimizing fuel economy & performance over drive cycles, implementing control algorithms and simulation models that run real-time and choosing the correct model complexities for vehicle and controller simulations. Thoughts about design automation for EV powertrains included:

- selecting the proper level of model fidelity for power electronics, passive components, battery packs, motors, and other components
- passing models along the V-model design between systems and subsystem designs
- capturing design data from hardware components when possible
- using simulation for controller design, system optimization, component selection and sizing
- considering the proper model fidelity that enables both desktop and real-time simulations.

According to Mentor, automotive IC suppliers like STMicroelectronics, ON Semiconductor, Toshiba and ROHM are using this tool to accurately assess heat impact on IC performance.

After presenting an overview of the Synopsys multi-domain mixed-signal simulator Saber, Alan Courtay of the Saber R&D group talked about power electronics circuit simulation challenges. These include lack of standard models, incomplete or inconsistent datasheets, qualifying SPICE models from manufacturers, and stiff systems being prone to convergence errors. Because Saber utilizes a datasheet driven modeling tool solution, it overcomes problems encountered with other simulators in the market, according to Synopsys.

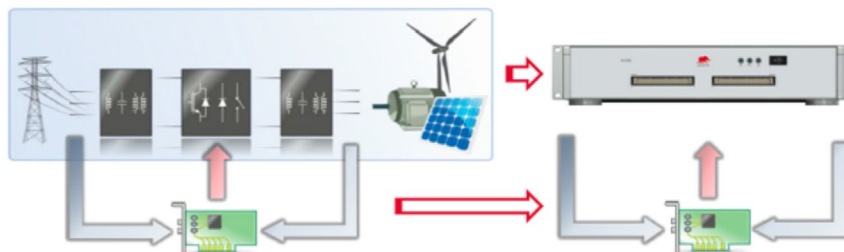


Figure 5: Controller hardware-in-the-loop (CHIL) is a new way of testing power electronics controllers.
(Figure courtesy of Typhoon HIL)

Aiming to empower test driven control development, Ivan Celanovic of Typhoon HIL gave a presentation on ultrahigh fidelity real-time simulation for power electronics and microgrids. During his talk, Celanovic said that with grid digitalization and exponential deployment of distributed energy resources, electrical grid is undergoing rapid transformation and power electronics is enabling this transformation. This system complexity is driving the need for new tools and methods. Towards that goal, he emphasized the need for model-based system engineering comprising model-based specification, design, development, testing (verification), certification, documentation and maintenance. According to Celanovic, controller hardware-in-the-loop (CHIL) offers that solution with guaranteed efficiency, productivity and test coverage (Figure 5). Key features of CHIL include low latency and high fidelity, long-term stability, ease of use and test automation. Speaking of tests, Celanovic indicated that CHIL allows a designer to define tests early on based on requirements and then run them from the very beginning of the development cycle. Other features include simulation, HIL simulation, lab bench test, as well as validating your models from one stage to the next.

He also stated that CHIL reduces software bugs by 38%, as well as the cost of fixing HIL test problems by 61% when compared to non-HIL tests. As a result, he concluded that in applications ranging from integrated power grid to microgrids and aerospace, designers are quickly embracing model based engineering and testing.

In summary, according to Prof. Mantooth's presentation, the DAPE workshop signals three trends emerging in the tools arena for design automation. These include MBE V-diagram, digital twins and multi-objective design optimization. While several versions of V-diagram were presented, the term "digital twins" was portrayed as a current and evolving practice. And multi-objective optimization is a multidisciplinary hierarchical method that guarantees full utilization of design space.

10.5 Compact Models of Wide Band Gap devices

One side effect of the increased complexity of device models has been the embedding of advanced design techniques such as modelling and simulation into the design process. This has become an integral aspect of the delivery of hardware to the extent that the major aircraft manufacturers have moved to become integrators of major systems and with the increase in electrification required to use simulation to validate the performance of all electrical equipment prior to integration of the hardware itself.

In order to achieve effective design of motor drives using modern wide band gap devices requires the implementation of modern compact simulation models for diodes and MOSFETs.

The Hefner model [5] has been applied in a number of power modules, and using parameter optimization can be very effective in accurately simulation the detailed effects of SiC devices in practice [5].

10.6 References

- [1] A. Bindra and A. Mantooth, "Modern Tool Limitations in Design Automation." *IEEE Power Electronics Magazine*, Vol. 6, March 2019, p. 28
- [2] H.A. Mantooth and Y. Peng, "Wide Bandgap Power Electronics: A New Trend on Design Automation," WiPDA 2018 workshop
- [3] T. M. Evans, Q. Le, S. Mukherjee, I. A. Razi, T. Vrotsos, Y. Peng, and H. A. Mantooth, "PowerSynth: A Power Module Layout Generation Tool," *IEEE Transactions on Power Electronics*, DOI 10.1109/TPEL.2018.2870346
- [4] H. A. Mantooth and M. Fiegenbaum, "Modeling with an Analog Hardware Description Language," Kluwer Academic Publishers, Norwell, MA, 1995.
- [5] T. R. McNutt, A. R. Hefner, H. A. Mantooth, D. Berning, and S. H. Ryu, "Silicon carbide power mosfet model and parameter extraction sequence," *IEEE Transactions on Power Electronics*, vol. 22, no. 2, pp. 353–363, March 2007
- [6] Z. Chen, R. Burgos, D. Boroyevich, F. Wang, and S. Leslie, "Modeling and simulation of 2 kv 50 a sic mosfet/jbs power modules," in 2009 13th European Conference on Power Electronics and Applications, Sept 2009, pp. 1–10.

Chapter 11: Standards

11.1 Background

The rapid pace of change in the wide band gap (WBG) power semiconductor area has led to an explosion in potential uses for WBG devices in a huge variety of applications. The applications include automotive, aerospace and traction applications, as well as grid related or charging systems, with the potential to provide paradigm shifts in performance and efficiency over Silicon devices in current use today. Despite these exciting developments, however, there are still many outstanding challenges for both researchers and industry to solve before WBG technology becomes pervasive. In this paper we will explore some of these challenges and highlight the strengths of WBG devices, some of the specific issues for machine drives and develop some potential solutions for future developments in power electronics.

There is a strong drive for an increase the electrification of transportation, whether that is regarding automotive, aerospace, trains or marine applications. For many years this has driven the use of increasing power density devices from Thyristors and GTOs to IGBTs and Silicon based technology such as MOSFETs. It is safe to say that we have reached a plateau of capability from these technologies as researchers and engineers have striven to develop optimal solutions that achieve close to the theoretical maximum possible performance from Silicon, and the packaging has been fine-tuned to such a degree that there is marginal gain to be made without a fundamental change.

With the development of wide band gap devices able to tolerate high voltages and currents using materials such as Silicon Carbide (SiC) and Gallium-Arsenide (GaN), this situation has changed and a new frontier of developments are possible due to the fundamental shift in what is now possible in these new devices. A particular area of interest for power electronics is in that of electric vehicles, where the demand for more compact, high power density, high temperature operation is of critical importance and is imminent on a mass scale [17-23]. The issue for implementation of practical electric drives systems is now a matter of efficient thermal design, advanced packaging and proximity to the electric motor itself [24-31].

11.2 Standards

Power Electronics is a rapidly growing field, with an accelerating pace of technological change. This combination is a challenge for industry professionals trying to keep up to date with the regulatory and standards requirements and also the standards bodies (such as the Institute of Electrical and Electronics Engineers or IEEE) to ensure that the published standards are up to date. Standards are vitally important for ensuring that products conform to standard levels of functional interoperability, performance, safety and reliability. This is essential at all levels of scale from consumer products such as electric toothbrushes to cars and aircraft. There are a plethora of standards and regulatory bodies and this can lead to confusion, ignorance and inadvertent problems. These could possibly be averted if designers and product engineers had a greater awareness of not only which standards were required, but also some idea of how to interpret and implement them. This article is intended to assist power electronics engineers navigate these difficult and complex issues, and give some insight on how best to handle the use of standards. There are in fact a number of very commonly asked questions by practicing engineers that have been posed within the IEEE Power Electronics Society (PELS), and hopefully the answers will be constructive and useful. In addition, the roadmap for wide bandgap power semiconductors (ITRW), while this is not directly a standard as such, is closely related to a number of standards activities.

As wide band-gap devices proliferate power electronics systems, it will become crucial to ensure that there is an adequate regulatory framework for suppliers from devices and modules to sub-systems and systems to ensure that resilience and performance is “designed-in” to complete power electronic systems. An important part of that effort will be to define standards and recommended practices for design engineers at all stages of systems development to ensure consistency of measurements, assessment of performance and reliability with these new devices.

For example, will it become necessary to carry out “double pulse” testing rather than “single pulse” testing to ensure validity of results? There is also a shift of emphasis from the device itself to the module and integration of gate drive,

heatsink and connectors to ensure that system integrators can have confidence that adequate testing and validation has been carried out with these new devices. This is particularly important for SiC devices where assumptions might be made that these devices can operate at significantly higher temperatures, without considering the impact on system reliability as a whole.

It is still early in the development process of standardization; however, the role of organizations such as ITRW (IEEE), SAE and JEDEC will be crucial in establishing a viable and common framework of standards that will serve the power electronics industry for the next several decades.

11.3 Which Standards do I Need to Know About?

This is a common question often asked by engineers struggling to make sure they adhere to the correct standards. This obviously depends on the industry sector you are operating in. In many industries there may be “known” key standards and the most relevant standards body will also often be highly industry dependent. The IEEE is one of the world’s most important standards organizations and in many cases it is true to say that the IEEE standard will be the de facto global standard. A very good example of this is the IEEE standard 802 (and its various constituent standards) relating to computer networking – that have become ubiquitous. The use of the phrase “de facto” is important, as it is often the adoption by an industry that drives the use of a particular standard, rather than legislation per se. Other standards bodies in addition to the IEEE have a major influence on industry adoption including the International Electrotechnical Commission (IEC), the American National Standards Institute (ANSI), the International Organization for Standardization (ISO), the Society of Automotive Engineers (SAE), European Committee for Standardization (CEN), and a large number of regional or national standards bodies worldwide. These are often closely linked to regulatory bodies at governmental level who will seek to enforce certain standards in legislation at a national level. An example of such a national approach is the use of the “Kitemark” in the UK, which is a product and service quality certification mark defined and authorized by the British Standards Institution (BSI). The role of this mark is to ensure that products sold in the UK conform to certain safety standards – particularly important for safety equipment such as helmets or protective clothing for example.

A nuance in the standards world is the use of the generic term “standard” to encompass not only mandatory activities, but also guidelines and recommended practices. These related standards are not necessarily binding or legally enforced, but rather provide a framework for the implementation of good practice or providing information to consumers to assist them in making informed choices. A good example of this is the recently published IEEE Std 1789 - IEEE Recommended Practices for Modulating Current in High-Brightness LEDs for Mitigating Health Risks to Viewers [1]. This recommended practice provides information that has been obtained by researchers and engineers in this technical area to enable some standardization of terminology and metrics for the comparison of LED lighting products in a sensible manner.

11.4 How are Standards Developed?

It is interesting to note that the vast majority of standards in electronics and electrical engineering were not developed by bureaucrats or legislators, but rather by engineers themselves. These same engineers usually provide their time and expertise as volunteers, working together for the benefit of the engineering community and the overall good of everyone. Of course, it would be naïve to assume that companies do not have a vested interest in ensuring they have an input to standards, however each standards body takes great care to ensure that no one company can dominate the process and there are usually strict rules to ensure that working groups and voting procedure cannot be skewed by one company or industry sector.

The approach within the IEEE to creating a new standard is shown diagrammatically in Figure 1.

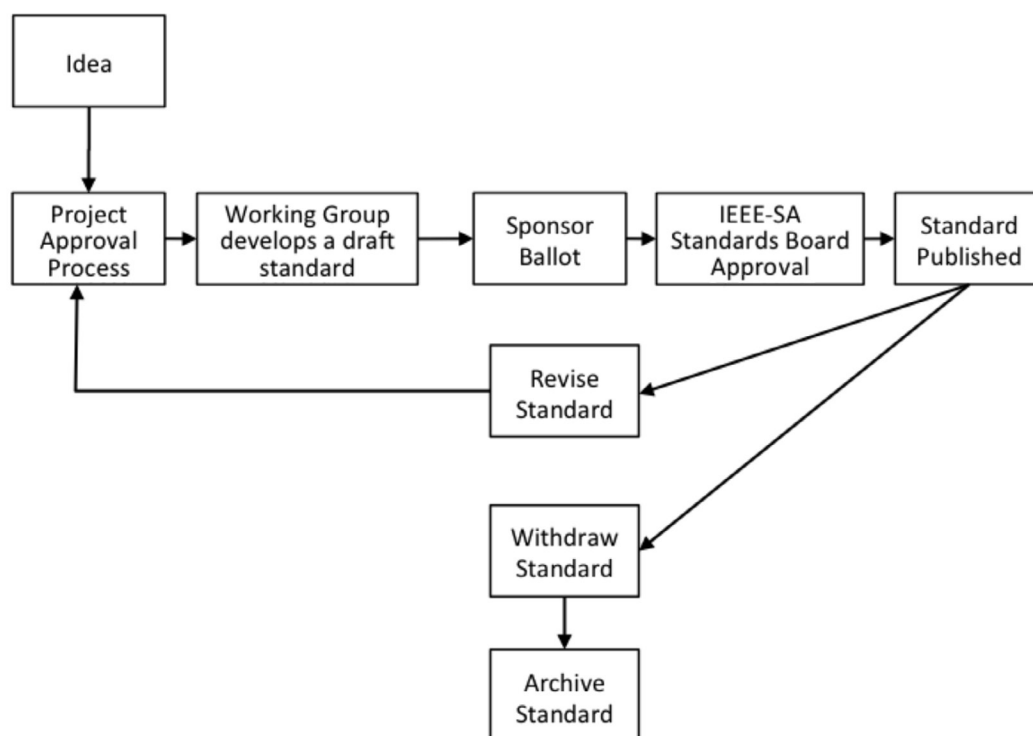


Figure 1: The IEEE standards process.

The first stage in any new standards process is the development of an idea. This can be an individual or a group, however once this has taken place the idea must be worked up into a proposal for a new standard. This can be relatively brief, however the scope of the proposed standard should be clear and the rationale for a new standard should also be clear. At the same time as the proposal has been developed, the appropriate technical society or societies to develop the standard should be identified, and they can shepherd the new draft through the process – this is termed the “sponsor”. This is not mandatory, however it is highly recommended to give the working group exposure among the technical community that is relevant to the proposed standard. At this point, the individual leading the draft standard should then create a “PAR” – Project Authorization Request. This is considered by the IEEE Standards Association and at this stage, if approved, a project number is assigned, which will become the number of the new standard if the process is completed.

At this stage, the working group needs to be formed, and this will generally consist of volunteers in that technical area who are willing and able to contribute to creating the standard draft. This can take quite a significant amount of time, however the overall time for the approval process is 4 years, and so it is not infinite. After the draft has been developed into a state where the working group is happy, a balloting pool can be formed whose members will vote on the new standard. The main role of the ballot is to ensure that a large enough majority of the community finds the new standard acceptable, and that all relevant concerns have been addressed. It is beyond the scope of this article to describe the full details of the process and these can be found on the IEEE Standards Association website [2].

Once the ballot has been passed, then the draft standard is passed to the IEEE Standards board for final approval and editorial review, and once this process has completed, the standard can be published. There is a mandatory period of review and update for the standard after 10 years (this used to be 5 years), at which point the standard can be renewed as is, revised, or withdrawn. If the latter, the standard will then be archived.

11.5 Establishing a Framework of Standard Metrics for Wide Bandgap Technology

One of the major challenges for the power electronics community in the comparison of power electronics devices and systems is being able to have a framework of standard metrics to enable this comparison to take place. The well known “Moore’s Law” is the observation that the number of transistors fabricated in a dense electronic circuit doubles every two years [12]. While this has been useful as a specific metric for the silicon device community that basically establishes a rule of thumb for the “cutting edge” of device technology based on dimension alone, and in fact it has led to a number of related trends in the Silicon world such as power loss, switching speed and complexity, these do not translate directly into the power electronics world and more specifically wide bandgap semiconductors such as silicon carbide (SiC) or gallium nitride (GaN). From the power electronics standpoint a key parameter is the $R_{ds(on)}$ resistance and this provides a suitable measure of the basic device performance in terms of the relationship between $R_{ds(on)}$ and the breakdown voltage. When the curves for silicon (Si), SiC and GaN are compared it can be seen that there is a fundamental measure of the limit for each technology as shown in Figure 2.

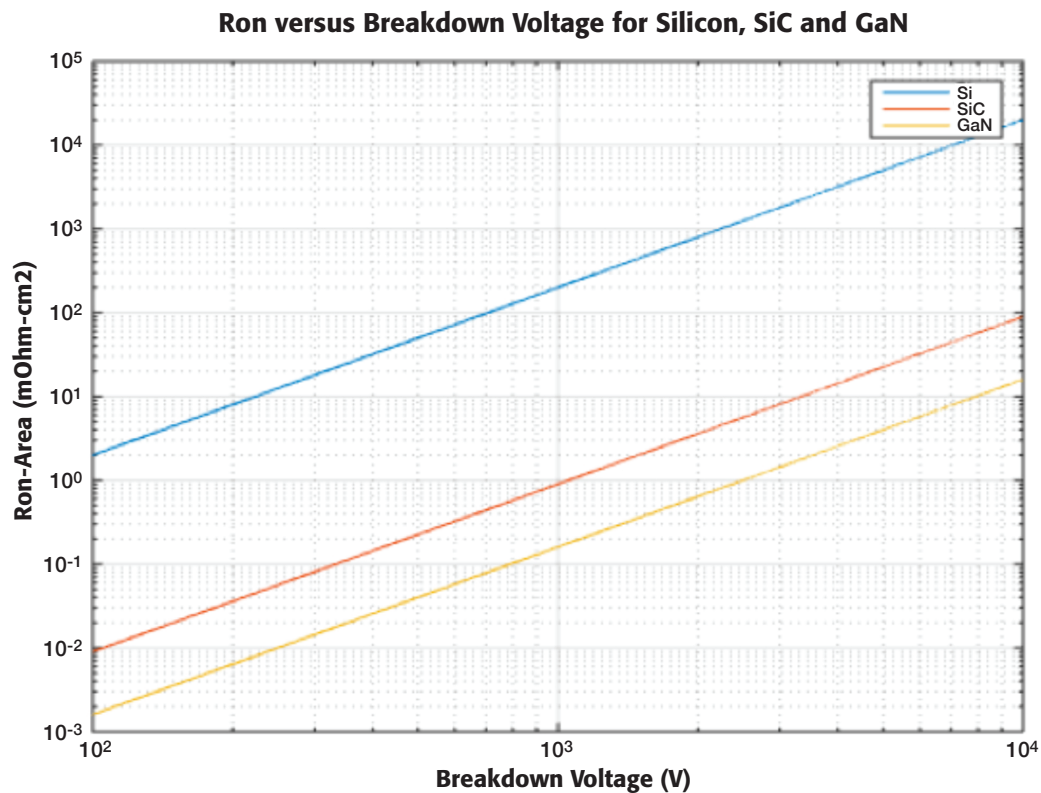


Figure 2: $R_{ds(on)}$ resistance versus breakdown voltage for Si, SiC and GaN technologies.

Useful though this metric is, in fact it is not the complete picture. If we compare the thermal performance of Si and SiC, for example, it is well known that SiC devices can operate at much wider temperature ranges than Si, and as such their range of operation is much wider. Also, their ability to tolerate higher temperatures makes it less important to ensure that the devices are maintained at a lower temperature (as would be the case for Si devices). This, in conjunction with a much lower on resistance (and consequently lower static power loss), makes temperature an orthogonal aspect of the metric framework potentially to be considered. Another issue that is of particular relevance to the power electronics community is that of reliability. Some wide bandgap devices are at a very early stage of commercialization and therefore the technology is immature enough to leave questions over reliability when deployed over a long period.

One of the particular difficulties in establishing this framework is also the wide range of application of each aspect of these devices, for example running at a high temperature, or perhaps extensive periods of high power operation. The other aspect is how can we predict the performance in a particular module, package or system, when those aspects may influence performance equally as much as the device itself? For example, a SiC device may be intrinsically robust,

but the driver may not be, especially if it is integrated using a wire-bonded package.

The role of the ITRW is to establish some of the key criteria in a framework of metrics for wide bandgap power semiconductors, in the context of power electronic systems, to enable specific technical work and standards activities to be undertaken.

11.6 What are Examples of Typical Standards Relevant for Power Electronics?

There are a large variety of IEEE standards for power electronics activities. IEEE Std 1573 [4] - Recommended Practice for Electronic Power Subsystems: Parameters, Interfaces, Elements, and Performance – has been developed for the overall design of power electronics modules and sub-systems and includes guidelines for effective and efficient design of power electronics. This standard provides guidance on interface definitions and application, including parametric values for power electronic subsystems consisting of single or multiple elements. The recommended practice applies to ac-dc and dc-dc electronic power subsystems. The range of power subsystems includes dc, single phase, and three-phase inputs, with elements having power levels from a fraction of a watt to 20 kW. The voltage range is 600 V and below, at a frequency or frequencies of dc -1 kHz (although switching frequencies can of course be much higher than this range). For higher power converters, there is 295-1969 - IEEE Standard for Electronics Power Transformers [5], which relates to sine wave or poly-phase voltages[AM1] , and this will be of relevance to those working in “Grid” related activities.

Of course, there are many specific technical areas where standards must be adhered to and a highly popular area at the moment is that of solar inverters. For example, for a solar inverter to be used in Europe it must conform to the following standards: EN 50524 [6], current version published in 2010 and covers the data sheet and name plate for photovoltaic inverters. EN 50530 [7], also current version published in 2010 defines the methods for testing the overall efficiency of grid connected photovoltaic inverters. IEC 61683, current version published in 1999 [8], defines the procedure for measuring efficiency and power conditioners. What this means in practice is how to assess the power level, input voltage, output voltage, power factor, harmonic content, load non-linearity and temperature. EN62109 [9] deals with the safety of the power conversion equipment for use in PV systems and defines the minimum requirements for protection against fire, mechanical, electric shock and other requirements.

In addition to the need for specific power electronics-related standards, there may also be module level or consumer product standards, particularly for safety. For example, if a dc-dc converter needs to be integrated in a communications system it will need to satisfy the safety requirements in telecommunications [10] or information technology equipment [11].

These standards are simply a fraction of the standards applicable to power electronics systems today, but will give the reader an idea of the scope of individual standards and highlight some of the key issues for power electronics designers.

11.7 Reliability

Another critical issue for WBG devices in power electronics relating to reliability is due to their ability to operate at much higher frequencies than their Silicon predecessors. At higher switching speeds, the intrinsic parasitic inductances can cause a significant problem of transients, which can adversely affect the reliability of the module [13].

The packaging of WBG devices using conventional wire bond techniques is also under scrutiny, with work being undertaken to employ approaches such as flip-chipping that have been in common usage in the high density VLSI industry for many years for power devices [14]. One of the major issues as highlighted already is that of both thermal management and lead inductance and this has been investigated extensively in last several years with some interesting solutions being proposed to integrate thermal structures into the package during the assembly process [15] to provide direct thermal links to the device inside the module package.

In order to understand the details of the effect on reliability it is essential to break down the individual mechanical elements and estimate their impact on reliability accelerators, in particular thermal effects. The work in [16] provides

a detailed insight into this and a comparison of IGBTs and SiC-MOSFETs that shows a potentially significant increase in both efficiency and reliability as a result of lower losses (and therefore heat) in the SiC device.

11.8 JEDEC

For over 50 years, JEDEC has been the global leader in developing open standards and publications for the microelectronics industry. JEDEC committees provide industry leadership in developing standards for a broad range of technologies. Current areas of focus include:

- Main Memory: DDR4 & DDR5 SDRAM
- Flash Memory: SSDs, UFS, eMMC
- Mobile Memory: LPDDR, Wide I/O, Memory MCP
- Memory Module Design File Registrations
- JEP30: Part Model Guidelines
- Registered Outlines: JEP95
- Memory Configurations: JESD21-C
- Lead-Free Manufacturing
- ESD: Electrostatic Discharge

JEDEC is the global leader in developing open standards for the microelectronics industry, with more than 3,000 volunteers representing nearly 300 member companies.

JEDEC brings manufacturers and suppliers together to participate in more than 50 committees and subcommittees, with the mission to create standards to meet the diverse technical and developmental needs of the industry.

JEDEC's collaborative efforts ensure product interoperability, benefiting the industry and ultimately consumers by decreasing time-to-market and reducing product development costs.

JEDEC publications and standards are adopted worldwide. JEDEC is accredited by ANSI and maintains liaisons with numerous standards bodies throughout the world.

11.9 Conclusions

The role of standards in power electronics systems has become revitalized with the development of major steps forward due to wide bandgap power semiconductor devices, particularly in SiC and GaN. The step change in performance and thermal tolerance in particular has led directly to the need for new standards and definitions to be produced, both for researchers and industry. It is an exciting time to be in the power electronics field and there are great opportunities for the power electronics community to come together to define a standard approach to manage the adoption of these exciting new technologies.

11.10 References

Note: not all standards documents are freely available

- [1] IEEE Std 1789 - IEEE Recommended Practices for Modulating Current in High-Brightness LEDs for Mitigating Health Risks to Viewers, 2015, <https://standards.ieee.org/findstds/standard/1789-2015.html> [last accessed 11/6/2016]
- [2] IEEE Standards Association Frequently Asked Questions, <https://standards.ieee.org/faqs/>
- [3] ITRS documents 2001-2015, <http://www.itrs2.net/itrs-reports.html>

- [4] P1573 - Recommended Practice for Electronic Power Subsystems: Parameters, Interfaces, Elements, and Performance, <https://standards.ieee.org/develop/project/1573.html>
- [5] 295-1969 - IEEE Standard for Electronics Power Transformers, <https://standards.ieee.org/findstds/standard/295-1969.html>
- [6] EN 50524 - Data sheet and name plate for photovoltaic inverters, <http://shop.bsigroup.com/ProductDetail/?pid=000000000030183362>
- [7] EN 50530 - Overall efficiency of grid connected photovoltaic inverters, <http://shop.bsigroup.com/ProductDetail/?pid=000000000030270551>
- [8] EN 61683 - Photovoltaic Systems – Power Conditioners – procedure for measuring efficiency, https://webstore.iec.ch/preview/info_iec61683%7Bed1.0%7Den.pdf
- [9] EN 62109 - Safety of power converters for use in photovoltaic power systems - Part 1: General requirements, <https://webstore.iec.ch/publication/6470>
- [10] ETS 300 132-2 “Environmental Engineering (EE); Power supply interface at the input to telecommunications and datacom (ICT) equipment; Part 2: Operated by -48 V direct current (dc)”, <http://www.etsi.org>.
- [11] IEC 60950-1, “Information technology equipment - Safety”, http://ulstandards.ul.com/standard/?id=60950-1_1
- [12] Gordon Moore, “Cramming more components onto integrated circuits”, *Electronics*, 1965
- [13] E. Velander, L. Kruse, S. Meier, A. Lfgren, T. Wiik, H. P. Nee, and D. P. Sadik, “Analysis of short circuit type ii and iii of high voltage sic mosfets with fast current source gate drive principle,” in 2016 IEEE 8th International Power Electronics and Motion Control Conference (IPEMC-ECCE Asia), May 2016, pp. 3392–3397.
- [14] S. Seal, M. D. Glover, A. K. Wallace, and H. A. Mantooth, “Flip-chip bonded silicon carbide mosfets as a low parasitic alternative to wire-bonding,” in 2016 IEEE 4th Workshop on Wide Bandgap Power Devices and Applications (WiPDA), Nov 2016, pp. 194–199.
- [15] N. Zhu, M. Chen, D. Xu, H. A. Mantooth, and M. D. Glover, “Design and evaluation of press-pack sic mosfet,” in 2016 IEEE 4th Workshop on Wide Bandgap Power Devices and Applications (WiPDA), Nov 2016, pp. 5–10.
- [16] J. Rice and J. Mookken, “Economics of high efficiency sic mosfet based 3-ph motor drive,” in PCIM Europe 2014; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management, May 2014, pp. 1–8.

Chapter 12: Future Challenges/Future Vision

12.1 Future Challenges

The International Technology Roadmap for Wide Bandgap Power Semiconductors (ITRW) has four distinct technical working groups each providing their own perspective on the technology and industrial drivers for the adoption of wide bandgap (WBG) power semiconductors in the sphere of power electronics. To understand the perspective of each of these distinct communities, a series of standard questions were asked of each and the responses captured in this chapter of the ITRW roadmap.

The questions were designed to gain clear opinions and are listed below:

- How should your area advance in the next 5, 10 and 20 year timeframes?
- What major roadblocks do you foresee? Any technological issues?
- Name one killer application for your area? (Opinion)
- Any specs that you see changing or are needed to advance your area?
- How disruptive will these new technologies be? What impact will they have? What is the broader impact?
- How do you see the cost structure changing?

Clearly, these questions are technological in nature, however they are useful in that they force all the working groups to provide a common set of responses to the same questions enabling common and distinct answers to be obtained and identified.

12.2 Future Vision – Working Group Opinion

ITRW SiC Materials and Devices

How should your area advance in the next 5, 10, 20 years?

Next 5 years

- 200 mm SiC wafer material will be commercially available. Defects and wafer planarity will still need improvement. Foundries will process 200 mm SiC wafers in their Si lines for economy of scale.
- The price of SiC MOSFET devices will be within 2 times that of its similarly rated Si counterparts. SiC devices will be inserted in 10% EV/HEV, and PV as the two top applications. 3.3 kV SiC MOSFET devices will become commercially available. 6.5 kV SiC MOSFET devices will be close to commercialization.

Next 10 years

- 200 mm SiC wafer material defects and planarity will have improved and not be a major issue. Several foundries will be processing in their SiC/Si lines. Financially viable dedicated SiC foundries will start to appear.
- The price of SiC MOSFET devices will be within 1.5 times of that of its similarly rated Si counterparts. SiC devices will capture ~15% of the total power device market. 6.5 and 10 kV SiC MOSFET devices will become commercially available. SiC superjunction MOSFETs will be close to becoming commercially available.

Next 20 years

- 200 mm SiC wafer material defects and planarity will be at Si levels. SiC wafers for power will be of same area as Si, likely 300 mm. SiC devices will be in production in dedicated high volume SiC foundries.
- SiC MOSFET devices will be comparable in price to Si. SiC will capture 40% of the power device market. SiC superjunction and 15–25 kV SiC IGBTs will be commercially available.

What major roadblocks do you foresee? Any technological issues?

Roadblocks:

Cost. Will come down as wafer quality improves and volume production is in place.

Ruggedness is inferior to comparable Si parts; requires design trade-offs.

Workforce not trained in SiC use limits adoption and is detrimental to volume production cost reductions; educational initiatives needed.

Technological issues

Lack of planarity and high defect density of SiC wafers complicate fabrication.

Lack of room temperature implantation, increases labor costs

Low quality gate oxide reduces mobility and leads to threshold instability

Name one killer application for your area? (Opinion)

EV/HEV

Any specs that you see changing or are needed to advance your area?

- Wafer planarity approaching that of Si.
- Defect density approaching that of Si.
- Device Ruggedness and Safe Operating Area similar to that of Si.
- Room temperature implantation in fabrication becomes available.
- Gate oxide improves allowing for higher mobility and threshold stability.

How disruptive will these new technologies be? What impact will they have? What is the broader impact?

- Strongly competing with GaN and Si in 600-900 V, not disruptive but competitive. Simplification due to high frequency and reduced cooling requirements.
- SiC devices will be the mainstream choice in 1.2-6.5 V for power conversion applications displacing Si. Disruptive will be circuit architectures enabled by the efficient high voltage capability and high frequency.
- SiC discrete devices capable of 10-50 kV will be disruptive in enabling novel circuit architectures and eliminating the multiple level topologies in Si.
- Impact is novel circuit topologies with lower complexity, new opportunities due to lower volume/weight/reduced cooling high voltage power electronics, and higher efficiency.
- Broader impact is manufacturing job creation, widespread novel high voltage applications, and a cleaner planet due to lower emissions (higher efficiency).

How do you see the cost structure changing?

- Cost reduction with SiC wafers of larger area
- Cost reduction with device fabrication in high volume foundries, and process streamlining to include room temperature implantation

ITRW GaN Materials and Devices

How should your area advance in the next 5, 10, 20 years?

Next 5 years

GaN power HEMTs in commercial production on 8" Si wafers processed on CMOS-compatible lines. Normally-off operation achieved by techniques such as p-type gates at the device level or cascode configuration at the package level. Voltage ratings up to 900 V available and in competition with Si and SiC devices at comparable cost. Research on devices with higher voltage ratings in progress, likely up to 1200 V, in competition with SiC. Research ongoing on package-level gate drive integration (heterogeneous), as well as chip-level fully integrated gate drives (homogeneous small-scale ICs). Research also ongoing on reliability issues such as dynamic Ron, gate breakdown, etc. Circuit designs are adapted to account for extremely fast switching (high dI/dt and dV/dt) using "quasi-RF" distributed circuit concepts. For higher voltages, basic research on vertical GaN materials and devices occurs, with a focus on native GaN substrates, growth of thick, low-doped drift regions, and effective processing techniques.

Next 10 years

GaN power HEMTs with voltage ratings ≤ 900 V are fully mature and capture significant market share, disrupting Si dominance. It is unlikely that Si disappears, rather each material is used as appropriate depending on the application. Cost of GaN is less than that of Si on a per-Amp basis. CMOS-compatible fabrication on Si wafers continues, with the wafer size dependent upon the status of the Si industry (i.e. the relative numbers of 8" vs. 12" fabs). Fabrication of GaN power HEMTs on other large-area substrates (e.g. those that better match the coefficient of thermal expansion of GaN) may be widespread. Reliability issues are largely solved, and circuit designers effectively design using GaN devices. "Quasi-RF" circuit design techniques are fully adopted to maximize the utility of GaN compared to Si. Heterogeneously and homogeneously integrated gate drives are available and widely utilized. Power HEMTs up to 1200 V are available, although market adoption is difficult to predict especially as SiC MOSFETs are available in the 650-1200 V range. Initial availability of vertical GaN devices for ≥ 1200 V voltages.

Next 20 years

Use of GaN power HEMTs for applications ≤ 650 V is standard, and circuit designers view GaN as the default for these applications. Optimal "quasi-RF" circuit design techniques are standard. Reliability is well established, and devices are robust and over-design is minimal. Cost is well below that of Si. Si substrate size still depends on the status of the Si industry. Alternate substrates are in use as well, such as SiC and sapphire. GaN power HEMTs with voltage ratings between 650 and 1200 V are commercially available, but market share is partial. Above 1200 V, vertical GaN devices are available, but market share is still relatively small (shared with SiC). Both vertical GaN and SiC may face a challenge at very high voltage ratings by "ultra" WBG materials such as Gallium Oxide, Diamond, and Aluminum Nitride, which are becoming viable.

What major roadblocks do you foresee? Any technological issues?

Roadblocks:

Limited breakdown voltage: GaN power HEMTs are generally limited in breakdown voltage to < 900 V due to the lateral device structure, although higher breakdown voltages have been reported in the literature. Vertical GaN devices should remove this limitation.

Reliability: Circuit designers need to be confident in the long-term reliability of GaN power devices. If they are not, adoption will be limited and slow.

Extremely fast switching: While the extremely fast switching of GaN power HEMTs is of course desirable, it exacerbates issues such as parasitics and EMI. "Quasi-RF" distributed circuit design techniques borrowed from the RF community will need to be employed.

Circuit design community: Circuit designers need to be familiar with GaN devices and need to design their circuits in a manner consistent with GaN properties. Education is key in achieving this.

Technological issues

Threshold voltage: GaN HEMTs are naturally normally-on, which is undesirable because power electronics engineers primarily require normally-off devices for safety reasons. While various approaches have been implemented to create normally-off devices, this is an on-going challenge.

Ruggedness: GaN power HEMTs do not exhibit avalanche breakdown and therefore do not possess avalanche ruggedness. This may result in their over-design, which limits the efficiency of the device, and may also limit their adoption in certain applications.

Reliability: Various degradation mechanisms exist in GaN power HEMTs such as dynamic on-resistance, threshold voltage shift, and time-dependent dielectric breakdown, and these may depend on the specifics of the epitaxial growth and device processing. Despite extensive research on this topic, work remains to be done to fully understand these mechanisms.

Vertical GaN: There is a lack of high-quality, large-area, low-cost substrates. The epitaxial growth of thick, low-doped drift regions is challenging, and difficulties in device processing (e.g. ion implantation) at present limit the number of available device architectures.

Name one killer application for your area? (Opinion)

Low-voltage power supplies for cell phones, laptops, and similar devices will be a very large market. However, a truly “killer app” is likely point-of-load power conversion for data centers and other computing applications, which could result in the explosion of GaN utilization.

Any specs that you see changing or are needed to advance your area?

- Reduced defect density in epitaxial growth
- Robust normally-off devices
- Device ruggedness and safe operating area comparable to that of Si
- Comprehensive understanding of reliability
- For vertical GaN, the availability of high-quality, large-area, low-cost substrates; the ability to grow thick, low-doped drift regions; and the ability to process a full range of device structures

How disruptive will these new technologies be? What impact will they have? What is the broader impact?

In the long run, GaN power HEMTs will significantly (though perhaps not completely) displace Si for numerous applications, especially those ≤ 900 V. Circuit design for power conversion will likewise be revolutionized by the need to deal with the extremely high switching frequencies enabled by GaN. Much as GaN has already revolutionized lighting and RF electronics, GaN will revolutionize power electronics. Highly efficient and compact GaN power converters will significantly impact a broad range of applications, from data centers to electric vehicles. This will complement the revolution in power electronics occurring due to parallel developments in SiC. The improvements in power conversion enabled by GaN will be necessary to fully move to a clean energy economy. This technology in itself will not only mitigate the environmental concerns, but will also underpin the significant economic growth enabled by these clean energy technologies. Significant high-quality manufacturing jobs will be created due not only to the production of the GaN technology itself, but also due to the broader new industries that it enables.

How do you see the cost structure changing?

The cost structure for GaN power HEMTs is already competitive due to the use of Si substrates and CMOS-compatible device processing. As device design improves, the cost on a per-Amp basis (i.e. normalized to die area) will continue to fall, and the cost will become significantly lower than that of Si.

System Integration and Application: SiC

How should your area advance in the next 5, 10, 20 years?

SiC will have large market shares in automotive, aviation, railway, industry drives, renewable energy and utility-based power electronics applications such as solid-state transformers.

What major roadblocks do you foresee? Any technological issues?

The major concerns and technological challenges are application specific qualifications, reliability, short circuit capability, and very importantly high temperature packaging.

Though high dv/dt and di/dt present as current challenges to power converter and electric machine designs, these challenges most possibly will gradually vanish.

Name one or two killer application(s) for your area? (Opinion)

- Traction drive inverters for electric vehicles for 900 V to 1.2 kV rated SiC devices
- Traction systems for high speed trains for 3.3 kV to 6.5 kV rated SiC devices
- In a long run, aviation industry will move from more electric aircrafts (MEA) to hybrid electric propulsion. SiC based motor drives with power rating ranging from tens of kilowatt for MEA to megawatt for hybrid propulsion will become popular.

Any specs that you see changing or are needed to advance your area?

Short circuit capability specification is based on motor drives with IGBTs and MOSFETs. With more reliable fast detection and protection methods, the short circuit time for SiC devices need to be revisited.

How disruptive will these new technologies be? What impact will they have? What is the broader impact?

SiC in above-mentioned applications will increase system efficiency thus lower operation costs. For electric vehicles, the increased efficiency of traction systems will also reduce the size of batteries onboard the vehicles.

How do you see the cost structure changing?

Cost reductions of SiC chips will happen gradually with more field implementations. But it would be system level cost saving including savings on both the capital cost and operation cost will make SiC more appealing.

System Integration and Application

How should your area advance in the next 5, 10, 20 years?

The power system will at least find SiC application in such demonstrations projects in less than 5 years as power electronics transformers, solid static substations or flexible substations.

PET or similar installation shall begin their massive SiC application in less than 10 years provided that

- 1) the penetration of various distributed generators in the distribution grid is comparative high;
- 2) high capacity SiC devices are available and not prohibitively high-cost;

The application of SiC devices will go high voltage, such as VSC-HVDC and FACTS in 20 years.

What major roadblocks do you foresee? Any technological issues?

The major roadblock for utility application is that there is no large SiC device (>10kV and >1kA) available at the moment.

Name one killer application for your area? (Opinion)

The killer application should be PET. There are many different names for it such as flexible substation, solid static substation.

Any specs that you see changing or are needed to advance your area?

There is no good suggestion at the moment.

How disruptive will these new technologies be? What impact will they have? What is the broader impact?

High operation frequency and low power losses resulted from SiC application will make the PET feasible from the engineering point of view.

How do you see the cost structure changing?

- The Wafer cost is about 50% for the SiC device, it will increase its percent as related technology improves and experiences is accumulated.
- Another factor is system design to take full advantages of SiC devices.

System Integration and Application: GaN

How should your area advance in the next 5, 10, 20 years?

In the next 5 years, GaN will have large market penetration in LIDAR, data center, consumer power electronics (charger and adapter), and drones.

In the next 10 years, GaN will be used in motor controls, EV – auxiliary power supplies, rooftop PV inverter, wireless power transfer, EV onboard charger, and aerospace;

In 20 years, GaN will see application in medical devices, and monolithic integration

What major roadblocks do you foresee? Any technological issues?

- Cost
- Reliability
- Supply chains
- Packaging challenges, esp. at higher power
- Thermal management (higher heat flux)

Name one or two killer application(s) for your area? (Opinion)

- 1) Charger in all types of applications including consumer power electronics, EV and aerospace. The low loss, high switching capability of GaN will help improve efficiency, size and overall cost.
- 2) PV inverter. The low loss, high switching capability of GaN will reduce loss, filter, and cooling, resulting in improved efficiency, size, and cost.

Any specs that you see changing or are needed to advance your area?

- Better model
- Better/cheaper instrumentation (probes, scopes, sensors)
- Faster and cheaper controllers
- Better magnetics
- Better packaging and thermal management
- More work on reliability testing and characterization (including radiation, cosmic ray characterization)

How disruptive will these new technologies be? What impact will they have? What is the broader impact?

Some of the technologies can be considered disruptive, as they are critical enablers for mass applications, for example EV and other future electrified transportation (e.g. electrified airplanes, drones). On border impact, the technologies will change how things are run and how people live, also have environmental impact.

How do you see the cost structure changing?

Similar to SiC, cost reduction of GaN will happen with increased mass adoption. On the other hand, the converter and system level cost benefit vs. Si can already be seen in some applications, e.g. charger and PV inverter. One key issue is to gain experience on reliability (both for device and converter).

ITRW Packaging and Integration

How should your area advance in the next 5, 10, 20 years?

5 years

Pressure from users for familiar packages in a like-for-like replacement format will continue to drive the bulk of packaging for modules (with consequent limitations in performance) and for discrete dies. Innovation in packages for discrete dies and modules, mainly for niche applications, will see more widespread introduction of direct bonded Cu foil as an interconnect, replacing wire bonds and providing lead-out connections. Integration of additional functionality will be limited to small-value ceramic decoupling capacitors, sensors and potentially gate drives (subject to temperature constraints). Chip-direct packages will be a focus for experienced users of GaN devices at lower power levels and for higher frequency switching. It is unlikely that there will be widespread use of chip-direct packages for SiC.

10 years

Further evolution of conventional packaging solutions will focus on achieving low-inductance through planar interconnects and alternative terminations. Integration of EMI suppression features, decoupling capacitors, sensors and gate drives will become more widespread, making use of adapted conventional assembly methods. There will be niche application of converter-cell architectures for high frequency switching, assembled using a mixture of power module and standard PCB processes. Embedded-die, integrated solutions will start to make an impact at low power (< 10 kW) levels, delivering an off-the-shelf solution at low cost. Specialist packages will emerge to support high-voltage SiC > 10kV MOSFET/IGBT applications and areas such as integrated drives where specific geometry, termination, cooling and environmental requirements are dominant.

20 years

Although conventional packaging will remain available and in widespread use for many legacy applications, there will be a strong push towards more integrated products, providing the end-user with an easy-to-use converter-cell solution. The final form of such products and their impact on the user community will pivot on the successful introduction of low-cost, volume scalable manufacturing routes and new materials that deliver high-performance, robust solutions. These may be based on additive or layer-based approaches, ultimately delivering complex 3D architectures with minimum volume, minimal use of resources and simplified recycling routes.

What major roadblocks do you foresee? Any technological issues?

Incompatibility of materials will remain the principal limiting factor, in particular mismatch in mechanical properties such as coefficient of thermal expansion. Manufacturing processes and designs can be optimized to overcome some of these limitations but the discovery and development of new materials will remain a major research objective. Integration at converter cell level could also be limited by lack of availability of suitable control elements (high temperature, high speed) and passive components, especially magnetics.

Name one killer application for your area? (Opinion)

Strictly no one killer application as all will require suitable packaging. The evolution of integrated solutions will

likely be driven by requirements for low-cost, easy-to-use, power-dense converter cells across a broad spectrum of application areas.

How disruptive will these new technologies be? What impact will they have? What is the broader impact?

Improved solutions for packaging and integration will unlock the full potential of wide bandgap devices through radically new approaches to integration, moving from assemblies of discrete components, to fully integrated assemblies comprising power devices, gate drives, filters, sensing, and control functions. Adoption of new materials and differentiated manufacturing routes will result in converter-cell solutions, which deliver higher efficiency, greater power density, ease-of-use and ease-of-recycling with an attractive cost-performance trade-off.

How do you see the cost structure changing?

Incremental reductions in cost will continue as existing technologies are adapted and developed. A step-change reduction in costs will be the ultimate outcome of the successful development and deployment of new manufacturing routes for converter-cell, integrated products.

12.3 Discussion of Future Vision Questions and Responses

In this section the responses from the working groups will be considered together and some conclusions drawn.

12.3.1 How should your area advance in the next 5, 10 and 20 year timeframes?

It was clear that the SiC devices and applications groups saw a trend towards higher voltage and power applications, with ever increasing power requirements, going hand in hand with greater production driving costs down. Key applications being the major market of transport electrification. In contrast, the perspective of the GaN community was in a variety of lower voltage markets mainly focused on consumer, data based platforms and charging. The packaging and integration team saw a demand for like-for-like replacement in the short term and moving towards much deeper levels of integration in the future and a drive towards new manufacturing techniques to support that greater level of integration. SiC was seen as becoming dominant in transport electrification, and utility-based applications, in the long term.

12.3.2 What major roadblocks do you foresee? Any technological issues?

Incompatibility of different materials was seen as a particular problem with new materials coming on stream and heterogeneous packaging having to manage this issue. Cost in the short run and also thermal densities were seen as important issues in most of the work groups. A particular issue for higher-level integration was the availability of suitable passives as the frequencies and power densities increase.

12.3.3 Name one killer application for your area? (Opinion)

There were a number of applications and potential “killer” applications identified by the working groups and these are given below. It was clear that some areas emerged as being of particular interest across the board and these are electric vehicle drives, chargers, and PV inverters. It was interesting that wind based renewable or conventional energy systems were not listed specifically, and perhaps this recognizes the sheer scale of the automotive sector in terms of potential market, although the opportunity for grid integration of SiC was identified if suitable high voltage (>10kV) devices could be eventually fabricated.

- Traction drive inverters for electric vehicles for 900 V to 1.2 kV rated SiC devices
- Traction systems for high speed trains for 3.3 kV to 6.5 kV rated SiC devices
- In a long run, aviation industry will move from more electric aircrafts (MEA) to hybrid electric propulsion.
- SiC based motor drives with power rating ranging from tens of kilowatt for MEA to megawatt for hybrid propulsion will become popular.

- Strictly no one killer application as all will require suitable packaging. The evolution of integrated solutions will likely be driven by requirements for low-cost, easy-to-use, power-dense converter cells across a broad spectrum of application areas.
- Charger in all types of applications including consumer power electronics, EV and aerospace. The low loss, high switching capability of GaN will help improve efficiency, size and overall cost.
- PV inverter. The low loss, high switching capability of GaN will reduce loss, filter, and cooling, resulting in improved efficiency, size, and cost.
- Low-voltage power supplies for cell phones, laptops, and similar devices will be a very large market. However, a truly “killer app” is likely point-of-load power conversion for data centers and other computing applications, which could result in the explosion of GaN utilization.
- EV/HEV

12.3.4 Any specs that you see changing or are needed to advance your area?

There were some specific tests, such as short circuit tests, especially for devices that will be direct replacements for Silicon MOSFETs or IGBTs. New technology devices will also need additional testing for reliability and basic behaviour as the speed increases.

12.3.5 How disruptive will these new technologies be? What impact will they have? What is the broader impact?

Adoption of new materials and differentiated manufacturing routes will result in converter-cell solutions, which deliver higher efficiency, greater power density, ease-of-use and ease-of-recycling with an attractive cost-performance trade-off.

Some of the technologies can be considered disruptive, as they are critical enablers for mass applications, for example EV and other future electrified transportation (e.g. electrified airplanes, drones). On border impact, the technologies will change how things are run and how people live, also have environmental impact.

SiC in above-mentioned applications will increase system efficiency thus lower operation costs. For electric vehicles, the increased efficiency of traction systems will also reduce the size of batteries onboard the vehicles. Much as GaN has already revolutionized lighting and RF electronics, GaN will revolutionize power electronics. Highly efficient and compact GaN power converters will significantly impact a broad range of applications, from data centers to electric vehicles. This will complement the revolution in power electronics occurring due to parallel developments in SiC.

12.3.6 How do you see the cost structure changing?

A common theme is the reduction in costs as wafer sizes become larger and the cost of individual die reduces. It is true to say that the fixed cost for SiC wafers are still much higher than Silicon, however the overall costs reduce dramatically as the wafer size moves from 200mm to 300mm.

Chapter 13: Roadmap Pathways

13.1 Introduction

The mechanism by which the ITRW can facilitate an acceleration of the R&D process is to signpost paths of technology development, using a series of cause and effect events that lead to a specific destination as shown in Figure 13.1. Providing a list of metrics and benchmarks is not enough in itself, as this is essentially static, and does not describe adequately the path to the desired outcomes. The participants in the ITRW process come from the world leading R&D groups that intimately involved in the dynamic path that the technology development has and will be following.

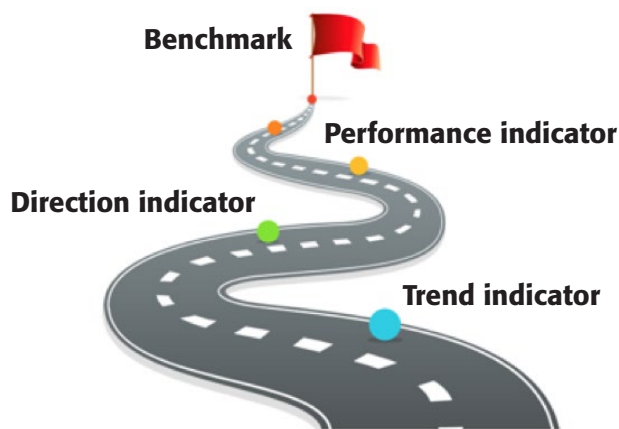


Figure 1: Roadmap Pathways and Indicators

As active experts, we can try to map some future paths, taking history and experience into account, and noting carefully current developments. These dynamic paths have a series of indicators based on the horizon distance, with short-term performance indicators and benchmarks being appropriate for existing products and devices, particularly useful for independent and objective comparison of technology. Direction indicators of medium term innovations of proven research show how these technologies could transition into the marketplace and become products of the future. Finally, indicators that are more “blue sky” in nature show longer term trends and the extent of the research landscape and highlight the potential for research that could lead to exciting new technology in the future.

This chapter will highlight the roadmap paths identified by the ITRW working groups, defining benchmarks, performance, direction and trend indicators.

Some examples are given, however they could definitely do with some tangible numbers to quantify the paths. Can we apply historical data to look at specific trends?

13.2 Path of Wide Bandgap Power Semiconductors Metrics

Benchmark

Obviously comparing devices is controversial, however being able to show the development of devices over time is important. Key aspects include not only fundamentals such as the breakdown voltage (and others in the performance indicators), but combined figures of merit (FOM), which relate to power density.

Performance Indicator

Specific Performance Indicators could include:

- Breakdown Voltage
- $R_{ds(on)}$

- dV/dt
- Power Density
- Thermal Capacity
- Peak Current
- Maximum Frequency

Direction Indicator

The drivers from different applications will push the direction of devices in different ways, therefore a number of application specific direction indicators are required, however being able to categorize the best in class devices across a number of indicators would be helpful.

Trend Indicator

Are there fundamental limits that can never be breached?
Can they be “bent” – such as the Silicon “limit”

13.3 Path of SiC Schottky Diodes

Benchmark

One of the trailblazer parts of WBG power electronics – how could these be benchmarked? Do we have historical data?

Performance Indicator

Specific Performance Indicators could include:

- Breakdown Voltage
- On resistance
- Thermal conductivity
- turn-on voltage
- recovery time
- junction capacitance

Direction Indicator

The drivers from different applications will push the direction of devices in different ways, therefore a number of application specific direction indicators are required, however being able to categorize the best in class devices across a number of indicators would be helpful.

Trend Indicator

Where are these devices going.

13.4 Path of SiC Robustness

Benchmark

Device FOM in combination with circuit FOM.
Proof by technology demonstrators.

Performance Indicator

Reasonable targets of WGB robustness, that may not necessary need to on par with Si, but overall converter robustness should meet requirements. Some specific tests using standard test configurations would provide metrics.

1. High Temperature Reverse Bias (HTRB): measurement of drain leakage current, I_{DSS}
2. High Temperature Gate Bias (HTGB): measurement of gate leakage current, I_{GSS}
3. Negative Bias Temperature Instability/ Positive Bias Temperature Instability (NBTI/PBTI) to spec'd gate bias extremes ($V_{GS_{op}}$); need to specify conditions: (threshold voltage)
4. Output Characteristics (I_D vs. V_{DS} step V_{GS} in 2V steps) of drain family vs. temp. up to rated T_{jmax}
5. High Humidity High Temperature Reverse Bias (H3TRB) – as per JEDEC: JESD-22
6. Short Circuit Tests
7. Include Body diode (forward characteristics) and unclamped inductive switching tests for devices.

Direction Indicator

After many years of technology improvement the overvoltage breakdown and overcurrent of Si devices are at a high level.

Achieving comparable robustness is currently a bottleneck for WBG devices.

Trend Indicator

Additional protection devices are included in the current WBG converter to compensate for the lack of robustness. Also, other circuit topologies seem to work better with WBG, for example quasi resonant circuits, and the circuit could take stresses and instead of relying on very robust devices.
Reliability studies are starting to be published

13.5 Path of GaN Heterogeneous Integrated Circuits

Benchmark

- Power densities eg. for benchmark 400V GaN converters.

Performance Indicator

- Power density of converters for predefined thermal and EMC environments.
- Capacitance.

Direction Indicator

High-speed microprocessors stimulated multilayer PCB technology and automotive converters stimulated the introduction of more copper in PCBs.

In the same way GaN power can stimulate 400V embedded PCB technology.

Trend Indicator

Reduction of circuit parasitics are essential if one wants to exploit the very high frequency properties of GaN power devices.

As a converter examples of many GaN device companies employ PCB embedded technology. GaN seems to have a definite competitive edge over Si in 400V very high frequency converters

13.6 Path of Power Electronic Transformers

Benchmark

A conventional transformer with the similar power capacity and voltage class could become a shadow benchmark.

Performance Indicator

High power conversion efficiency, high power density or small footprint, high operation reliability, reasonable overload capability, acceptable cost and maintenance requirements.

Direction Indicator

Multi physics in the overall assembly is important. The system integrators want more control over shape, size and stresses. Modular design with essential and reliable status indicators is preferred. Achieving comparable operation reliability is currently a bottleneck even for Si devices.

Trend Indicator

Several PET demonstration projects began to accumulate their experiences. High power high frequency transformers are essential if one wants to make the most of the high voltage SiC devices. A Si and SiC hybrid topology could be a transitional choice.

13.7 Path of Integrated Power Electronics for Integrated Power Electronics and Modules

Benchmark

A FOM could be developed based on performance indicators. The level of design automation could become the benchmark. Low-cost flexible manufacturing also.

Benchmark Metrics:

- Power Density
- Thermal Resistance
- Loss density

Performance Indicator

Specific Performance Indicators could include:

- Thermal resistance
- Parasitic inductances
- Substrate to heatsink capacitance.

Direction Indicator

Multi physics in the overall assembly is important with integrated drive systems. The system integrator want more control over shape, size and stresses. Not just drives but a host of integrated solutions where added functionality/customization is enabled by flexible manufacturing and integrated design.

Trend Indicator

A number of automotive drivetrain manufacturers already own their module designs.

13.8 Paths for Simplifying the Engineering Effort of Custom Design

Benchmark

The development cost or time to do a custom design. This can be reduced by introducing design automation platforms or defining power electronic building blocks. The ability to build integrated power electronic circuits is an important technology enabler.

Performance Indicator

The design complexity cannot be passed on the design engineer. You should not need a PhD to design a WBG circuit.

Direction Indicator

Integrated gate drive circuits are introduced in some cases; new EMI filter designs are introduced; GaN device manufacturers provide reference designs; SiC devices are slowed down to reduce over short and EMI.

Trend Indicator

Wide bandgap power electronic circuits are more complex, due to (1) the fast switching, (2) complex gate drive circuits and (3) vulnerability of the devices to overvoltage and over current.

Chapter 14: Conclusions

The ITRW effort has received enthusiastic participation; very successful in engaging the community. The engagement has been global with well attended meetings across the world, including Europe, North America, and Asia.

ITRW has engaged academia and industry in shared goals; industry advisory board brought leading companies together in a non-competitive setting, with collaborative work and exchange of good practice being presented.

ITRW has clearly shown its potential and expectation to provide guidance; a process has been set in motion. A key aspect of this has been the engagement with the stakeholders in the power electronics community and how it has influenced the discussion of wide bandgap power electronics and its adoption.

ITRW has Identified weaknesses to be addressed for example the need to better embed expected module performance/ specification in applications.

The ITRW roadmap has provided examples of “best in class” and state of the art designs in the included case studies. These demonstrate the impact of wide bandgap power electronics in both academia and industry.

ITRW has benefitted from the large industry presence at APEC and strengthened the hosting workshop WiPDA resulting in a second yearly mirrored WiPDA in Asia.

ITRW has shown that wide bandgap power electronics is taking off, some devices are reliable and good enough and WBG receive growing orders. However, at the same time reliability and lifetime needs to be improved and improved to before large scale adoption becomes a reality.

Appendix A: Product Showcases

SiC based 6500-V/200-A SuperCascode Module

- 200A - 6500V Half-bridge Module
- Uses 5.7 mOhm, 1700V SiC JFETs (lowest Rds single 1700V chip in the world!)
- Standard Silicon Gate Drive (0-12 V) and 0.7 V Diode knee
- Applications include Solid State Transformers, MV-XFC fast chargers, Wind, Traction, HVDC

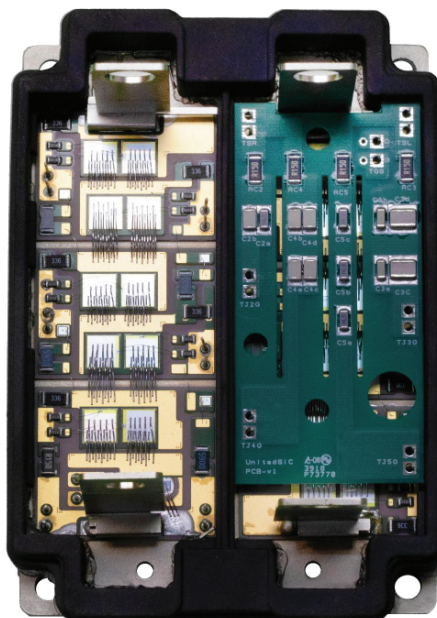


Photo Courtesy
of UnitedSiC

UnitedSiC's SiC Cascode technology can be leveraged to build much higher voltage power modules. The technique uses a series and parallel connected stack of 5.7 mΩ 1700 V normally-on SiC JFETs terminated with a single 30 V silicon MOSFET creating a composite enhancement-mode 6500 V switch. Using this technology, UnitedSiC is developing a 6500 V, 200 A half-bridge module at a fraction of the cost of a module using 6500 V SiC MOSFETs with 1.8X the threshold voltage, and 3X lower diode knee voltage.

SiC Active Harmonic Filter for Variable Frequency Drives

- Power stage electrical and mechanical design
- Implementation of one AHF prototype rated at 150A, 480V and a power density target value of 1.2 kW/liter
- Signal level verification between the controller and power stage
- Demonstrate high power test at nominal voltage (480V) and current (150A) according to design specifications with a 40 kHz minimum switching frequency

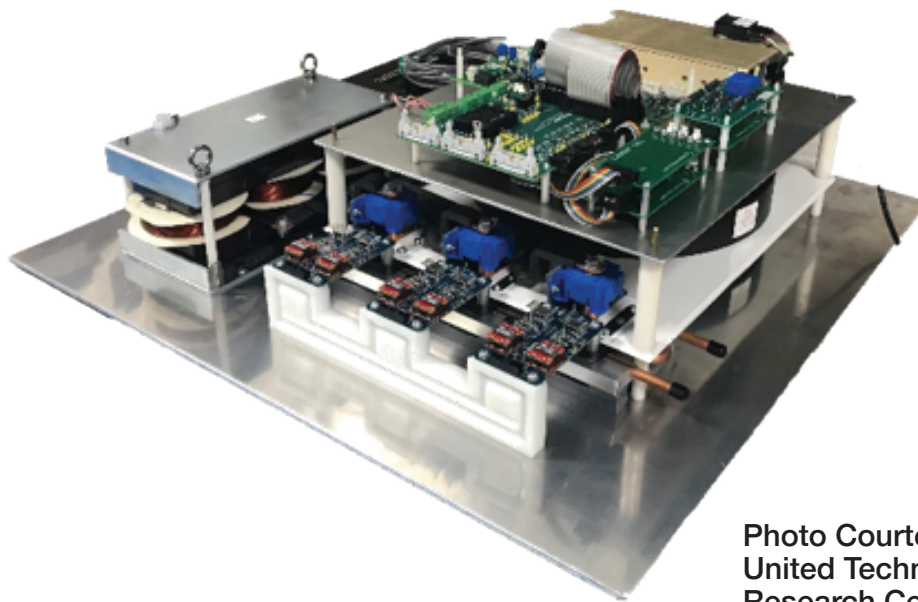


Photo Courtesy of
United Technologies
Research Center

This project will accelerate the adoption of wide band gap devices for commercial applications by enabling higher power density and lower cost active harmonic filters that can be used as power quality point of load solution. In addition to active harmonic filter applications, this technology could be used for high speed variable frequency drives and grid tie inverters. Providing a significant market opportunity for the application of WBG devices.

GaN based 22 kW On Board Charger

- GaN Systems-based 22 kW On Board Charger (OBC)
- Record setting power density >3.5 kW/liter
- 5 times smaller than Silicon-based OBC units
- 3 times lower power losses compared to Silicon-based OBC units



Photo courtesy
of GaN Systems

Size and weight of electronics in electric vehicles (EV) are some of the most important design criteria car manufacturers have, and speed and convenience of charging batteries is paramount to the EV consumer. On Board Chargers (OBC) entered the EV market at power levels of 3.3 kW and power densities of 0.7 kW/liter (4.7 liters) but the 8 to 12-hour long charging time has driven the market towards faster charging 22 kW OBCs. GaN-based solutions like this are the only way to achieve the very high-power density required to keep the size and weight manageable.

Design of a 250 kW, 1200 V SiC MOSFET-Based Three-Phase Inverter

Key Advantages in Using SiC Technology Being :

- Higher Voltages
- Switching Speeds
- Operating Temperatures

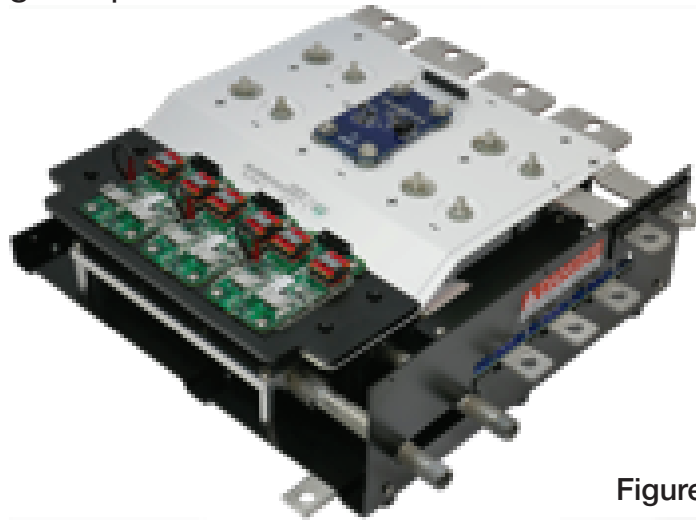


Figure 1

With some of the key advantages in using SiC technology being potentially higher voltages, switching speeds and operating temperatures; these make the use of SiC MOSFETs a candidate for high power, high speed and high temperature power converters, especially inverters.

Nevertheless, great care is required not only on the circuit design, but the integration and packaging to ensure that these advantages are realized in the ultimate design. In this case study a 250kW SiC three phase inverter is described (full details can be found in [2]). Wolfspeed's CAS325M12HM2 is an example of power module designed to utilize these key advantages of SiC.

This all-SiC power module is rated at 1200 V and 445 A, with a 3.6 mΩ on-resistance and a maximum operating temperature of 175°C [3], [4]. Detailed electromagnetic and thermal analyses were carried out to design the internal bus structure and gate drive circuit layout, resulting in the compact layout shown in Figure 1.

Dr Ty McNutt
Wolfspeed, Inc., USA

GaN-based High Density Power Package for Next Generation Power Conversion

- Vertical integration of GaN switching devices, driver and critical passive components inside a single package
- 60% reduction of parasitic parameters and form factor
- 2X improvement of driving speed
- Patent-pending

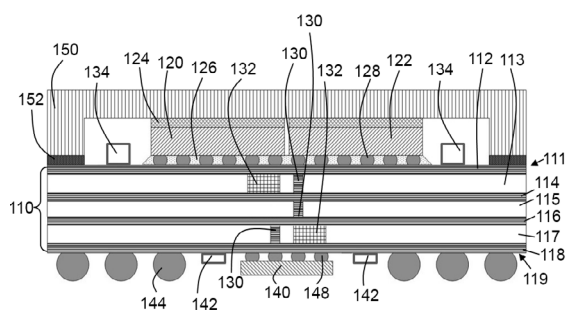
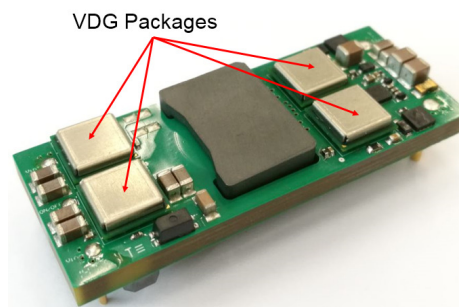


Photo Courtesy of VDG Solution

Photo Courtesy of
1/8 brick converter



The “Vertical-Driver-GaN (VDG)” power packaging technology is developed to solve the concurrent high speed & high efficiency driving of GaN devices for the next generation power conversion applications, such as DC-DC converters widely used in the telecommunication equipment, data centers, medical, military and aerospace industries. Example isolated 1/8 brick converter showed a maximum output power level of 500W and a power conversion efficiency of 96% were achieved, respectively.

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GaN-based High Density Power Module for Next Generation Power Conversion

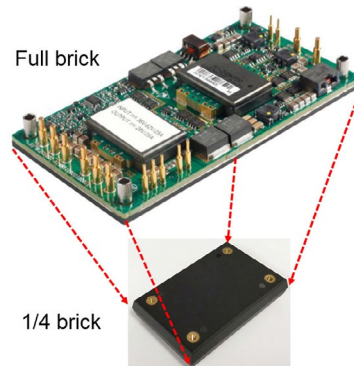


Photo Courtesy of 1KW 1/4 brick converter prototype

- Dual-side-cooled (DSC) packaging solution with GaN devices and integrated magnetic components
- 1KW DC-DC converter in a 1/4 brick format (power density 800W/in³)
- 97% power conversion efficiency
- Additional thermal interface to improve 50% heat dissipation limit

This technology platform aims to deliver an ultra-high density and efficiency power converter by developing the following key technologies to solve the electrical, thermal and packaging challenges for next generation power conversion: (1) “VDG” power packaging solution to achieve a reliable switching at ~MHz; (2) Magnetic integration to improve power density & efficiency; (3) Sandwiched package structure + Dual-side-cooled (DSC-SiP) thermal interface to breakthrough the heat dissipation limit. Example converter will exhibit a maximum power level of 1KW in a standard 1/4 brick with a power conversion efficiency up to 97%.

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GaN-based High Density Power Module for Next Generation Power Conversion

- A patent-pending solution with the adoption of GaN switching devices and “3PI” magnetic packaging technology
- 3% higher of power conversion efficiency over the full range
- Improved power density up to 100W/cm³
- 30% reduction of thermal resistance

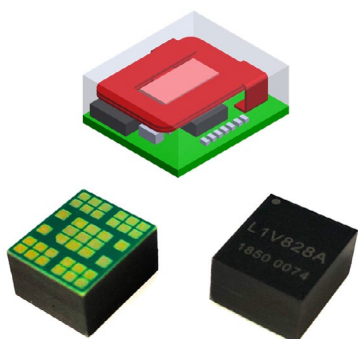
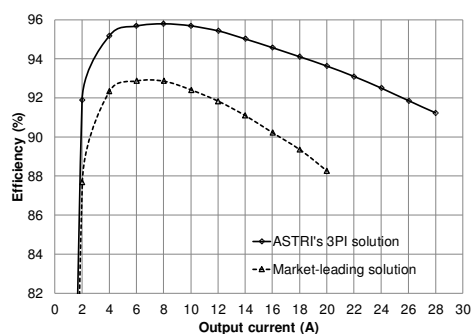


Photo Courtesy of 3PI POL Converter

Photo Courtesy of Benchmark with Market-leading Solution



By leveraging the high operation frequency and low power loss of GaN switching devices, a novel “3D Package-embedded Inductor (3PI)” magnetic packaging technology platform is developed to dramatically improve the electrical and thermal performance of key power conversion modules in data centers and telecommunication equipment. Example POL converter product (5V to 1.8V) showed a maximum power conversion efficiency of 96% and a power density of 100W/cm³ were achieved at the same time.

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Smart Power Hub (SPH) with SiC Power Module

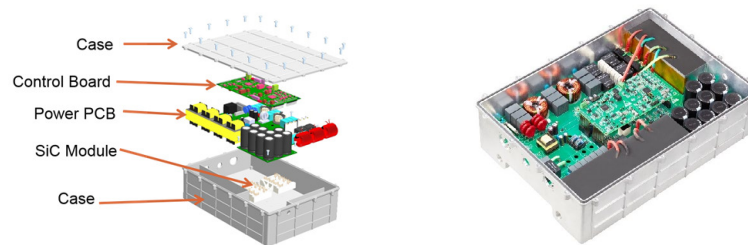


Photo Courtesy of 11KW SPH with SiC module

- A 11KW Smart Power Hub (SPH) integrates Electric Vehicle (EV) into power grid, home appliance and renewable energy
- Single phase ac grid and three-phase ac grid compatible
- Max efficiency 96.5%
- Power density 1.5KW/L with water cooling

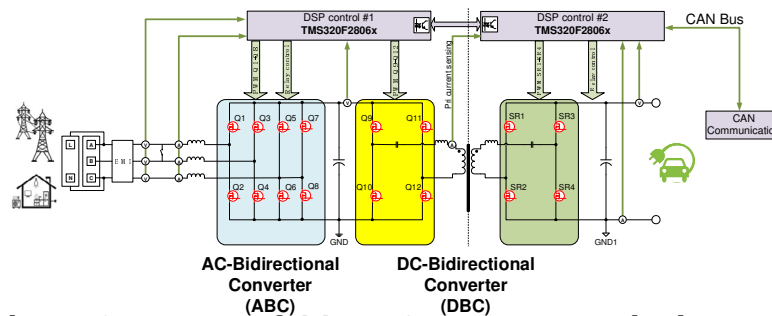


Photo Courtesy of 11KW SPH System Block Diagram

Towards the next generation power system, the SPH with SiC Module is developed for renewable energy applications, especially EV. With advanced SiC module and new topology, it brings the outstanding features of high integration, high density, high efficiency and bi-directional smart changing for power system. The SPH can support different operation modes with G2V, V2G and V2L from single phase ac grid to three-phase ac grid connections. Thanks to SiC module with high performance and fast switching, the 11KW SPH can achieve peak efficiency to 96.5% with power density up to 1.5KW/L.

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Affiliate Member

Project Title: Design, Fabrication, and Vehicular Testing of SiC Inverter for Heavy-Duty Vehicles

Objectives: 200 kW 1050 V DC bus SiC dual manufacturing and commercialization

Major Milestones: Deployment of engine coolant (up to 115°C max coolant high power density (> 20 kW/Liter) Gen-2 SiC dual inverter in JD 644K Hybrid Loader (upper right picture)

Deliverables: Dedicated cooling system for inverter eliminated in 644K Hybrid Loader

SOPO Task No.: **BP3-4.1**

TPOC/PI: Dr. Brij N. Singh

Email: SinghBrijN@JohnDeere.com

Phone: (701) 552-8516



WBG Technology Impact

1. Higher DC bus voltage, higher switching frequency, higher junction temp and heat flux, higher kW/Liter and kW/kg, smaller passives and system level advantages.
2. Heavy-duty off-highway and on-highway vehicles.
3. Production intent technology ready by 2020
4. Switching frequency (> 15 kHz with SiC versus 8 kHz with Si). DC bus cap (400 μ F with SiC versus 2000 μ F with Si). Inverter power density (> 20 kW/Liter with SiC versus ~11 kW/L with Si)

Additional Impacts of WBG Technology

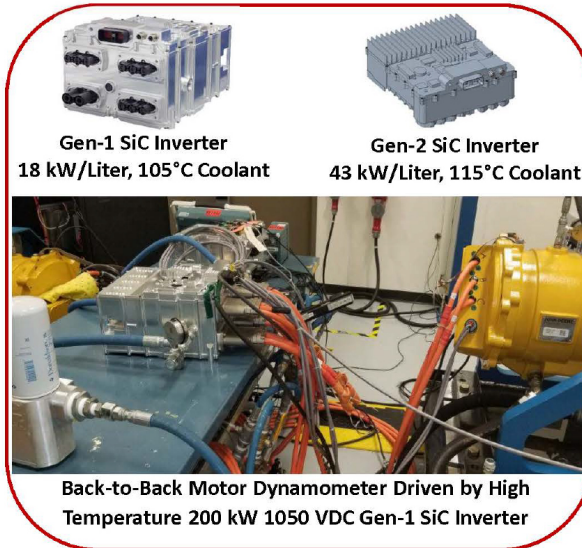
1. System approach could allow WBG inverter that will cost competitive with silicon device based inverter.
2. John Deere will designs and manufacturers SiC inverter in Fargo, North Dakota and heavy-duty vehicle will be manufactured in Iowa.
3. JDES hires numerous interns provides them WBG PE experience.
4. Cost competitive and performance superior SiC inverter can be realized by innovations by 9 C solutions: Capacitor, Coolant, Connector, Case, Control, coreless current sensor, Copper, Cable, and Cost.
5. Technology maturity : TRL 3/4 to TRL 6/7

PowerAmerica

Public Release

2018

Design, Fabrication, and Vehicular Testing of SiC Inverter for Heavy-Duty Vehicles



Project Objectives:

- 200 kW 1050 V DC bus SiC dual manufacturing and commercialization
 - WBG technology development and make strides for SiC inverter commercialization in heavy-duty vehicles

Achievements:

- Proposed power density of SiC inverter: > 20 kW/Liter
 - Power density achieved: 43 kW/Liter
- SiC inverter efficiency: 97%
 - > 98% efficiency achieved over coolant temperature
- Develop engine coolant (maximum 115°C WEG coolant) SiC inverter
 - Gen-1 SiC inverter proven with 105°C WEG coolant
 - Gen-2 SiC inverter deigned for 105°C WEG coolant

Project impact:

- Accelerating commercialization of the SiC inverter in heavy-duty off-highway and on-highway vehicles
- Power-dense, engine-coolant, and high-efficiency SiC inverter simplifies vehicle architecture and easier for system level integration

2



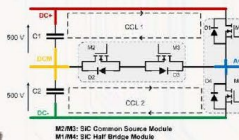
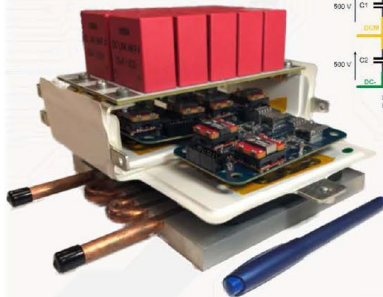
UA High Density PEBB-based Drives



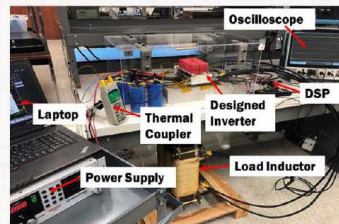
Industry Liaison:



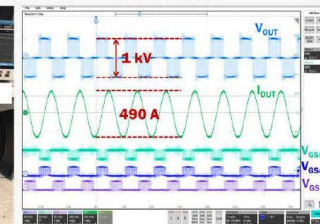
Achievements:
 Weight^{A1}: 2.1 kg
 Volume^{A1}: 12" x 6" x 4.5"
 Specific Weight Density[#]: 27.7 kW/kg
 Power Density[#]: 308.61 W/in³
 *Excluding EMI Filters & Cold plate



Gen 1: Si IGBT- SiC MOSFET Hybrid 100 kW PEBB



Continuous test setup



Captured Waveform

Gen 2: SiC MOSFET 150 kW PEBB

- Specific Power = 35.36 kW/kg
- Optimized for less EMI emission
- Continuous test at 64.5 kVA, $V_{Dlink} = 1$ kV, $f_{sw} = 30$ kHz
- Measured efficiency of 99.45%

CENTER FOR POWER OPTIMIZATION OF ELECTRO-THERMAL SYSTEMS

1

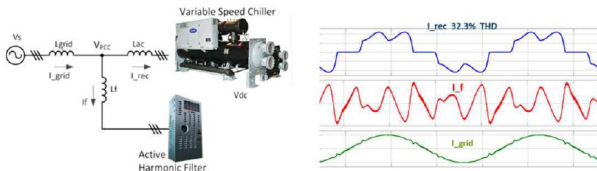
United Technologies Research Center



SiC Active Harmonic Filter for Variable Frequency Drives

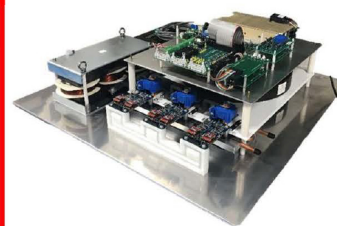
Objectives:

- Increase the adoption of WBG devices by developing and demonstrating on a commercial chiller a 150 Amps rated active harmonic filter that is 50% smaller and 20% projected lower cost (TRL 6)



WBG Technology Impact

This project will accelerate the adoption of wide band gap devices for commercial applications by enabling higher power density and lower cost active harmonic filters that can be used as power quality point of load solution. In addition to active harmonic filter applications, this technology could be used for high speed variable frequency drives and grid tie inverters. Providing a significant market opportunity for the application of WBG devices



- 4x lower inductance value
- 2x inductor volume reduction



- Integrated bus-bar DC Link Capacitor
- 10 nH parasitic inductance
- No need for snubber Capacitors



✓ 2x Higher power density - 1 kW/liter

Accomplishments

- ✓ Power stage electrical and mechanical design
- ✓ Implementation of one AHF prototype rated at 150A, 480V and a power density target value of 1.2 kW/liter
- ✓ Signal level verification between the controller and power stage
- ✓ Demonstrate high power test at nominal voltage (480V) and current (150A) according to design specifications with a 40 kHz minimum switching frequency.

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Phone: (860) 610-7163

PowerAmerica

For Public Release

2018



**International Technology
Roadmap for Wide Bandgap
Power Semiconductors**